

SMS++ File Format Reference

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Contents

1	Introduction	2
2	General netCDF file structure	3
3	General SMS++ file structure	4
3.1	Block description	5
3.2	Configuration description	5
3.2.1	BlockConfig description	5
3.2.2	BlockSolverConfig description	6
3.2.3	ComputeConfig description	7
3.3	Auxiliary structures formats	7
3.3.1	AbstractPath	7
3.3.2	SimpleDataMapping	11
4	File formats of existing :Block	15
4.1	SimpleMILPBlock	15
4.2	StructuredMILPBlock	17
4.3	MCFBlock	19
4.4	UCBlock	21
4.5	UnitBlock	26
4.5.1	ThermalUnitBlock	27
4.5.2	HydroUnitBlock	30
4.5.3	BatteryUnitBlock	35
4.5.4	IntermittentUnitBlock	38
4.5.5	SlackUnitBlock	39
4.6	HydroSystemUnitBlock	41
4.7	PolyhedralFunctionBlock	42
4.7.1	PolyhedralFunction	42
4.8	NetworkBlock	43
4.8.1	NetworkData	44
4.8.2	BusNetworkBlock	45
4.8.3	DCNetworkBlock	45
4.9	HeatBlock	46
4.10	SDDPBlock	49
4.11	StochasticBlock	52
4.12	BendersBlock	54
4.13	BendersBFunction	55
5	BlockConfig for existing :Block	57
5.1	Parameters of MCFBlock	57
6	ComputeConfig for existing :Solver	58
6.1	Standard parameters in Solver	58
6.2	Standard parameters in CDASolver	59
6.3	Standard parameters in Function	60
6.4	Standard parameters in C05Function	61
6.5	Parameters of MCFSolver<MCFC>	61
6.5.1	Parameters of MCFSolver<MCFSimplex>	62
7	ComputeConfig for existing :Function	63
7.1	Parameters of BendersBFunction	63

8 Examples	64
8.1 Seasonal Storage Valuation	64
8.1.1 BendersBlock	65
8.1.2 StochasticBlock	68
8.1.3 SDDPBlock	72
9 Conclusion	74

1 Introduction

This document describes the format of input files for the SMS++ components of the PLAN4RES project. These are the components tasked with performing the actual solution of the different optimization problems involved in the process. The production of these files possibly requires:

- a sophisticated data extraction and preparation phase;
- a tuning phase aimed at finding appropriate parameters for the different modelling and algorithmic options of the solution process.

The description of both phases is outside of the scope of this document, which only deals with the required final form of the input.

Insomuch as the set of types of optimization problems (`Block` in SMS++ parlance) to be solved within the project, and the set of available solution methods for doing it (`Solver` in SMS++ parlance) may be expanding over time, this document is intended as dynamic: as new components will added, the document will grow to cover the corresponding input formats. Non-backward-compatible modification of existing input formats will be kept to a minimum, and always agreed upon with the involved partners before being put into production. Appropriate versioning of both the software and this document will ensure consistency between the data preparation and data consumption processes.

The self-documenting nature of `NETCDF`, together with its portable readability and the schema definition of the various `Block`, will make it possible to re-use all or some blocks of one SMS++-generated model in other contexts.

2 General netCDF file structure

NETCDF (network Common Data Form) [1] is a set of interfaces for array-oriented data access and a freely distributed collection of data access libraries for C, Fortran, C++, Java, and other languages. The NETCDF libraries support a machine-independent format for representing scientific data. Together, the interfaces, libraries, and format support the creation, access, and sharing of scientific data.

NETCDF data is:

- Self-Describing. A NETCDF file includes information about the data it contains.
- Portable. A NETCDF file can be accessed by computers with different ways of storing integers, characters, and floating-point numbers.
- Scalable. A small subset of a large dataset may be accessed efficiently.
- Appendable. Data may be appended to a properly structured NETCDF file without copying the dataset or redefining its structure.
- Sharable. One writer and multiple readers may simultaneously access the same NETCDF file.
- Archivable. Access to all earlier forms of NETCDF data will be supported by current and future versions of the software.

For the purpose of this document, only a relatively small set of NETCDF concepts need be discussed.

- *NETCDF file*: is the basic container of NETCDF, e.g. corresponding to a standard file in the filesystem of the host machine. In the C++ interface used by SMS++ it corresponds to an object of type `netCDF::NcFile`. A typical constructor of the object requires the pathname of the file (a standard `char *`) and the mode, which can be e.g. `netCDF::NcFile::read` or `netCDF::NcFile::replace` for reading and writing, respectively.
- *NETCDF group*: a NETCDF file is organised into several *groups*, which can be nested inside each other to an arbitrary degree. In the C++ interface used by SMS++, each group corresponds to an object of type `netCDF::NcGroup`; note that `netCDF::NcFile` derives from `netCDF::NcGroup`, and therefore all capabilities of the group are shared by the file (but not vice-versa). Each *group* has a unique string name by which it can be retrieved; `netCDF::NcGroup` (and, therefore, `netCDF::NcFile`) has a method `getGroup()` taking as input a `std::string` with the group name and returning an `netCDF::NcGroup` object containing the *child group* with the given name (if any) of that group.
- *NETCDF attribute*: is a simple data point specific of a `netCDF` group, having a unique string name by which it can be retrieved and a simple value (`int`, `double`, `std::string`, ...). In the C++ interface used by SMS++, it is represented by a `netCDF::NcGroupAtt` object, which can be obtained by its enclosing `netCDF::NcGroup` via the method `getAtt()` taking in input its name. Then, the method(s) `getValue()` (available with different signatures corresponding to the different supported basic data types) allows to retrieve its value.
- *NETCDF dimension*: it basically represent the different indices over which the actual data is indexed, each of which is given a name. In the C++ interface used by SMS++, it is represented by a `netCDF::NcDim` object, which can be obtained by its enclosing `netCDF::NcGroup` via the method `getDim()` taking in input its name. The *dimension* (a `size_t` value) is returned by the method `getSize()`.
- *NETCDF variable*: this is the container for the “complex” data of a *groups*. It has a name (`std::string`), a basic data type (`int`, `double`, `std::string`, ...) and in principle any number of *dimensions*. In the C++ interface used by SMS++, it is represented by a `netCDF::NcVar` object, which can be obtained by its enclosing `netCDF::NcGroup` via the method `getVar()` taking in input its name. Once the `netCDF::NcVar` object is obtained, individual values of the *variable* can be read by its method(s) `getVar()` (available with different signatures corresponding to the different supported basic data types), taking a `std::vector<size_t>` as long as the the number of *dimensions* of the *variable* (obtainable e.g. with its method `NgetDimCount()`).

The above components allow to read and write structured data files with basically any nested format, which makes NETCDF ideal for representing the data characterizing SMS++, as discussed below.

3 General SMS++ file structure

In SMS++, files can describe two different kinds of objects:

1. Block objects, representing the data characterizing a specific instance of a (structured) optimization problem to be solved;
2. Configuration objects, representing parameters of the solution process, either pertaining to a Block (say, which of the different available formulations of the optimization problem should be used), or to a Solver used to actually perform the optimization process.

Both objects have `serialize()` and `deserialize()` methods allowing to save the current state to a given NETCDF *file/group*, as well as *factories* allowing to create a new object from scratch out of a given NETCDF *file/group*.

The issue of SMS++ is that an optimization problem is represented by a Block, which can have any number of sub-Block (recursively) representing parts of the problem with a specific structure. Yet, “specific structure” here may mean that the enclosing Block can make some assumptions on what kind of sub-Block it contains, but it may not know exactly their type. In C++ parlance, Block is an abstract base class, and specific optimization problems are represented by specific derived classes (:Block). A given :Block can make assumption on the type of its sub-Block, but there can be many different variants of it, represented by different derived classes from the same base one. Thus, clearly the data-reading must be organized in such a way that each :Block is responsible for retrieving that which is necessary to it without making assumptions about the global organization of the file. This does not only apply to the data of the instance, but also to *configuration* options that a :Block may have. In fact, several aspect of the :Block representing a given problem (which formulation is used, if and how variables and constraint of the formulation are dynamically generated, which parts of the primal/dual solution are saved, ...) can be implemented in different ways. Note that these decisions are in principle independent from the specific instance, and therefore are conceptually to be stored in, and retrieved from, a separate container than that holding the data of the Block; this is why the Configuration class is provided. Since Block is a tree-shaped nested data structure, Configuration also have to be such. Clearly, NETCDF perfectly suit this kind of requirement; in particular, the standing assumption for both classes is that

*the data of one object (Block/Configuration) is all to be found in the same NETCDF group;
the data of its “sub-objects” (sub-Block/sub-Configuration) is to be found in child groups of the group.*

Hence, the format of a SMS++ NETCDF file depends on whether it holds Block information, Configuration information, or both. For the latter, there are actually two separate kind of Configuration information that can be attached to a :Block. The first describes the above-mentioned options that are inherent to the :Block itself, which are described by the specific derived class BlockConfig of Configuration. However, a :Block need be solved, which is done by Solver objects; each Block can have an arbitrary number of Solver “registered” with it. A Solver can be an arbitrarily complex solution process; in particular, the structure of Block immediately implies that one can register Solver with any sub-Block, which is actually useful e.g. to decomposition algorithms. Hence, not only each :Solver can have an arbitrarily large set of algorithmic configuration options; an arbitrarily large set of Solver can be attached to a Block and to all of its sub-Block, recursively. Thus, the different derived class BlockSolverConfig of Configuration is provided to describe all this information. Note, again, that this is conceptually separate from both the specific instance encoded in the :Block and the choice of the Block-specific options encoded in the BlockConfig, which justifies why a separate container is used.

To allow holding all this different information, a SMS++ NETCDF file has the following general structure. It must contain (in the `netCDF::NcFile` object, which is also a `netCDF::NcGroup`) an *attribute* `SMS++_file_type` of `int` type which specifies which of three different kinds of file it is. The values are encoded in the enum `smpp-netCDF_file_type` defined in `SMSTypedefs.h`, as follows:

- `eProbFile = 0`: the *file* (which is also a *group*) has any number of child *groups* with names `Prob_0`, `Prob_1`, ... In turn, each child *group* has *exactly three* child *groups* with names `Block`, `BlockConfig` and `BlockSolver`, respectively. The first is intended to contain the serialization of a :Block, the second the serialization of a :BlockConfig of the same :Block, and the third the serialization of a :BlockSolverConfig of the same :Block, although any of the three can in principle be empty. If any of the child is not empty, it *must* necessarily contain all the information necessary to reconstruct the specific instance.

- eBlockFile = 1: the *file* (which is also a *group*) has any number of child *groups* with names Block_0, Block_1, ... Each child *group* contains the serialization of a :Block.
- eConfigFile = 2: the *file* (which is also a *group*) has any number of child *groups* with names Config_0, Config_1, ... Each child *group* contains the serialization of a :Configuration. Note that no distinction is required between a :BlockConfig and a :BlockSolverConfig, since they are both derived from Configuration.

Note that (some variants of) the `deserialize()` methods of Block/ Configuration allow to directly specify which of the different objects stored in the file to read via its index i in the “name”.

To completely describe the file formats we now have to describe the format of the NETCDF *group* as far as the base Block and Configuration (for the latter, actually both sub-classes BlockConfig and BlockSolverConfig) are concerned, i.e., the (possibly, small) part of the contents that do not depend on the specific derived class.

3.1 Block description

The only content that any NETCDF *group* describing a :Block must necessarily have is a string *attribute* with name type containing the classname of the object. This must be *exactly* what is returned by the protected virtual method `classname()`, since it is used in the *factory* when deserializing the object. Note that template classes derived from Block are allowed, with the standard notation `class_name<template_paramaters>`. However, due to a technical issue about using macros (see the comments to `SMSpp_insert_in_factory_cpp_*` in `SMSTypedefs.h`), the name given to the class must not contain any “,” (comma). That is, something like “myBlock< std::pair< int , int > >” cannot be used when defining the Block. The typical way is to resort to a using declaration, i.e.,

```
using myBlock_ii = myBlock< std::pair< int , int > >;
SMSpp_insert_in_factory_cpp_1.t( myBlock_ii );
```

This implies that the type of that Block as returned by `classname()`, and therefore the one that has to be used in the NETCDF file, must necessarily be “myBlock_ii”. The exact string will typically be defined in the .cpp file where the Block is implemented, precisely in the corresponding invocation of the macro `SMSpp_insert_in_factory_cpp_*`. In all non-complicated-template cases, the name should just be the only possible form in which the classname can be written.

Besides the mandatory type *attribute*, all Block support an optional name *attribute* of string type. If present, this is supposed to contain the “name” of the Block, as returned by the `name()` method; if not present (and not set in-memory via `set_name()`), `name()` will return an empty string. This is not a classname, but rather a string describing something about the Block that is supposedly useful when printing/debugging it.

3.2 Configuration description

The only content that any NETCDF *group* describing a :Configuration must necessarily have is a string *attribute* with name type containing the classname of the object. This must be exactly what is returned by the protected virtual method `classname()`, since it is used in the *factory* when deserializing the object. Note that template classes derived from Configuration are allowed, with the standard notation `class_name<template_paramaters>`; see for instance the template `SimpleConfiguration<>` class. However, the same issue with “,” (commas) not being allowed to the class name as in Block applies here, see §3.1 for more details.

3.2.1 BlockConfig description

Besides the mandatory type *attribute* of any :Configuration, a SMS++ NETCDF *group* for a BlockConfig can contain the following:

- the *group* `static_constraints` containing a Configuration object for the static constraints of the Block;
- the *group* `dynamic_constraints` containing a Configuration object for the dynamic constraints of the Block;
- the *group* `static_variables` containing a Configuration object for the static variables of the Block;
- the *group* `dynamic_variables` containing a Configuration object for the dynamic variables of the Block;

- the *group* *objective* containing a Configuration object for the objective of the Block;
- the *group* *is_feasible* containing a Configuration object for the *is_feasible()* method of the Block;
- the *group* *is_optimal* containing a Configuration object for the *is_optimal()* method of the Block;
- the *group* *solution* containing a Configuration object for the methods of the Block dealing with solutions (*get_Solution()* and *map_[back/forward]_solution()*);
- the *group* *extra* containing a Configuration object, which has no direct use in the base Block class, but is added so that derived classes can put there any configuration information without having to define further derived classes form BlockConfig (which, however, they can still do if they want);
- the *dimension* *n_sub_Block* containing the number of BlockConfig descriptions for the sub-Block of the current :Block;
- with *n* being the size of *n_sub_Block*, *n groups*, with name *sub-BlockConfig_i* for all $i = 0, \dots, n - 1$, containing each the description of a BlockConfig for one of the sub-Block of the current :Block.

Of all these, only the name *attribute* and the *n_sub_Block dimension* are mandatory (they must exist, although they may be empty/zero). All other groups may not exist, in which case the corresponding field of the class is filled with a *nullptr*, indicating that the “default” configuration (whatever that may mean for the :Block in question) have to be used. Note that that the matching between the sub-BlockConfig and the sub-Block is positional: the BlockConfig found in the group “sub-BlockConfig_i” is that for the *i*-th sub-Block. Note that the vector of sub-BlockConfig is allowed to be of different size than the number of sub-Block; if it is larger any extra BlockConfig is simply ignored, if it is shorter then all missing sub-BlockConfig are treated as *nullptr* (default configuration).

3.2.2 BlockSolverConfig description

Besides the mandatory type attribute of any :Configuration, a SMS++ NETCDF *group* for a BlockSolverCon fig can contain the following:

- the *dimension* *n_SolverConfig* containing the number of Solver that are to be attached to this Block, and therefore the (expected) number of their :Configuration objects;
- the *variable* *SolverNames*, of type string and indexed over the *dimension* *n_SolverConfig*; the *i*-th entry of the variable is assumed to contain the classname of a :Solver object to be attached to the Block (this must be exact, e.g., as returned by the protected virtual method *name()*, since it is used in the *factory* when creating the object);
- with *n* being the size of *n_SolverConfig*, *n groups*, with name *SolverConfig_i* for all $i = 0, \dots, n - 1$, containing each the description of a :Configuration object for the *i*-th :Solver;
- the *dimension* *n_BlockSolverConfig* containing containing the number of BlockSolverConfig descriptions for the sub-Block of the current :Block;
- with *m* being the size of *n_BlockSolverConfig*, *m groups*, with name *BlockSolverConfig_i* for all $i = 0, \dots, m - 1$, containing each the description of a BlockSolverConfig for one of the sub-Block of the current :Block.

Of all these, only the *n_SolverConfig* and *n_BlockSolverConfig dimensions* and the *SolverNames variable* are mandatory; the individual *groups* may not exist if the corresponding *dimension* is 0. Note that that the matching between the sub-BlockSolverConfig and the sub-Block is positional: the BlockSolverConfig found in the group “BlockSolverConfig_i” is that for the *i*-th sub-Block. Note that the vector of sub-BlockSolverConfig is allowed to be of different size than the number of sub-Block; if it is larger any extra BlockSolverConfig is simply ignored, if it is shorter then all missing sub-BlockConfig are treated as *nullptr* (default configuration).

Note that the :Configuration objects to be found in the *groups* *SolverConfig_i* are not “any” :Configura tion; rather, they are assumed to be of the specific type ComputeConfig, whose format is described in the next subsection.

3.2.3 ComputeConfig description

ComputeConfig (defined in ThinComputeInterface.h) is a class of :Configuration objects specifically tailored for representing sets of algorithmic parameters of solution algorithms, like the ones presumably required by a :Solver. Besides the mandatory type attribute of any :Configuration, a SMS++ NETCDF *group* for a ComputeConfig can contain the following:

- the *attribute* `diff` of `int` type containing the value for the `f_diff` field of the ComputeConfig (basically, a `bool` telling if the information in it has to be taken as “the configuration to be set” or as “the changes to be made from the current configuration”);
- the *dimension* `num_int_par` containing the number of `int` parameters;
- the *variable* `int_par_names`, of type `string` and indexed over the *dimension* `num_int_par`; the *i*-th entry of the variable is assumed to contain the string name of an `int` parameter;
- the *variable* `int_par_vals`, of type `int` and indexed over the *dimension* `num_int_par`; the *i*-th entry of the variable is assumed to contain the value of the `int` parameter whose string name is to be found in the *i*-th entry of `int_par_names`;
- the *dimension* `num_dbl_par` containing the number of `double` parameters;
- the *variable* `dbl_par_names`, of type `string` and indexed over the *dimension* `num_dbl_par`; the *i*-th entry of the variable is assumed to contain the string name of a `double` parameter;
- the *variable* `dbl_par_vals`, of type `double` and indexed over the *dimension* `num_dbl_par`; the *i*-th entry of the variable is assumed to contain the value of the `double` parameter whose string name is to be found in the *i*-th entry of `dbl_par_names`;
- the *dimension* `num_str_par` containing the number of `string` parameters;
- the *variable* `str_par_names`, of type `string` and indexed over the *dimension* `num_str_par`; the *i*-th entry of the variable is assumed to contain the string name of a `string` parameter;
- the *variable* `str_par_vals`, of type `string` and indexed over the *dimension* `num_str_par`; the *i*-th entry of the variable is assumed to contain the value of the `string` parameter whose string name is to be found in the *i*-th entry of `str_par_names`;
- the *group* `extra` containing a Configuration object, which has no direct use in the base ComputeConfig class, but is added so that derived classes can put there any configuration information without having to define further derived classes from ComputeConfig (which, however, they can still do if they want).

The three *dimensions* `num_*_par` are mandatory. If any of these is zero, the corresponding variables `*_par_names` and `*_par_vals` may not be defined. The extra *group* may not exist, in which case the corresponding Configuration object is set to a `nullptr`.

3.3 Auxiliary structures formats

This section presents the description of some auxiliary structures that are necessary to describe some :Block. The AbstractPath, presented in §3.3.1, is used to specify a path from a Block to some SMS++ element (Block, Constraint, Variable, Objective, or Function) defined in that Block or in any of their nested Block, recursively. The SimpleDataMapping, presented in §3.3.2, offers a mechanism for changing the data of a Block.

3.3.1 AbstractPath

The AbstractPath represents a path from a Block, here referred to as the reference Block, to one of its elements (the target element): a Block, a Constraint, a Variable, an Objective, or a Function. The target element can directly belong to the reference Block itself (even be the reference Block itself) or belong to any of its sons, recursively. The reference Block is not explicitly represented in the path. In fact, the representation of the path is independent from the reference Block. For the path to be meaningful, the reference Block should be clear from the context. The reference Block must be available when the path is constructed and when the target element is

retrieved. Furthermore, the type of the target element cannot always be inferred from the path. The type of the target element, therefore, must also be known from the context.

The `AbstractPath` is particularly useful for the serialization and deserialization of pointers to objects. If an object stores pointers to other objects (for example, a `Function` has a set of pointers to `Variable`, others have pointers to `Block`), these pointers cannot be serialized and deserialized as such. Rather, an “abstract representation” of these pointers has to be serialized, from which the pointers can be reconstructed at deserialization time; this is the facility that `AbstractPath` offers. The fact that the representation of the path is independent from the reference `Block` facilitates its serialization and deserialization, and makes it possible to use the same path to target different objects (say, two `Constraint` “in the same position” in two “twin” `Block` can be represented by the same `AbstractPath`).

The path is defined as a sequence of nodes. Each node has one of the following types:

- ‘O’, if the node is associated with an `Objective`;
- ‘B’, if the node is associated with a `Block`;
- ‘C’, if the node is associated with a static `Constraint`;
- ‘c’, if the node is associated with a dynamic `Constraint`;
- ‘V’, if the node is associated with a static `Variable`;
- ‘v’, if the node is associated with a dynamic `Variable`.

Notice that, for `Constraint` and `Variable`, an upper case letter indicates that the element is static, while a lower case letter indicates that the element is dynamic. Also notice that there is no node associated with a `Function`, but this does not prevent one from constructing a path to a `Function`. Although the nodes in the path are stored in forward order, i.e., the first node is the origin of the path and the last one is the destination, it is easier to understand the path if we look at it backwards.

Consider the path from some reference `Block` to some `Variable`. The last node in this path necessarily has the ‘V’ or ‘v’ type, indicating this is a path to a `Variable`. This `Variable` belongs to some `Block` and it is either static or dynamic. This last node has all information needed to retrieve this `Variable` from its father `Block`: an indication of whether the `Variable` is static or dynamic, the index of the group to which it belongs, and the index of the `Variable` within that group.

Note: The index of a `Variable` (or `Constraint`) within a group is a single number and may not be entirely obvious which number it should be (specially for multidimensional arrays and (multidimensional) arrays of lists). Please refer to the §3.3.1.1 for an explanation of the index of a `Variable` (or `Constraint`) within a group.

A ‘V’ or ‘v’ node is always preceded by a ‘B’ node, unless it is the only node in the path. If the path has only a single node, which is a ‘V’ or ‘v’ node, then the `Variable` this path refers to is defined in the reference `Block`. In other words, if the target `Variable` of the path belongs to the reference `Block`, then the path is formed by a single node whose type is either ‘V’ or ‘v’.

A ‘B’ node is associated with a `Block` and may contain the index of this `Block` in the vector of nested `Blocks` of its father `Block`. There is one case in which this node does not have this index, which is when the node is associated with the reference `Block`. Since the reference `Block` is the root of the tree that contains the path, no allusion to the father of the reference `Block` must be made. In this case, the index has the value $+\infty$. If the index of a ‘B’ node is not $+\infty$, then this node is necessarily preceded by another ‘B’ node, which is associated with the father of that `Block`.

An ‘O’ node, which is associated with an `Objective`, is either preceded by a ‘B’ node (which is associated with the `Block` that owns that `Objective`) or is the only node in the path. If it is the last node in the path, then the target element is either this `Objective` or the `Function` in that `Objective` (in which case that `Objective` is an `FRealObjective`). The type of the target element must be known from the context. If this is not the last node in the path, then the type of the next node in the path is ‘B’ and it is associated with the inner `Block` of either a `BendersBFunction` or a `LagBFunction` which is the `Function` of that `Objective` (and thus that `Objective` is actually an `FRealObjective`).

Finally, a ‘C’ or ‘c’ node, which is associated with a `Constraint`, has characteristics pertaining both the ‘V’ (and ‘v’) and the ‘O’ nodes. Like the ‘V’ (and ‘v’) node, it has an indication of whether it is static or not, the index of the group to which it belongs, and the index of the `Constraint` within that group (exactly as defined for `Variable`). Like an ‘O’ node, a ‘C’ or ‘c’ node is either preceded by a ‘B’ node (which is associated with the `Block` that owns that `Constraint`) or is the only node in the path. If it is the last node in the path, then the target element is either this `Constraint` or the `Function` in that `Constraint` (in which case that `Constraint` is an `FRowConstraint`). The type of the target element must be known from the context. If this is not the last node in the path, then the type of the next node in the path is ‘B’ and it is associated with the inner `Block` of either a `BendersBFunction` or a `LagBFunction` which is the `Function` of that `Constraint` (and thus that `Constraint` is actually an `FRowConstraint`).

Besides the description of an `AbstractPath`, we also provide a description of a vector of `AbstractPath`. In §3.3.1.1 we present the description of a single `AbstractPath` and in §3.3.1.2 we present the description of a vector of `AbstractPath`.

3.3.1.1 A single `AbstractPath`

An `AbstractPath` is described by the following `NETCDF` *variables* and *dimensions*. These do not need to be in a specific `NETCDF` *group*, unless of course one has to specify more than one `AbstractPath` in a *group*, in which case it is necessary to make specific *groups* because the name of a *variable/dimension* cannot be duplicated in a given *group*.

- the *dimension* `TotalLength` containing the number N of nodes in the path;
- the *variable* `PathNodeTypes`, of type `netCDF::NcChar` and indexed over the *dimension* `TotalLength`, contains the types of the nodes in the path;
- the *variable* `PathGroupIndices`, of type `netCDF::NcUInt64` and indexed over the *dimension* `TotalLength`, contains the group indices associated with each node in the path;
- the *variable* `PathElementIndices`, of type `netCDF::NcUInt64` and indexed over the *dimension* `TotalLength`, contains the element indices associated with each node in the path. This *variable* is optional. If it is not provided, then `PathElementIndices[i]` is assumed to be `inf` for all $i \in \{0, \dots, \text{TotalLength} - 1\}$.

Note: All indices mentioned here belong to zero-based numbered sequences, that is, sequences whose first element is 0.

The i -th element in each of these arrays is associated with the i -th node in the path, for $i \in \{0, \dots, N - 1\}$. The type of a node may be indicated by any of the following letters:

- ‘O’, if the node is associated with an `Objective`;
- ‘B’, if the node is associated with a `Block`;
- ‘C’, if the node is associated with a static `Constraint`;
- ‘c’, if the node is associated with a dynamic `Constraint`;
- ‘V’, if the node is associated with a static `Variable`;
- ‘v’, if the node is associated with a dynamic `Variable`.

Notice that, for `Constraint` and `Variable`, an upper case letter indicates that the element is static, while a lower case letter indicates that the element is dynamic.

For a node whose type is ‘O’, i.e., representing a node associated with an `Objective`, the values stored in the variables `PathGroupIndices` and `PathElementIndices` are meaningless. If it is the last node in the path, then the path refers to an `Objective`. Otherwise, the next node in the path must be of type ‘B’.

For node whose type is ‘B’, i.e., representing a node associated with a `Block` B , the values stored in the variable `PathElementIndices` are meaningless. If this is the i -th node in the path with $i < N - 1$, i.e., it is not the last

node of the path, the value stored in `PathGroupIndices[i]` is the index of the sub-Block of Block B which is the next node in the path. If $i = N - 1$, i.e., it is the last node in the path, then the destination of the path is a Block which must be

- the Block B itself if `PathGroupIndices[i] = +∞`;
- the j -th sub-Block of Block B , where $j = \text{PathGroupIndices}[i]$.

If the i -th node has type ‘C’ or ‘c’, representing a Constraint, or ‘V’ or ‘v’, representing a Variable, the variable `PathGroupIndices[i]` stores the index of the (static or dynamic) group to which the element belongs. The variable `PathElementIndices[i]` stores the index of the element in that group.

A static group can be one of three types:

1. It is a single Constraint or Variable;
2. It is a vector of Constraint or Variable;
3. It is a multidimensional array of Constraint or Variable.

In the first case, in which the group is a single Constraint or Variable, the value of `PathElementIndices[i]` is 0. In the second case, in which the group is a vector of Constraint or Variable, `PathElementIndices[i]` is the index of the element in that vector. In the last case, in which the group is a multidimensional array of Constraint or Variable, `PathElementIndices[i]` is the index of the element in the vectorized multidimensional array in row-major layout. For instance, if the multidimensional array has two dimensions with sizes m and n , respectively, then the element at position (p, q) would have an element index equal to $np + q$ (recall the indices start from 0). In general, for a multidimensional array with k dimensions with sizes $(n_0, \dots, n_{k-1})$, the element at position $(i_0, \dots, i_{k-1})$ would have an element index equal to

$$\sum_{r=0}^{k-1} \left(\prod_{s=r+1}^{k-1} n_s \right) i_r. \quad (1)$$

A dynamic group can be one of three types:

1. It is list of Constraint or Variable;
2. It is a vector of lists of Constraint or Variable;
3. It is a multidimensional array of lists of Constraint or Variable.

In the first case, in which the group is a list of Constraint or Variable, `PathElementIndices[i]` is the index of the element in that list. In the second case, in which the group is a vector of lists of Constraint or Variable, for an element at position j of the k -th list of the vector, `PathElementIndices[i]` is given by

$$j + \sum_{t=0}^{k-1} s_t$$

where s_t is the number of elements in the t -th list of the vector. The last case is analogous.

3.3.1.2 Vector of AbstractPath

A vector of `AbstractPath` is described by the following NETCDF *variables* and *dimensions*. These do not need to be in a specific NETCDF *group*, unless of course one has to specify more than one vector of `AbstractPath` in a *group*, in which case it is necessary to make specific *groups* because the name of a *variable/dimension* cannot be duplicated in a given *group*.

- the *dimension* `PathDim`;
- the *dimension* `TotalLength`;

- the *variable* PathStart, of type `netCDF::NcUint64` and indexed over PathDim;
- the *variable* PathNodeTypes, of type `netCDF::NcChar` and indexed over the *dimension* TotalLength;
- the *variable* PathGroupIndices, of type `netCDF::NcUint64` and indexed over the *dimension* TotalLength;
- the *variable* PathElementIndices, of type `netCDF::NcUint64` and indexed over the *dimension* TotalLength. This *variable* is optional. If it is not provided, then PathElementIndices[i] is assumed to be inf for all $i \in \{0, \dots, \text{TotalLength} - 1\}$.

The format of a vector of AbstractPath is similar to that of a single AbstractPath with a few differences. Firstly, there is a *dimension* called PathDim which contains the number of paths that are described in that group. Secondly, there is a one-dimensional *variable* called PathStart, indexed over PathDim, which contains the index where the description of each AbstractPath starts. Then there are the one-dimensional *variables* PathNodeTypes, PathGroupIndices, and PathElementIndices, which store the types of nodes in the paths, the group indices, and the element indices, respectively. The description of the k -th AbstractPath, for $k < \text{PathDim} - 1$ is given in those arrays between the indices PathStart[k] and PathStart[k+1], i.e., in

(PathNodeTypes[PathStart[k]], ..., PathNodeTypes[PathStart[k+1] - 1]),

(PathGroupIndices[PathStart[k]], ..., PathGroupIndices[PathStart[k+1] - 1])

and

(PathElementIndices[PathStart[k]], ..., PathElementIndices[PathStart[k+1] - 1]).

The description of the last path is given between the indices PathStart[PathDim - 1] and TotalLength - 1, where TotalLength is a *dimension* containing the size of the arrays PathNodeTypes, PathGroupIndices, and PathElementIndices (and, therefore, these arrays are indexed over TotalLength).

The paths are represented in the same format as specified for a single AbstractPath, except that here they are concatenated in the arrays PathNodeTypes, PathGroupIndices, and PathElementIndices.

3.3.2 SimpleDataMapping

DataMapping defines an interface for all types of data mappings. The idea of a data mapping is to allow, in particular, to map the values given by a vector of double into the data of some object. Among its functions, it has the set_data() pair of functions, which have the following signature:

```
virtual void set_data( const std::vector< double > & data ,
                      c_ModParam issueMod = eModBlck ,
                      c_ModParam issueAMod = eModBlck ) const;

virtual void set_data( const Eigen::ArrayXd & data ,
                      c_ModParam issueMod = eModBlck ,
                      c_ModParam issueAMod = eModBlck ) const;
```

The idea of these functions is that the values of some data of an object can be modified considering the given data parameter. Typically, a DataMapping could be used to set the data of a Block. In this case, a pointer to that Block must be available. Pointers to a Block can be serialized and deserialized considering its AbstractPath, which is relative to some reference Block.

SimpleDataMapping is a template class that derives from DataMapping and is used to define some common kinds of data mapping. We define two vectors: the “large” one and the “small” one. The “large” vector refers to the vector that is given as input to the set_data() method. This is the vector containing all the data that can be used by the SimpleDataMapping. The “small” vector is a vector formed by a subset of the elements of the “large” vector. This is the vector that will effectively be used to perform some computation. This computation is typically the task of changing the data of some object based on this “small” vector. There is a mapping defined by the SetFrom set that specifies which elements of the “large” vector are used to compose the “small” vector. This SetFrom set

contains the indices of these elements in the “large” vector. The “small” vector is the one that will typically impact the data of some object. The `SetTo` set can be used to specify which part of this data is affected.

As an example, consider the case in which the “large” vector contains data related to costs and capacities of arcs of a network. Suppose this network is represented by a fictitious class `Network`. A `SimpleDataMapping` could be used to set the capacities of the arcs of a `Network` object considering the data provided by this “large” vector. Some elements of this “large” vector would be extracted and form a “small” vector containing the capacities of some arcs. The indices of the elements that are extracted from the “large” vector are specified by the `SetFrom` set. This set could be, for instance, the set `{0,3,8,11}`. This means that the elements at positions 0,3,8, and 11 in the “large” vector are selected to form a “small” vector with four elements. This “small” vector would then be used to change the capacities of some arcs of the `Network` object. The arcs whose capacities would be modified could be specified by the `SetTo` set. This could be the set `[2,6)`, for instance, stating that the arcs with indices 2,3,4, and 5 would have their capacities changed according to the “small” vector.

Besides the `SetFrom` and `SetTo` sets, the `SimpleDataMapping` also has a pointer to a function, which is invoked within the `set_data()` method. This is a function that receives, in particular, a pointer to a `Block`, the “small” vector, and the `SetTo` set. If the `SetTo` set is a `Block::Subset`, then the type of this function is

```
Block::FunctionType< typename std::vector< DataType >::const_iterator ,
                    SetTo && , bool >
```

If the `SetTo` set is a `Block::Range`, then the type of this function is

```
Block::FunctionType< typename std::vector< DataType >::const_iterator ,
                    const SetTo & >
```

A `Range` represents the closed-open interval $[a, b)$ set of indices (integers), determined by the integers a and b . A `Subset` represents a set of indices by explicitly listing its elements. Please refer to the definition of `Block::FunctionType` for completely understanding the type of this function.

In the network example above, this function could be, for instance,

```
void set_capacities( Network * network ,
                    std::vector<double>::const_iterator capacities ,
                    const Range & indices ,
                    c_ModParam , c_ModParam );
```

Ignoring the details of the type of this function, this is a function that receives a pointer to a `Network` object, a vector of capacities, and a `Range` of indices. This function could be responsible for changing the capacities of the arcs (of the given `Network` object) specified by the `indices` parameter according to the given capacities.

As you can see, the function associated with a `SimpleDataMapping` receives a pointer to a `Block` as its first parameter. This is a pointer to the caller object; the object that “invokes” the function. In the network example, this would be a pointer to a `Network` object whose capacities would be modified.

Finally, the `SimpleDataMapping` is also determined by the type of the data of the “small” vector, the `DataType`.

Notice that a `SimpleDataMapping` is general enough in that it is not only meant to change the data of some object, but perform arbitrary computation defined by the function associated with this `SimpleDataMapping`.

In summary, a `SimpleDataMapping` has the following template parameters:

- `SetFrom`: This is the type of the set that selects the appropriate data from data vector. It must be either `Block::Range` or `Block::Subset`.
- `SetTo`: This is the type of the set that indicates which part of the data of the caller object that is affected. It must be either `Block::Range` or `Block::Set`.
- `DataType`: This is the type of the data of the “small” vector (typically the type of the data that will be set in the caller object).
- `Caller`: This is the type of the caller object, which is the object that will “invoke” the function. By default, `Caller` is `Block`.

Besides the description of a single `SimpleDataMapping`, we also provide a description of a vector of `SimpleDataMapping`. In §3.3.2.1 we present the description of a single `SimpleDataMapping` and in §3.3.2.2 we present the description of a vector of `SimpleDataMapping`.

3.3.2.1 A single SimpleDataMapping

A SimpleDataMapping is described by the following NETCDF *variables* and *dimensions*. These do not need to be in a specific NETCDF *group*, unless of course one has to specify more than one SimpleDataMapping in a *group*, in which case it is necessary to make specific *groups* because the name of a *variable/dimension* cannot be duplicated in a given *group*.

- The one-dimensional *variable* SetSize, an array of type netCDF::NcUInt64 with two elements indicating the sizes (or types) of the SetFrom and SetTo sets. SetSize[0] indicates the size (or type) of the SetFrom set and SetSize[1] indicates the size (or type) of the SetTo set. For each $i \in \{0, 1\}$, if SetSize[i] == 0, then the corresponding set is a Range. Otherwise, if SetSize[i] ≠ 0, then the corresponding set is a Subset whose size is SetSize[i]. Notice, therefore, that SetSize[i] is not the size of the corresponding set when SetSize[i] == 0. In this case, it only indicates that the set is a Range, whose size (and elements) can be determined by the SetElements *variable*. This *variable* is optional. If it is not provided, then the SetFrom and SetTo sets are assumed to be Range.
- The one-dimensional *variable* SetElements, of type netCDF::NcUInt64, containing the concatenation of the representations of the sets SetFrom and SetTo. A Subset is represented by a sequence of indices (which are the elements of the Subset); while a Range is represented by two indices a and b such that the Range set is given by the integers in the closed-open interval $[a, b)$. For instance, if SetFrom is the Subset {3, 6, 8} and SetTo is the Range [2, 5), then SetElements would be the array (3, 6, 8, 2, 5).
- The *variable* FunctionName, whose type is netCDF::NcString, containing the name of the function as it is registered in the methods factory.
- The *group* AbstractPath containing the description of the AbstractPath representing the path to the caller object.

3.3.2.2 Vector of SimpleDataMapping

A vector of SimpleDataMapping is described by the following NETCDF *variables* and *dimensions*. These do not need to be in a specific NETCDF *group*, unless of course one has to specify more than one vector of SimpleDataMapping in a *group*, in which case it is necessary to make specific *groups* because the name of a *variable/dimension* cannot be duplicated in a given *group*.

- The NumberDataMappings *dimension*, indicating the number of SimpleDataMappings that is present in the vector of SimpleDataMapping. This *dimension* is optional. This dimension is optional. If it is not provided, then NumberDataMappings = 0 is assumed and every element in this *group* is ignored.
- The one-dimensional *variable* DataType, of type netCDF::NcChar and indexed over the NumberDataMappings *dimension*, specifying the type of the data that is associated with each SimpleDataMapping of the vector. This is the type of the data that can be set by the SimpleDataMapping (i.e., the DataType template parameter of SimpleDataMapping). This variable is optional. If it is not present, then the data type associated with each SimpleDataMapping in this vector is assumed to be double. If it is present then, for each $i \in \{0, \dots, \text{NumberDataMappings} - 1\}$, DataType[i] is the type of the data associated with the i -th SimpleDataMapping and can be either 'I' or 'D', indicating that the type of the data is int or double, respectively.
- The one-dimensional *variable* SetSize, an array of type netCDF::NcUInt64 with size given by $2 \times \text{NumberDataMappings}$ indicating the size of the sets that define each SimpleDataMapping (the SetFrom and SetTo sets). This variable is optional. If it is not present, then all sets are assumed to be Range. If it is present, then SetSize[2i + k] is the size of the SetFrom set of the i -th SimpleDataMapping if $k = 0$ or the size of the SetTo set of the i -th SimpleDataMapping if $k = 1$. If SetSize[j] == 0, then the corresponding set is a Range. Otherwise, the corresponding set is a Subset of size SetSize[j].
- The one-dimensional *variable* SetElements, of type netCDF::NcUInt64, containing the concatenation of the representations of the sets SetFrom and SetTo. A Subset is represented by a sequence of indices (which are the elements of the Subset); while a Range is represented by two indices a and b such that the Range set is

given by the integers in the closed-open interval $[a, b)$. If we let SetFrom_i and SetTo_i denote the representations of the SetFrom and SetTo sets of the i -th SimpleDataMapping , then SetElements is the array

$$(\text{SetFrom}_0, \text{SetTo}_0, \text{SetFrom}_1, \text{SetTo}_1, \dots, \text{SetFrom}_{N-1}, \text{SetTo}_{N-1})$$

where $N = \text{NumberDataMappings}$.

- The one-dimensional variable FunctionName , of type `netCDF::NcString` and indexed over $\text{NumberDataMappings}$, containing the names of the functions associated with each SimpleDataMapping . $\text{FunctionName}[i]$ gives the name of the function (as registered in the methods factory) associated with the i -th SimpleDataMapping .
- The one-dimensional variable Caller , of type `netCDF::NcChar` and indexed over $\text{NumberDataMappings}$, containing the types of the caller objects associated with each SimpleDataMapping . $\text{Caller}[i]$ gives the type of the caller object associated with the i -th SimpleDataMapping and can be either 'B', indicating that the caller is a `Block`, or 'F', indicating that the caller is a `Function`. This *variable* is optional. If it is not provided, then we assume that $\text{Caller}[i] = \text{'B'}$ for each $i \in \{0, \dots, \text{NumberDataMappings} - 1\}$, that is, we assume that all callers are `Blocks`.
- The *group* AbstractPath , containing a vector of AbstractPath with the paths to the `Blocks`. The number of AbstractPath that this vector must contain is $\text{NumberDataMappings}$. The i -th AbstractPath in this vector of AbstractPath is the path to the `Block` associated with the i -th SimpleDataMapping .

4 File formats of existing :Block

This section collects the description of the NETCDF file formats for all the implemented :Block developed for PLAN4RES. As such, the section will be continuously updated over the course of the project as new :Block will be added, although as usual non-backward-compatible changes to existing input formats will be kept to a minimum, and always agreed upon with the involved partners before being put into production.

4.1 SimpleMILPBlock

SimpleMILPBlock is a simple class, only having the “abstract representation”, for describing Mixed-Integer Linear Programs of the generic form

$$\min\{\bar{c} + cx : \underline{b} \leq Ax \leq \bar{b}, l \leq x \leq u\}$$

where $c, l, u \in \mathbb{R}^n$, $\underline{b}, \bar{b} \in \mathbb{R}^m$, $A \in \mathbb{R}^{m \times n}$, and $c^0 \in \mathbb{R}$. Actually, entries of $\underline{b}$ and l can be $-\infty$, and entries of $\bar{b}$ and u can be $+\infty$. The coefficient matrix A is usually very sparse, which justifies the choice of a standard sparse (row-wise) format to represent it.

Besides the mandatory type *attribute* of any :Block, the group will contain the following:

- the *attribute* `infinity` containing the number that is used to specify that one of the bounds (entries of $\underline{b}$, $\bar{b}$, l or u) is $\pm\infty$;
- the *dimension* `num_vars` containing the number n of Variable (columns in the matrix A) of the SimpleMILPBlock;
- the *dimension* `num_constraints` containing the number m of Constraint (rows in the matrix A) of the SimpleMILPBlock;
- the *dimension* `nzcount` containing the total number of nonzero coefficients in the Constraint of the SimpleMILPBlock;
- the *variable* `rmatval`, of type `double` and indexed over the *dimension* `nzcount`, that contains the nonzeros in the coefficient matrix A of the MILP, arranged row-wise (see `rmatbeg`);
- the *variable* `rmatind`, of type `UInt64` and indexed over the *dimension* `nzcount`, that contains the column indices of the nonzeros in the coefficient matrix A of the MILP, arranged row-wise; that is, the i -th entry of `rmatind` indicates to which variable (column) the (allegedly, nonzero) coefficient to be found in the i -th entry of `rmatval` refers;
- the *variable* `rmatbeg`, of type `UInt64` and indexed over the *dimension* `num_constraints`, that contains the indices into the `rmatval/rmatind` vectors where each constraint (row of the coefficient matrix A of the MILP) begins: all the nonzeros corresponding to constraint $h = 0, \dots, \text{num_constraints} - 1$ are to be found in `rmatval[ rmatbeg[ h ] ], \dots, rmatval[ rmatbeg[ h+1 ] - 1 ]`, with their column (variable) indices to be found in the corresponding entries of `rmatind[ ]`;
- the *variable* `x_lb`, of type `double` and indexed over the *dimension* `num_vars`, whose i -th entry is the (possibly, $-\infty$) lower bound on the feasible value of the i -th variable (the vector l);
- the *variable* `x_ub`, of type `double` and indexed over the *dimension* `num_vars`, whose i -th entry is the (possibly, $+\infty$) upper bound on the feasible value of the i -th variable (the vector u);
- the *variable* `row_lb`, of type `double` and indexed over the *dimension* `num_constraints`, whose i -th entry is the (possibly, $-\infty$) lower bound on the feasible value that the linear expression of the i -th constraint (as specified by the coefficients in `rmatval`, `rmatind`, and `rmatbeg`) can take (the vector $\underline{b}$);
- the *variable* `row_ub`, of type `double` and indexed over the *dimension* `num_constraints`, whose i -th entry is the (possibly, $+\infty$) upper bound on the feasible value that the linear expression of the i -th constraint (as specified by the coefficients in `rmatval`, `rmatind`, and `rmatbeg`) can take (the vector $\bar{b}$);
- the *variable* `obj`, of type `double` and indexed over the *dimension* `num_vars`, whose i -th entry is the coefficient of the i -th variable in the linear objective function of the MILP (the vector c);

- the scalar *variable* `obj_offset`, of type `double` (and, since it is a scalar, not indexed over any *dimension*) giving the fixed offset in the linear objective function of the MILP (the constant $\bar{c}$).

All these are mandatory.

4.2 StructuredMILPBlock

While SimpleMILPBlock is a “leaf” Block just representing a generic MILP, StructuredMILPBlock derives from SimpleMILPBlock and extends it to represent “structured” MILP of the format

$$\begin{aligned} \min \bar{c} + c^0 x^0 + \sum_{h=1}^k c^h x^h \\ x^h \in X^h \quad h = 1, \dots, k \\ \underline{b}^0 \leq A^0 x^0 \leq \bar{b}^0, \quad l^0 \leq x^0 \leq u^0 \\ \underline{r} \leq B^0 x^0 + \sum_{h=1}^k B^h x^h \leq \bar{r} \end{aligned}$$

where X^h denotes the feasible region of the h -th sub-Block of the StructuredMILPBlock, which must be a SimpleMILPBlock, x^h its Variable, and c^h the coefficients of its objective. Thus, a StructuredMILPBlock has its own MILP constraints on its own variable x^0 , represented exactly as in SimpleMILPBlock (indeed, a StructuredMILPBlock is a SimpleMILPBlock), plus the constraints of an arbitrary numbers of SimpleMILPBlock, plus constraints “linking” the variable x^0 of the “father” with those of the “sons”. Note that the “sons” of a StructuredMILPBlock are SimpleMILPBlock, which means they can actually be other StructuredMILPBlock. Thus, StructuredMILPBlock allows to represent any block-structured MILP *provided that the linking constraints only link variables of the father to variables of its immediate sons, and not to variables of further descendants.*

Besides the mandatory type attribute of any :Block, and whatever has to be found in the netCDF::NcGroup of a SimpleMILPBlock (since a StructuredMILPBlock is a SimpleMILPBlock), the group should contain the following:

- the *dimension* linking_num_constraints containing the number of “linking” Constraint (rows of the matrices B^h) of the StructuredMILPBlock;
- the *dimension* linking_nzcount containing the total number of nonzero coefficients in the “linking” Constraint (the matrices B^h) of the StructuredMILPBlock;
- the *variable* linking_rmatval, of type double and indexed over the *dimension* linking_nzcount, that contains the nonzeros in the coefficient matrix of the “linking” Constraint (the matrices B^h), arranged row-wise (see linking_rmatbeg);
- the *variable* linking_rmatind, of type UInt64 and indexed over the *dimension* linking_nzcount, that contains the column indices of the nonzeros in the coefficient matrix of the “linking” Constraint (the matrices B^h), arranged row-wise; this has to be used together with linking_rmatblk, see below;
- the *variable* linking_rmatblk, of type UInt64 and indexed over the *dimension* linking_nzcount, that contains the block indices of the variables corresponding to nonzeros in the coefficient matrix of the “linking” Constraint (the matrices B^h); that is, the i -th entry of linking_rmatblk indicates the index h of the Block to which the Variable belongs, with $h = 0$ used for the “father” StructuredMILPBlock and $h = 1, 2, \dots$ used for the Variable of the 1st, 2nd, ... of its SimpleMILPBlock sons, while the i -th entry of linking_rmatind is the index of the Variable in the h -th Block;
- the *variable* linking_rmatbeg, of type UInt64 and indexed over the *dimension* linking_num_constraints, that contains the indices into the linking_rmatval/linking_rmatind vectors where each “linking” Constraint (row of the matrices B^h) begins: all the nonzeros corresponding to constraint $h = 0, \dots, \text{linking_num_constraints} - 1$ are to be found in linking_rmatval[linking_rmatbeg[h]], ..., linking_rmatval[linking_rmatbeg[$h + 1$] - 1], with their column (variable) and Block indices to be found in the corresponding entries of linking_rmatind and linking_rmatblk, respectively;
- the *variable* linking_row_lb, of type double and indexed over the *dimension* linking_num_constraints, whose i -th entry is the (possibly, $-\infty$) lower bound on the feasible value that the linear expression of the i -th “linking” Constraint (as specified by the coefficients in linking_rmatval, linking_rmatind, linking_rmatbeg, and linking_rmatblk) can take;
- the *variable* linking_row_ub, of type double and indexed over the *dimension* linking_num_constraints, whose i -th entry is the (possibly, $+\infty$) upper bound on the feasible value that the linear expression of the

i -th “linking” Constraint (as specified by the coefficients in `linking_rmatval`, `linking_rmatind`, `linking_rmatbeg`, and `linking_rmatblk`) can take.

All these are mandatory.

4.3 MCFBlock

The MCFBlock class implements the Block concept for the (linear) Min-Cost Flow (MCF) problem.

The data of the problem consist of a (directed) graph $G = (N, A)$ with $n = |N|$ nodes and $m = |A|$ (directed) arcs. Each node i has a deficit $b[i]$, i.e., the amount of flow that is produced/consumed by the node: source nodes (which produce flow) have negative deficits and sink nodes (which consume flow) have positive deficits. Each arc (i, j) has an upper capacity $U[i, j]$ and a linear cost coefficient $C[i, j]$. Flow variables $X[i, j]$ represents the amount of flow to be sent on arc (i, j) . Parallel arcs, i.e., multiple copies of the same arc (i, j) are in general allowed; it is expected that they have different costs (for otherwise they can be merged into a unique arc), but this is not strictly enforced. Multiple copies of some arc (i, j) can be seen as “total” flow cost on that arc being a piecewise-linear convex function. The formulation of the problem is:

$$\begin{aligned} \min \quad & \sum_{(i,j) \in A} C[i, j] X[i, j] \\ \text{subject to} \quad & \sum_{(j,i) \in A} X[j, i] - \sum_{(i,j) \in A} X[i, j] = b[i] & i \in N \\ & 0 \leq X[i, j] \leq U[i, j] & (i, j) \in A \end{aligned}$$

The n equations are the flow conservation constraints and the $2m$ inequalities are the flow nonnegativity and capacity constraints. At least one of the flow conservation constraints is redundant, as the demands must be balanced ($\sum_{i \in N} b[i] = 0$); indeed, exactly $n - \text{ConnectedComponents}(G)$ flow conservation constraints are redundant, as demands must be balanced in each connected component of G .

The graph G is allowed to be “partly dynamic”. The set of nodes and arcs that are input at the beginning are assumed not to be changed (save for changing costs, capacities, and deficits, and for arcs to be closed or opened). Then, new arcs and nodes, up to a set maximum, can be dynamically added or deleted. This means that the graph can be fully static (if the maximum number of dynamic arcs and nodes is set to zero) as well as fully dynamic (if the initial graph is empty).

Besides the mandatory `type` attribute of any `:Block`, the *group* representing the MCFBlock should contain the following:

- the *dimension* `NNodes` containing the number of nodes in the graph (n);
- the *dimension* `NArCs` containing the number of arcs in the graph (m);
- the *variable* `C`, of type `double` and indexed over the *dimension* `NArCs`; the i -th entry of the variable is assumed to contain the upper capacity of the i -th arc in the graph, that whose starting node and ending node are specified by the *variable* `SN` and `EN` (below);
- the *variable* `U`, of type `double` and indexed over the *dimension* `NArCs`; the i -th entry of the variable is assumed to contain the cost of the i -th arc in the graph (the lower capacity being fixed to 0), that whose starting node and ending node are specified by the *variable* `SN` and `EN` (below);
- the *variable* `B`, of type `double` and indexed over the *dimension* `NNodes`; the i -th entry of the variable is assumed to contain the deficit of the i -th node in the graph (note that node names here go from 0 to $n - 1$);
- the *variable* `SN`, of type `int` and indexed over the *dimension* `NArCs`; the i -th entry of the variable is assumed to contain the starting node of the i -th arc in the graph (note that node names here go from 1 to n , they are shifted by 1 w.r.t. to the indices used in the `B` variable);
- the *variable* `EN`, of type `int` and indexed over the *dimension* `NArCs`; the i -th entry of the variable is assumed to contain the ending node of the i -th arc in the graph (note that node names here go from 1 to n , they are shifted by 1 w.r.t. to the indices used in the `B` variable);
- the *dimension* `DynNNodes` containing the current number of dynamic nodes: all the nodes between 0 and `NNodes` - `DynNNodes` - 1 are static, i.e., they cannot be deleted (and re-created), whereas all those from `NNodes` + `DynNNodes` to `NNodes` - 1 are dynamic, i.e., they can be deleted and later on re-created;
- the *dimension* `DynNArCs` containing the current number of dynamic arcs: all the arcs between 0 and `NArCs` - `DynNArCs` - 1 are static, i.e., they cannot be deleted (and re-created), whereas all those from `NArCs` + `DynNArCs` to `NArCs` - 1 are dynamic, i.e., they can be deleted and later on re-created;

- the *dimension* MaxDynNNodes containing the maximum number of dynamic nodes in the graph (the maximum value that n can take); data in the MCFBlock is allocated to accommodate for the fact that further MaxDynNodes – DynNNodes nodes can later on be dynamically created (and deleted); if MaxDynNNodes < DynNNodes the *dimension* is ignored and NNodes is taken as the maximum overall number of nodes;
- the *dimension* MaxDynNArcs containing the maximum number of dynamic arcs in the graph (the maximum value that m can take); data in the MCFBlock is allocated to accommodate for the fact that further MaxDynNArcs – DynNArcs arcs can later on be dynamically created (and deleted); if MaxDynNArcs < DynNArcs the *dimension* is ignored and NArcs is taken as the maximum overall number of arcs.

The two *dimensions* NNodes and NArcs are mandatory, such as are the two *variables* SN and EN. The three other *variables* are optional. If C is missing, all arc costs are assumed to be 0. If U is missing, all arc capacities are assumed to be infinite. If B is missing, all node deficits are assumed to be 0. Finally, all the *dimensions* DynNNodes, DynNArcs, MaxDynNNodes and MaxDynNArcs are optional: if they are missing they are treated as being 0 (this happening for all four means that the graph is “fully static” and cannot be changed).

4.4 UCBlock

The class UCBlock, implements the Block concept [see Block.h] for the Unit Commitment (UC) problem in electrical power production. This is typically a short-term (across for instance one week or one day time horizon) **deterministic** problem regarding finding an optimal schedule of the production of electrical generators satisfying a (large) set of technical constraints. The model is quite flexible due to the fact that different types of units and network constraints can be used by means of the fact that the class manages son Block of type UnitBlock and NetworkBlock. Also UCBlock handles a reasonably large variety of constraints, regarding not only active power but also primary and secondary reserve and inertia. Admittedly, some choices in UCBlock (like HeatBlock, pollution constraints, ...) are quite specific of the UC of the plan4res project; however, all the nonstandard aspects of UC can be switched away from the model (by simply not providing the data describing them). The main elements that UCBlock handles are:

- The time horizon of the problem, i.e., a discrete set of (typically, equally-spaced) time instants at which decisions are made (like, the 24 hours in a day);
- A set of electricity generating units, represented by derived classes of the base class UnitBlock;
- A set of NetworkBlock, one for each time instant in the time horizon, which represent the constraints on the electricity demand satisfaction and the technical constraints on the transmission network. These can be basically empty if the capacity of the transmission network is such as to never really impact generation decisions (a bus);
- An optional set of HeatBlock, each representing the satisfaction of some specific type of heat on a close geographical area by heat-generating units possibly coupled with a heat storage. The link with the rest of the UC model lies in the fact that some of the heat-generating units in a HeatBlock may also be electricity generating ones (i.e., a UnitBlock); actually, the same UnitBlock can generate heat of different types, and therefore appear as a heat-generating units in more than one HeatBlock.

Besides the mandatory type attribute of any :Block, the *group* should contain the following:

- the *dimension* TimeHorizon containing the number of time steps in the problem;
- the *dimension* NumberUnits containing the number of units (UnitBlock) in the problem;
- the *groups* UnitBlock_0, UnitBlock_1, ..., UnitBlock_n with $n == \text{NumberUnits} - 1$, containing each one UnitBlock corresponding to one or more electricity generating unit (electrical generator). It is an error if the corresponding groups are not there. Each UnitBlock can have more than one electrical generator (cf. UnitBlock::get_number_generators()), a value that is useful in the following (cf. GeneratorNode) is the total number of those, which we will refer to as NumberElectricalGenerators. This is computed by just calling get_number_generators() on each of the UnitBlock and summing all the results. Clearly, NumberElectricalGenerators $\geq$ NumberUnits and indeed, most of the UnitBlock can be expected to have just one electrical generator. If this happens for all the units then the NumberElectricalGenerators == NumberUnits. If, instead, some UnitBlock (like cascades of hydro generators or combined cycle plants) actually has more than one electrical generator, then NumberElectricalGenerators $>$ NumberUnits (each UnitBlock must have at least one). This is because some UnitBlock (like cascades of hydro generators or combined cycle plants) can actually have more than one electrical generator in it (but each UnitBlock must have at least one). It is then useful (cf. GeneratorNode) to be able to assign a unique index $g = 0, 1, \dots, \text{NumberElectricalGenerators} - 1$ to each of the electrical generators in the UCBlock. When NumberElectricalGenerators == NumberUnits, the index is the same as $i = 0, 1, \dots, \text{NumberUnits} - 1$ (there is a one-to-one correspondence between UnitBlock and electrical generators). When, instead, NumberElectricalGenerators $>$ NumberUnits, a mapping must be defined. The mapping is the obvious one: UnitBlock have an ordering $i = 0, 1, \dots, \text{NumberUnits} - 1$ (cf. the groups UnitBlock_0, UnitBlock_1, ..., above), and the electrical generators into each UnitBlock also have some natural ordering (corresponding to the columns of the matrices of variables, cf. e.g. UnitBlock::get_commitment()). Thus, in general the mapping is:

- electrical generator 0 = first generator of UnitBlock_0
- electrical generator 1 = second generator of UnitBlock_0

```

- ...
- electrical generator k = k-th generator of UnitBlock_0
  (k = UnitBlock_0->get_number_generators())
- electrical generator k + 1 = first generator of UnitBlock_1
- electrical generator k + 2 = second generator of UnitBlock_1
- ...

```

which of course boils down to $g = i$ when each UnitBlock has exactly one electrical generator;

- the *dimension* NumberHeatBlocks containing the number of heat blocks in the problem. The *dimension* is optional: if it is not provided then it is taken to be 0, which means that there is no heat block in the problem;
- the *groups* HeatBlock_0, HeatBlock_1, ..., HeatBlock_n with $n == \text{NumberHeatBlocks} - 1$, containing each one a HeatBlock. When NumberHeatBlocks == 0, these groups need not be there since they are not read. If, instead, NumberHeatBlocks > 0, it is an error if the corresponding groups are not there. Since each HeatBlock can have more than one heat generator (cf. HeatBlock::get_number_heat_generators()), a value that is useful in the following (cf. HeatSet) is the total number of those. We will refer to such number as NumberHeatGenerators, which is computed by just calling get_number_heat_generators() on each of the HeatBlock and summing all the results. Clearly, NumberHeatGenerators >= NumberHeatBlock. Some of the HeatBlock may have just one heat generator; if this happens for all the heat blocks (but this is not likely), then NumberHeatGenerators == NumberHeatBlocks. It is then useful (cf. HeatNode) to be able to assign a unique index $h = 0, 1, \dots, \text{NumberHeatGenerators} - 1$ to each of the heat generators in the UCBLOCK. When NumberHeatGenerators == NumberHeatBlocks the index is the same as $b = 0, 1, \dots, \text{NumberHeatBlock} - 1$ (there is a one-to-one correspondence between HeatBlock and heat generators, but this is not likely to happen). When, instead, NumberHeatGenerators > HeatBlock, a mapping must be defined. The mapping is the obvious one: HeatBlock have an ordering $b = 0, 1, \dots, \text{NumberHeatBlocks} - 1$ (cf. the groups HeatBlock_0, HeatBlock_1, ..., above), and the heat generators into each HeatBlock also have some natural ordering (corresponding to the columns of the matrices of variables, cf. e.g. HeatBlock::get_heat()). Thus, in general the mapping is:

```

- heat generator 0 = first generator of HeatBlock_0
- heat generator 1 = second generator of HeatBlock_0
- ...
- heat generator k = k-th generator of HeatBlock_0
  (k = HeatBlock_0->get_number_heat_units())
- heat generator k + 1 = first generator of HeatBlock_1
- heat generator k + 2 = second generator of HeatBlock_1
- ...

```

which of course boils down to $h = b$ when each HeatBlock has exactly one heat generator (but this is not assumed to happen);

- optionally, the *dimensions* and variables necessary to deserialize a NetworkData object that describes the transmission network; see §4.8.1 for details. If that is not provided (basically, NumberNodes is not provided or it is == 1), then the transmission network is taken to have only one node (a bus). If the data of a NetworkData is specified, the NetworkData is passed to each of the NetworkBlock (see below) of the UCBLOCK, if any. However, if a NetworkBlock also has a NetworkData specified in its own group, then the NetworkData inside the NetworkBlock overrules that inside the UCBLOCK, which is ignored by that NetworkBlock;
- The *variable* ActivePowerDemand, of type netCDF::NcDouble, and indexed both over the dimensions NumberNodes and TimeHorizon. This variable is optional if
 - a NetworkBlock is defined for each time instant (see below), and
 - each of the defined NetworkBlock has the ActiveDemand variable specified in the corresponding group.

Otherwise it is mandatory. When it is defined, the entry `ActivePowerDemand[ n , t ]` is assumed to contain the active power demand of node n of the transmission network at the given time instant t , where the first dimension `NumberNodes` can be read via `NetworkBlock::NetworkData::get_number_nodes()` from either the `NetworkData` object in `UCBlock`, or these in the `NetworkBlock`. When `ActivePowerDemand` is defined, and also the `ActiveDemand` variable is defined in the group of some `NetworkBlock`, then `ActiveDemand` overrules the value in the corresponding row of `ActivePowerDemand`, which is ignored.

- the *groups* `NetworkBlock_0, NetworkBlock_1, ..., NetworkBlock_t` with $t = \text{TimeHorizon} - 1$, containing each the constraints on the transmission network at time t . The `NetworkBlocks` are optional, but if any of them are missing, then
 - `ActivePowerDemand` (see above) is mandatory in `UCBlock`.
 - also the `NetworkData` (see above) is mandatory in `UCBlock`, unless the transmission network is a bus (that is, `NumberNodes` is not provided or it is `== 1`).

In particular, when a `NetworkBlock` is not defined for a given time instant t then one of these happen:

- If `NumberNodes == 1` (or it is not provided) then a `BusNetworkBlock` is automatically constructed for that time instant, and the entry `ActivePowerDemand[ 0 , t ]` contains the active demand for t .
- If `NumberNodes > 1`, then a `DCNetworkBlock` is automatically constructed for that time instant, it is provided with the `NetworkData` object (which must be present in `UCBlock`) and the row `ActivePowerDemand[ ... , t ]` contains the active demand of each node at time instant t .
- the *variable* `GeneratorNode`, of type `netCDF::NcUInt64` and indexed over the set $\{0, \dots, \text{NumberElectricalGenerators} - 1\}$; `GeneratorNode[ g ]` tells to which node of the transmission network, the specified electrical generator g belongs. Note that this means that different electrical generators in the same `UnitBlock` can belong to different nodes of the transmission network. This is justified e.g. by hydro cascade units where different turbines can be rather far apart geographically, but still linked by (long) stretches of rivers. If `NumberElectricalGenerators == NumberUnits` (all `UnitBlock` have exactly one electrical generator), then this variable is indexed over `NumberUnits`. If `NumberNodes == 1` (say, it is not provided at all), then this variable need not be defined, since it is not loaded.
- the *variable* `HeatNode`, of type `netCDF::NcUInt64` and indexed over the *dimension* `NumberHeatBlocks`; the entry `HeatNode[ h ]` tells to which node of the transmission network **all** the heat generators that are also electrical generators in the `HeatBlock h` belong. Note that this implies that, unlike electrical generators in a `UnitBlock`, heat generators in a `HeatBlock` are always “geographically near to each other” so that they are necessarily attached to the same node of the transmission network (which is a reasonable assumption since heat, unlike say water, usually cannot travel much far). If `NumberHeatBlocks == 0` (say, it is not provided at all), then this variable need not be defined, since it is not loaded. This information is actually only used to determine in which Pollutant Zone a `HeatBlock` is located, in order to add the corresponding heat generators (that are not also electricity generators) to the pollution constraints. This means that also if `NumberPollutants ==` (say, it is not provided at all) this variable is useless and therefore need not be defined, since it is not loaded. Finally, notice that for heat generators into a `HeatBlock` that also are electrical generators, this variable provides again an information that is already known, i.e., to which node they belong to (cf. `GeneratorNode` above). Of course **the two information must agree**, otherwise the input file is ill-defined and exception is thrown.
- the *variable* `HeatSet`, of type `netCDF::NcUInt64` and indexed over the set $\{0, \dots, \text{NumberHeatGenerators} - 1\}$. If `HeatSet[ h ] = k < NumberElectricalGenerators`, then k is the unique name of the electrical generator corresponding to the heat generator h ; otherwise the heat generator h is not also an electrical generator. If `NumberHeatBlocks == 0` (there is no `HeatBlock`) then this variable need not be defined, since it is not loaded. It is possible that different heat generators correspond to the same electrical generator (that is, `HeatSet[ h1 ] == HeatSet[ h2 ]` for some $h1 \neq h2$), but not vice-versa: a heat generator either corresponds to a specific electrical generator, or to no electrical generator (it is an heat-only generator);
- the *variable* `PowerHeatRho`, of type `netCDF::NcDouble` and indexed over the *dimension* `NumberUnits`; entry `PowerHeatRho[ i ]` is assumed to contain the electrical-power-to-heat ratio for **all generators** within unit i (most likely, a single generator). This is only used in the constraints linking electrical units to heat units, hence if `NumberHeatBlocks == 0` (there are no `HeatBlock`) then this variable need not be defined, since it is not loaded;

- the *dimension* `NumberPrimaryZones` tells how many primary spinning reserve zones are there in the problem. The *dimension* is optional, if it is not provided then it is taken to be 0, which means that no primary reserve constraints are present in the problem;
- the *variable* `PrimaryZones`, of type `netCDF::NcUInt64` and indexed over the *dimension* `NumberNodes`. The entry `PrimaryZones[ i ]` tells to which primary zone the node *i* belongs: if `PrimaryZones[ i ] >= NumberPrimaryZones`, this means that node *i* does not belong to any primary zone, and hence the corresponding electrical generators are not involved into the primary reserve constraints. If `NumberPrimaryZones == 0` (say, it is not provided at all) then this variable need not be defined, since it is not loaded. If `NumberPrimaryZones == 1` and this variable is not defined, then there is only one primary zone and all the nodes belong to it;
- the *variable* `PrimaryDemand`, of type `netCDF::NcDouble` and indexed both over the *dimensions* `NumberPrimaryZones` and `TimeHorizon`: entry `PrimaryDemand[ i , t ]` is assumed to contain the primary reserves requirement which are specified on the primary reserve zone *i* in the time *t*. If `NumberPrimaryZones == 0` (say, it is not provided at all), then this variable need not be defined, since it is not loaded;
- the *dimension* `NumberSecondaryZones` tells how many secondary spinning reserve zones are there in the problem. The *dimension* is optional, if it is not provided then it is taken to be 0, which means that no secondary reserve constraints are present in the problem;
- the *variable* `SecondaryZones`, of type `netCDF::NcUInt64` and indexed over the *dimension* `NumberNodes`. The entry `SecondaryZones[ i ]` tells to which secondary zone the node *i* belongs: if `SecondaryZones[ i ] >= NumberSecondaryZones`, this means that node *i* does not belong to any secondary zone, and hence the corresponding electrical generators are not involved into the secondary reserve constraints. If `NumberSecondaryZones == 0` (say, it is not provided at all) then this variable need not be defined, since it is not loaded. If `NumberSecondaryZones == 1` and this variable is not defined, then there is only one secondary zone and all the nodes belong to it;
- the *variable* `SecondaryDemand`, of type `netCDF::NcDouble` and indexed both over the *dimensions* `NumberSecondaryZones` and `TimeHorizon`: entry `SecondaryDemand[ i , t ]` is assumed to contain the secondary reserves requirement which are specified on the secondary reserve zone *i* in the time *t*. If `NumberSecondaryZones == 0` (say, it is not provided at all), then this variable need not be defined, since it is not loaded;
- the *dimension* `NumberInertiaZones` tells how many inertia constraints zones are there in the problem. The *dimension* is optional, if it is not provided then it is taken to be 0, which means that no inertia constraints are present in the problem;
- the *variable* `InertiaZones`, of type `netCDF::NcUInt64` and indexed over the *dimension* `NumberNodes`. The entry `InertiaZones[ n ]` tells to which inertia zone the node *n* belongs: if `InertiaZones[ n ] >= NumberInertiaZones`, this means that node *n* does not belong to any inertia zone, and hence the corresponding units are not involved into the inertia reserve constraints. If `NumberInertiaZones == 0` (say, it is not provided at all) then this variable need not be defined, since it is not loaded. If `NumberInertiaZones == 1` and this variable is not defined, then there is only one inertia zone and all the nodes belong to it;
- the *variable* `InertiaDemand`, of type `netCDF::NcDouble` and indexed both over the *dimensions* `NumberInertiaZones` and `TimeHorizon`: entry `InertiaDemand[ i , t ]` is assumed to contain the inertia reserves requirement which are specified on the inertia reserve zone *i* in the time *t*. If `NumberInertiaZones == 0` (say, it is not provided at all), then this variable need not be defined, since it is not loaded;
- the *dimension* `NumberPollutants` containing the number of pollutants in the problem. The *dimension* is optional, if it is not provided then it is taken to be 0, which means that no pollutants constraints are present in the problem;
- the *variable* `NumberPollutantZones` of type `netCDF::NcUInt64` and indexed over the *dimension* `NumberPollutants`: the entry `NumberPollutantZones[ p ]` is assumed to contain the number of pollutant zones associated with pollutant *p*. If `NumberPollutants == 0` (say, it is not provided) then this variable need not be defined, since it is not loaded. Then, all number of pollutant zones associated with each pollutant

p is computed as: `TotalNumberPollutantZones == NumberPollutantZone[ 0 ] + ... + NumberPollutantZone[ NumberPollutants - 1 ]`;

- the *variable* `PollutantZones`, of type `netCDF::NcUInt64` and indexed over the *dimensions* `NumberPollutants` and `NumberNodes`: the entry `PollutantZones[ p , n ]` tells to which pollutant zone associated with pollutant p the node n belongs. If `PollutantZones[ p , n ] >= NumberPollutantZones[ p ]`, this means that node n does not belong to any pollutant zone, and hence the corresponding units are not involved into the pollutant budget constraints associated with pollutant p . If `NumberPollutants == 0` (say, it is not provided) then this variable need not be defined, since it is not loaded;
- the *variable* `PollutantBudget`, of type `netCDF::NcDouble` and indexed over the set $\{ 0, \dots, \text{TotalNumberPollutantZones} - 1 \}$: the entry `PollutantBudget[ n ]` for $n = 0, \dots, \text{TotalNumberPollutantZones} - 1$ is assumed to contain the pollutant budget (across all the time horizon) for the pair (zone of the pollutant, pollutant) corresponding to n . In another word, since the number of pollutant zones of each pollutant may not be equal with each other it is useful (to avoid having to store `PollutantBudget` as an irregular matrix) to be able to assign a unique index $n = 0, 1, \dots, \text{TotalNumberPollutantZones} - 1$ to each pollutant budget of each pollutant zone. A mapping must be defined between each entry n and the pair (zone of the pollutant, pollutant). The mapping is the obvious one: `UCBlock` has a set of pollutants $p = 0, 1, \dots, \text{NumberPollutants} - 1$, and each pollutant p may have several pollutant zones (see comments of variable `NumberPollutantZones` above). Thus, in general the mapping is:

- $n = 0$ corresponds to the zone 0 of pollutant 0
- $n = 1$ corresponds to the zone 1 of pollutant 0
- ...
- $n = \text{NumberPollutantZone}[0] - 1$ corresponds to the zone `NumberPollutantZone[ 0 ] - 1` of pollutant 0
- $n = \text{NumberPollutantZone}[0]$ corresponds to the zone 0 of pollutant 1
- $n = \text{NumberPollutantZone}[0] + 1$ corresponds to the zone 1 of pollutant 1
- ...

If `NumberPollutants == 0` (say, it is not provided) then this variable need not be defined, since it is not loaded;

- the *variable* `PollutantRho`, of type `netCDF::NcDouble` and indexed over three *dimensions* which are `TimeHorizon` and `NumberPollutants` and the set $\{ 0, \dots, \text{NumberElectricalGenerators} - 1 \}$ (see comments above). The first *dimension* can have size either 1 or `TimeHorizon`. In the former case the entry `PollutantRho[ 0 , p , g ]` is assumed to contain the conversion factor of pollutant p due to the electrical generator g which is equal for all time instants t . Otherwise, the first *dimension* has full size `TimeHorizon` and the entry `PollutantRho[ t , p , g ]` gives the conversion factor of pollutant p due to the electrical generator g for time t . If `NumberPollutants == 0` (it is not provided) then this variable need not be defined, since it's not loaded.
- the *variable* `PollutantHeatRho`, of type `netCDF::NcDouble` and indexed over three *dimensions*. The first *dimension* can have size 1 or `TimeHorizon`. The second and third *dimensions* have sizes `NumberPollutants` and `NumberHeatBlocks`, respectively. If the first *dimension* has size 1, then the entry `PollutantRho[ 0 , p , h ]` is assumed to contain the conversion factor of pollutant p due to the generation of **all** heat generators in `HeatBlock h`, which is equal for all time instants t . If, instead, the first *dimension* has full size `TimeHorizon`, then the entry `PollutantRho[ t , p , h ]` gives the conversion factor of pollutant p due to the generation of **all** heat generators in `HeatBlock h` for time instant t . If `NumberPollutants == 0` (say, it is not provided) then this variable does not need be defined, since it is not loaded. If `NumberHeatBlocks == 0` (there is no `HeatBlock`) then this variable need not be defined, since it is not loaded.

4.5 UnitBlock

The `UnitBlock` class, which derives from the `Block`, defines a base class for any possible unit that can be attached to a `UCBlock`. A unit is in general a set of electrical generators tied together by some technical constraints, although many units actually correspond to only one generator. The base `UnitBlock` class only has very basic information that can characterize almost any different kind of unit:

- The time horizon of the problem;
- The number of generators in the unit (1 by default, see `get_number_generators()`);
- Four `boost::multi_array< ColVariable, 2 >` objects, that are used to store the information regarding:
 - the commitment of the generators in the unit;
 - the primary spinning reserve of the generators in the unit;
 - the secondary spinning reserve of the generators in the unit;
 - the active power produced by the generators in the unit;

Each of the `boost::multi_array< ColVariable, 2 >` has as first *dimension* the time horizon and as second *dimension* the number of generators (as returned `get_number_generators()`). There are two possible cases:

- if some `boost::multi_array< ColVariable, 2 >` is `empty()`, then the corresponding variable does not exist (for instance, the generator in the unit may not have reserve);
- otherwise, the `boost::multi_array< ColVariable, 2 >`, say M , must have `fltime_horizon` rows and `get_number_generators()` columns: each element $M[t, g]$ gives the variable for time step t of generator g .

The class also outputs some general information regarding how the active power and/or commitment status of each generator of the unit at a given time instant impact the unit's capability of satisfying inertia constraints, and the fixed consumption (if any) of each generator in the unit when it is off. Besides the mandatory `type` attribute of any `:Block`, the *group* representing the `UnitBlock` should contain the following:

- the *dimension* `TimeHorizon` containing the time horizon. The *dimension* is optional because the same information may be passed via the method `set_time_horizon()`, or directly retrieved from the father if it is a `UCBlock`; see the comments to `set_time_horizon()` for details.
- the *dimension* `NumberIntervals` that is provided to allow that all time-dependent data in the `UnitBlock` can only change at a subset of the time instants of the time interval, being therefore piecewise-constant (possibly, constant). `NumberIntervals` should therefore be $\leq \text{TimeHorizon}$, with three distinct cases:
 - `NumberIntervals` ≤ 1 , which is taken to mean `NumberIntervals` $= 1$; this is what is assumed if the *dimension*, that is optional, is not there. This means that the value of each relevant data in the `UCBlock` is the same for each time instant $0, \dots, \text{TimeHorizon} - 1$ in the time horizon. In this case, the variable `ChangeIntervals` (see below) is ignored;
 - $1 < \text{NumberIntervals} < \text{TimeHorizon}$, which means that in some time instants, **but not all of them**, the values of some of the relevant data are changing; the intervals are then described in variable `NumberIntervals`;
 - `NumberIntervals` $= \text{TimeHorizon}$, which means that values of the relevant data changes at every time interval (in principle; of course there is nothing preventing the same value to be repeated in the `netCDF` input). Also in this case the variable `NumberIntervals` is ignored, since it is useless.

Note that this (together with `NumberIntervals`, if defined) obviously sets the maximum frequency at which data can change; if some data changes less frequently (say, it is constant), then the same value will have to be repeated. Individual data can also have specific provisions for the case where the data is all equal despite `NumberIntervals` saying differently.

- the *variable* `ChangeIntervals`, of type integer and indexed over the *dimension* `NumberIntervals`. The time horizon is subdivided into `NumberIntervals` = k of the form $[0, i_1], [i_1+1, i_2], \dots, [i_{k-1}+1, \text{TimeHorizon}-1]$; `ChangeIntervals` then has to contain $[i_1, i_2, \dots, i_{k-1}]$. Note that, therefore, `ChangeIntervals` has one significant value less than `NumberIntervals`, which means that `ChangeIntervals[ NumberIntervals - 1 ]` is ignored. Anyway, the whole variable is ignored if either `NumberIntervals` ≤ 1 (such as if it is not defined), or `NumberIntervals` $\geq \text{TimeHorizon}$.

4.5.1 ThermalUnitBlock

The ThermalUnitBlock class derives from UnitBlock and implements a reasonably standard thermal unit of a Unit Commitment Problem. That is, the class is designed in order to give mathematical formulation to describe the operation of large set of conventional power plants (such as nuclear, hard coal, gas turbine, gas, combined cycle, oil, ...) which are directly connected to the transmission grid. Besides the mandatory type attribute of any :Block, the group must contain all the data required by the base UnitBlock, as described in the comments to UnitBlock::deserialize(netCDF::NcGroup). In particular, we refer to that description for the crucial *dimensions* TimeHorizon, NumberIntervals and ChangeIntervals. The netCDF::NcGroup must then also contain:

- the *variable* MinPower, of type netCDF::NcDouble and either of size 1 or indexed over the *dimension* NumberIntervals. This is meant to represent the vector $MnP[t]$ that, for each time instant t , contains the minimum active power output value of the unit for the corresponding time step. If MinPower has length 1 then $MnP[t]$ contains the same value for all t . Otherwise, $MinPower[i]$ is the fixed value of $MnP[t]$ for all t in the interval $[ChangeIntervals[i - 1], ChangeIntervals[i]]$, with the assumption that $ChangeIntervals[-1] = 0$. Note that it must be $MnP[t] \geq 0$ for all t . If NumberIntervals ≤ 1 or NumberIntervals $\geq$ TimeHorizon, then the mapping clearly does not require ChangeIntervals, which in fact is not loaded;
- the *variable* MaxPower, of type netCDF::NcDouble and either of size 1 or indexed over the *dimension* NumberIntervals. This is meant to represent the vector $MxP[t]$ that, for each time instant t , contains the maximum active power output value of the unit for the corresponding time step. If MaxPower has length 1 then $MxP[t]$ contains the same value for all t . Otherwise, $MaxPower[i]$ is the fixed value of $MxP[t]$ for all t in the interval $[ChangeIntervals[i - 1], ChangeIntervals[i]]$, with the assumption that $ChangeIntervals[-1] = 0$. Note that it must be $MxP[t] \geq 0$ for all t . If NumberIntervals ≤ 1 or NumberIntervals $\geq$ TimeHorizon, then the mapping clearly does not require ChangeIntervals, which in fact is not loaded;
- the *variable* DeltaRampUp, of type netCDF::NcDouble and either of size 1 or indexed over the *dimension* NumberIntervals. This is meant to represent the vector $DP[t]$ that, for each time instant t , contains the ramp-up value of the unit for the corresponding time step, i.e., the maximum possible increase of active power production w.r.t. the power that had been produced in time instant $t - 1$, if any. This variable is optional; if it is not provided then it is assumed that $DP[t] == MxP[t]$, i.e., the unit can ramp up by an arbitrary amount, i.e., there are no ramp-up constraints. If DeltaRampUp has length 1 then $DP[t]$ contains the same value for all t . Otherwise, $DeltaRampUp[i]$ is the fixed value of $DP[t]$ for all t in the interval $[ChangeIntervals[i - 1], ChangeIntervals[i]]$, with the assumption that $ChangeIntervals[-1] = 0$. If NumberIntervals ≤ 1 or NumberIntervals $\geq$ TimeHorizon, then the mapping clearly does not require ChangeIntervals, which in fact is not loaded.
- the *variable* DeltaRampDown, of type netCDF::NcDouble and either of size 1 or indexed over the *dimension* NumberIntervals. This is meant to represent the vector $DM[t]$ that, for each time instant t , contains the ramp-down value of the unit for the corresponding time step, i.e., the maximum possible decrease of active power production w.r.t. the power that had been produced in time instant $t - 1$, if any. This variable is optional; if it is not provided then it is assumed that $DM[t] == MxP[t]$, i.e., the unit can ramp down by an arbitrary amount, i.e., there are no ramp-down constraints. If DeltaRampDown has length 1 then $DM[t]$ contains the same value for all t . Otherwise, $DeltaRampDown[i]$ is the fixed value of $DM[t]$ for all t in the interval $[ChangeIntervals[i - 1], ChangeIntervals[i]]$, with the assumption that $ChangeIntervals[-1] = 0$. If NumberIntervals ≤ 1 or NumberIntervals $\geq$ TimeHorizon, then the mapping clearly does not require ChangeIntervals, which in fact is not loaded.
- the *variable* PrimaryRho, of type netCDF::NcDouble and either of size 1 or indexed over the *dimension* NumberIntervals. This is meant to represent the vector $PR[t]$ that, for each time instant t , contains the maximum possible fraction of active power that can be used as primary reserve value of the unit for the corresponding time step. This variable is optional; if it is not provided then it is assumed that this unit may not be capable of producing any primary reserve, which correspond to $PR[t] == 0$ for all t . If PrimaryRho has length 1 then $PR[t]$ contains the same value for all t . Otherwise, $PrimaryRho[i]$ is the fixed value of $PR[t]$ for all t in the interval $[ChangeIntervals[i - 1], ChangeIntervals[i]]$, with

the assumption that $\text{ChangeIntervals}[-1] = 0$. If $\text{NumberIntervals} \leq 1$ or $\text{NumberIntervals} \geq \text{TimeHorizon}$, then the mapping clearly does not require ChangeIntervals , which in fact is not loaded.

- the *variable* `SecondaryRho`, of type `netCDF::NcDouble` and either of size 1 or indexed over the *dimension* `NumberIntervals`. This is meant to represent the vector $\text{SR}[t]$ that, for each time instant t , contains the maximum possible fraction of active power that can be used as secondary reserve value of the unit for the corresponding time step. This variable is optional; if it is not provided then it is assumed that this unit may not be capable of producing any secondary reserve, which correspond to $\text{SR}[t] == 0$ for all t . If `SecondaryRho` has length 1 then $\text{SR}[t]$ contains the same value for all t . Otherwise, $\text{SecondaryRho}[i]$ is the fixed value of $\text{SR}[t]$ for all t in the interval $[\text{ChangeIntervals}[i-1], \text{ChangeIntervals}[i]]$, with the assumption that $\text{ChangeIntervals}[-1] = 0$. If $\text{NumberIntervals} \leq 1$ or $\text{NumberIntervals} \geq \text{TimeHorizon}$, then the mapping clearly does not require ChangeIntervals , which in fact is not loaded.
- the *variable* `QuadTerm`, of type `netCDF::NcDouble` and either of size 1 or indexed over the *dimension* `NumberIntervals`. This is meant to represent the vector $A[t]$ that, for each time instant t , contains the quadratic term of power cost function of the unit for the corresponding time step. This variable is optional; if it is not provided then it is assumed that $A[t] == 0$, i.e., the cost of the unit is linear in the produced power. If `QuadTerm` has length 1 then $A[t]$ contains the same value for all t . Otherwise, $\text{QuadTerm}[i]$ is the fixed value of $A[t]$ for all t in the interval $[\text{ChangeIntervals}[i-1], \text{ChangeIntervals}[i]]$, with the assumption that $\text{ChangeIntervals}[-1] = 0$. If $\text{NumberIntervals} \leq 1$ or $\text{NumberIntervals} \geq \text{TimeHorizon}$, then the mapping clearly does not require ChangeIntervals , which in fact is not loaded.
- the *variable* `StartUpCost`, of type `netCDF::NcDouble` and either of size 1 or indexed over the *dimension* `NumberIntervals`. This is meant to represent the vector $\text{SC}[t]$ that, for each time instant t , contains the start up cost value of the unit for the corresponding time step. This variable is optional; if it is not provided then it is assumed that $\text{SC}[t] == 0$, i.e., this unit may not have any start up cost. If `StartUpCost` has length 1 then $\text{SC}[t]$ contains the same value for all t . Otherwise, $\text{StartUpCost}[i]$ is the fixed value of $\text{SC}[t]$ for all t in the interval $[\text{ChangeIntervals}[i-1], \text{ChangeIntervals}[i]]$, with the assumption that $\text{ChangeIntervals}[-1] = 0$. If $\text{NumberIntervals} \leq 1$ or $\text{NumberIntervals} \geq \text{TimeHorizon}$, then the mapping clearly does not require ChangeIntervals , which in fact is not loaded.
- the *variable* `LinearTerm`, of type `netCDF::NcDouble` and either of size 1 or indexed over the *dimension* `NumberIntervals`. This is meant to represent the vector $B[t]$ that, for each time instant t , contains the linear term of power cost function of the unit for the corresponding time step. This variable is optional; if it is not provided then it is assumed that $B[t] == 0$, i.e., the cost of the unit has no linear dependence on the produced power (say, only the quadratic one). If `LinearTerm` has length 1 then $B[t]$ contains the same value for all t . Otherwise, $\text{LinearTerm}[i]$ is the fixed value of $B[t]$ for all t in the interval $[\text{ChangeIntervals}[i-1], \text{ChangeIntervals}[i]]$, with the assumption that $\text{ChangeIntervals}[-1] = 0$. If $\text{NumberIntervals} \leq 1$ or $\text{NumberIntervals} \geq \text{TimeHorizon}$, then the mapping clearly does not require ChangeIntervals , which in fact is not loaded.
- the *variable* `ConstTerm`, of type `netCDF::NcDouble` and either of size 1 or indexed over the *dimension* `NumberIntervals`. This is meant to represent the vector $C[t]$ that, for each time instant t , contains the const term of power cost function of the unit for the corresponding time step. This variable is optional; if it is not provided then it is assumed that $C[t] == 0$, i.e., the cost of the unit has no fixed term, only those depending (linearly or quadratically) on the produced power. If `ConstTerm` has length 1 then $C[t]$ contains the same value for all t . Otherwise, $\text{ConstTerm}[i]$ is the fixed value of $C[t]$ for all t in the interval $[\text{ChangeIntervals}[i-1], \text{ChangeIntervals}[i]]$, with the assumption that $\text{ChangeIntervals}[-1] = 0$. If $\text{NumberIntervals} \leq 1$ or $\text{NumberIntervals} \geq \text{TimeHorizon}$, then the mapping clearly does not require ChangeIntervals , which in fact is not loaded.
- the scalar *variable* `InitialPower`, of type `netCDF::NcDouble` and not indexed over any *dimension*. This variable indicates the amount of the power that the unit was producing at time instant -1 , i.e., before the start of the time horizon; this is necessary to compute the ramp-up and ramp-down constraints. Clearly, it must be that $\text{MaxPower} \geq \text{InitialPower} \geq \text{MinPower}$ if the unit was “on” at time instant -1 , and it must be that $\text{InitialPower} == 0$ if the unit was “off” at time instant -1 . The on/off status of the unit is also encoded by the scalar variable `InitUpDownTime`: in particular, $\text{InitUpDownTime} > 0$ then the unit was

on at time instant -1 , and therefore $\text{InitialPower} \geq \text{MinPower}$ must hold, while if $\text{InitUpDownTime} \leq 0$ then the unit was off at time instant -1 , and therefore $\text{InitialPower} == 0$ by definition. In fact, if $\text{InitUpDownTime} \leq 0$ then this variable need not be defined since it is not loaded.

- the scalar *variable* InitUpDownTime , of type `netCDF::NcInt64` and not indexed over any *dimension* and indicates the initial time to generating the unit. If $\text{InitUpDownTime} > 0$, this means that the unit has been on for InitUpDownTime time stamps prior to time stamp 0 (the beginning of the horizon). If, instead, $\text{InitUpDownTime} \leq 0$, this means that the unit has been off for $-\text{InitUpDownTime}$ time stamps prior to time stamp 0; note that $\text{InitUpDownTime} == 0$ means that the unit has been just shut down at the end of time instant -1 , i.e., the beginning of time instant 0.
- the positive scalar *variable* MinUpTime , of type `netCDF::NcUInt64` and not indexed over any *dimension*, which indicates the minimum allowed up time in this unit. This variable is optional, if it is not provided it is taken to be $\text{MinUpTime} == 0$, which mean that the unit can shut down in the very same time stamp in which it starts up.
- the positive scalar *variable* MinDownTime , of type `netCDF::NcUInt64` and not indexed over any *dimension*, which indicates the minimum allowed down time in this unit. This variable is optional, if it is not provided it is taken to be $\text{MinDownTime} == 0$, which mean that the unit can start up in the very same time stamp in which it starts up.
- the *variable* FixedConsumption , of type `netCDF::NcDouble` and either of size 1 or indexed over the *dimension* NumberIntervals . This is meant to represent the vector $\text{FC}[t]$ that, for each time instant t , contains the fixed consumption of the power plant if it is OFF at time t . The variable is optional; if it is not defined, $\text{FC}[t] == 0$ for all time instants. If it has size 1, then $\text{FC}[t] == \text{FixedConsumption}[0]$ for all t , regardless to what NumberIntervals says. Otherwise, $\text{FixedConsumption}[i]$ is the fixed value of $\text{FC}[t]$ for all t in the interval $[\text{ChangeIntervals}[i-1], \text{ChangeIntervals}[i]]$, with the assumption that $\text{ChangeIntervals}[-1] = 0$.
- the *variable* InertiaCommitment , of type `netCDF::NcDouble` and either of size 1 or indexed over the *dimension* NumberIntervals . This is meant to represent the vector $\text{IC}[t]$ that, for each time instant t , contains the contribution that the unit can give to the inertia constraint for the sole fact that is is on (basically, the constant to be multiplied to the commitment variable) at time t . The variable is optional; if it is not defined, $\text{IC}[t] == 0$ for all time instants. If it has size 1, then $\text{IC}[t] == \text{InertiaCommitment}[0]$ for all t , regardless to what NumberIntervals says. Otherwise, $\text{InertiaCommitment}[i]$ is the fixed value of $\text{IC}[t]$ for all t in the interval $[\text{ChangeIntervals}[i-1], \text{ChangeIntervals}[i]]$, with the assumption that $\text{ChangeIntervals}[-1] = 0$.

4.5.2 HydroUnitBlock

The `HydroUnitBlock` class derives from `UnitBlock` and implements a reasonably standard hydro unit of a Unit Commitment Problem. That is, the class is designed in order to give mathematical formulation to describe the operation of large set of hydro storage. To model complex reservoir systems several technical parameters have to be considered. These are divided into reservoir-specific parameters, the hydro links connecting the reservoirs and finally the turbine/pump parameters. The values are collected within a reservoir database, a hydro-link database and a turbine/pump-database. Besides the mandatory `type` attribute of any `:Block`, the group must contain all the data required by the base `UnitBlock`, as described in the comments to `UnitBlock::deserialize(netCDF::NcGroup)`. In particular, we refer to that description for the crucial *dimensions* `TimeHorizon`, `NumberIntervals` and `ChangeIntervals`. The `netCDF::NcGroup` must then also contain:

- The *dimension* `NumberReservoirs` containing the number of all reservoirs (or nodes) in the `HydroUnitBlock`. The *dimension* is optional, if it is not provided then it is taken to be `== 1`, which means that the (in principle) cascading hydro system is actually single hydro reservoir. Note, however, that a single reservoir can still have multiple hydro generating units (see `NumberArcs` below).
- The *dimension* `NumberArcs` containing the set of arcs (or units) connecting the reservoirs in cascading system. Each arc represents either a turbine generating electricity by converting potential energy of water going downhill, or a pump consuming electricity for moving water uphill.
- The *variable* `StartArc`, of type `netCDF::NcUInt64` and indexed over the *dimension* `NumberArcs`; the r -th entry of the variable is the starting point of the arc (a number in $0, \dots, \text{NumberReservoirs} - 1$). Note that arcs are oriented; that is, a positive flow along arc r (turbine) means that water is being taken away from `StartArc[ r ]` and delivered to `EndArc[ r ]` (see next), a negative flow (pump) means vice-versa. Note that reservoir names here go from 0 to $\text{NumberReservoirs} - 1$.
- The *variable* `EndArc`, of type `netCDF::NcUInt64` and indexed over the *dimension* `NumberArcs`; the r -th entry of the variable is the ending point of the arc; this is a number in $0, \dots, \text{NumberReservoirs}$. Note: this is `NumberReservoirs` and **not** `NumberReservoirs - 1`, because arcs can end in the “fake” reservoir `NumberReservoirs`. This indicates that water that flows along that arc “goes away from the system” and it is no longer counted, because it can no longer be used to produce electricity further down the river, or pumped back into one of its reservoirs. Indeed, there will be something like “the most downstream turbines”: after water has been used there, it just goes away down some river and does not go to any other reservoir. Arcs are oriented (see above); `StartArc[ r ] == EndArc[ r ]` (a self-loop) is not allowed, but multiple arcs between the same pair of reservoirs are. Indeed, often the same physical equipment can be used both as a turbine and as a pump; in our model these are represented as two parallel arcs (but with different upper and lower flow capacity, see `MinFlow` and `MaxFlow` below).
- The *variable* `MinFlow`, of type `netCDF::NcDouble` and indexed over both *dimensions* `NumberIntervals` and `NumberArcs`. The first *dimension* may have either size 1 or size `NumberIntervals` whereas the second one always has size `NumberArcs` (if the variable is provided at all). This is meant to represent the matrix `MinF[ t , 1 ]` which, for each time instant t and each arc l , contains the minimum flow value of the unit. This variable is optional; if it is not provided then it is assumed that `MinF[ t , 1 ] == 0`, i.e., the minimum flow of the unit is zero. If the first *dimension* has size 1 then the entry `MinF[ 0 , 1 ]` gives the fixed minimum flow value of the unit for all time steps and each arc l . Otherwise, `MinFlow[ i , 1 ]` is the fixed value of `MinF[ t , 1 ]` for all time t and arc l in the interval $[\text{ChangeIntervals}[i - 1] , \text{ChangeIntervals}[i]]$, with the assumption that `ChangeIntervals[ - 1 ] = 0`. If `NumberIntervals <= 1` or `NumberIntervals >= TimeHorizon`, then the mapping clearly does not require `ChangeIntervals`, which in fact is not loaded.
- The *variable* `MaxFlow`, of type `netCDF::NcDouble` and indexed over both *dimensions* `NumberIntervals` and `NumberArcs`. The first *dimension* may have either size 1 or size `NumberIntervals` whereas the second one always has size `NumberArcs` (if the variable is provided at all). This is meant to represent the matrix `MaxF[ t , 1 ]` which, for each time instant t and each arc l , contains the maximum flow value of the unit. This variable is optional; if it is not provided then it is assumed that `MaxF[ t , 1 ] == 0`, i.e., the maximum flow of the unit is zero. If the first *dimension* has size 1 then the entry `MaxF[ 0 , 1 ]` gives the fixed maximum flow value of the unit for all time steps and each arc l . Otherwise, `MaxFlow[ i , 1 ]` is the fixed value of `MaxF[ t , 1 ]` for all time t and arc l in the interval $[\text{ChangeIntervals}[i - 1] , \text{ChangeIntervals}[i]]$, with

the assumption that $\text{ChangeIntervals}[-1] = 0$. If $\text{NumberIntervals} \leq 1$ or $\text{NumberIntervals} \geq \text{TimeHorizon}$, then the mapping clearly does not require ChangeIntervals , which in fact is not loaded.

Note: MinFlow and MaxFlow values can be either positive or negative (or zero); whenever $\text{MinF}[t, l] < \text{MaxF}[t, l] \leq 0$ for each t and l , the unit is considered a pump and whenever $0 \leq \text{MinF}[t, l] < \text{MaxF}[t, l]$, the unit is considered a turbine. Note that an arc must **always** be the same kind for **all** instants, i.e., it is not allowed that a unit suddenly changes between a turbine and a pump, or vice-versa. This is because the flow-to-active-power function of turbines is a convex piecewise function with possibly many pieces (see NumberPieces , LinearTerm , ConstantTerm below), whereas the flow-to-active-power function of a pump is a simple linear function. In other words, the “number of pieces” of a turbine is ≥ 1 , whereas the “number of pieces” of a pump is necessarily equal to 1. In reality, the same equipment can sometimes be used both as a pump and as a turbine. In our model this is accounted for by artificially splitting the unit into two units, a pump one and a turbine one, which must be done at the data processing stage. This causes the possible problem that at some time instant both the pump and the turbine be active, which is not possible in practice. This is unlikely to happen (because pumps consume more than turbines produce for the same amount of water, so this would be uneconomical), but should it ever happen, this occurrence is not handled in our model (which lets it happen).

- The *variable* MinVolumetric , of type $\text{netCDF}::\text{NcDouble}$ and indexed over both *dimensions* NumberReservoirs and NumberIntervals . The first *dimension* always has size NumberReservoirs (if it is provided at all), whereas the second one may have size one or size NumberIntervals . This is meant to represent the matrix $\text{MinV}[r, t]$ which, for each reservoir r at each time instant t contains the minimum volumetric value of the unit for each reservoir and corresponding time step. It must be that $0 \leq \text{MinV}[r, t] < \text{MaxV}[r, t]$ for all r and t and . This variable is optional; if it's not provided then it is assumed that $\text{MinV}[r, t] == 0$, i.e., the minimum volumetric of the unit is zero. If the second *dimension* has size 1 then the entry $\text{MinV}[r, 0]$ gives the fixed minimum volumetric value of the unit for each reservoir r along all the time horizon. Otherwise, $\text{MinVolumetric}[r, i]$ is the fixed value of $\text{MinV}[r, t]$ for reservoir r and all t in the interval $[\text{ChangeIntervals}[i-1], \text{ChangeIntervals}[i]]$, with the assumption that $\text{ChangeIntervals}[-1] = 0$. If $\text{NumberIntervals} \leq 1$ or $\text{NumberIntervals} \geq \text{TimeHorizon}$, then the mapping clearly does not require ChangeIntervals , which in fact is not loaded.
- The *variable* MaxVolumetric , of type $\text{netCDF}::\text{NcDouble}$ and indexed over both *dimensions* NumberReservoirs and NumberIntervals . The first *dimension* always has size NumberReservoirs (if it is provided at all), whereas the second one may have size one or size NumberIntervals . This is meant to represent the matrix $\text{MaxV}[r, t]$ which, for each reservoir r at each time instant t contains the maximum volumetric value of the unit for each reservoir and corresponding time step. It must be that $0 \leq \text{MinV}[r, t] < \text{MaxV}[r, t]$ for all r and t and . This variable is optional; if it's not provided then it is assumed that $\text{MaxV}[r, t] == 0$, i.e., the maximum volumetric of the unit is zero. If the second *dimension* has size 1 then the entry $\text{MaxV}[r, 0]$ gives the fixed maximum volumetric value of the unit for each reservoir r along all the time horizon. Otherwise, $\text{MaxVolumetric}[r, i]$ is the fixed value of $\text{MaxV}[r, t]$ for reservoir r and all t in the interval $[\text{ChangeIntervals}[i-1], \text{ChangeIntervals}[i]]$, with the assumption that $\text{ChangeIntervals}[-1] = 0$. If $\text{NumberIntervals} \leq 1$ or $\text{NumberIntervals} \geq \text{TimeHorizon}$, then the mapping clearly does not require ChangeIntervals , which in fact is not loaded.

Note: It may happen that $\text{MinV}[r, t] == \text{MaxV}[r, t]$, but only **in a subset of the time instants**. For instance, the user may want to fix the final value of the reservoir, for whatever reason. So, if MinV and MaxV are independent of t , then $\text{MinV}[r] < \text{MaxV}[r]$ must surely happen. If, instead, they depend on t , then equality can be accepted at some instants (but not all of them). Besides the mandatory type attribute of any :Block , the group must contain all the data required by the base UnitBlock , as described in the comments to $\text{UnitBlock}::\text{deserialize}(\text{netCDF}::\text{NcGroup})$. In particular, we refer to that description for the crucial *dimensions* TimeHorizon , NumberIntervals and ChangeIntervals . The $\text{netCDF}::\text{NcGroup}$ must then also contain:

- The *variable* Inflows , of type $\text{netCDF}::\text{NcDouble}$ and indexed over both *dimensions* NumberReservoirs and TimeHorizon . This is meant to represent the matrix $\text{InF}[r, t]$ which, for each reservoir r , contains the amount of water that “naturally” enters into reservoir r (because of rain, ice melting, non-controlled rivers flowing, and of course net of water leaving by evaporation, human consumption etc.) during the time interval t , and therefore that is available in the reservoir at the end of time step t (hence, the beginning of time step $t+1$,

if any). This variable is optional; if it isn't defined, it is taken to be zero. Inflows can be either positive or negative.

- The *variable* `MinPower`, of type `netCDF::NcDouble` and indexed over both *dimensions* `NumberIntervals` and `NumberArcs`. The first *dimension* may have either size 1 or size `NumberIntervals`, whereas the second one always has size `NumberArcs` (if it is provided at all). This is meant to represent the matrix $\text{MinP}[t, l]$ which, for each time instant t at each arc l contains the minimum power value of the unit; it must be that $\text{MinP}[t, l] < \text{MaxP}[t, l]$ for each time instant t and each arc l . This variable is optional; if it is not provided then it is assumed that $\text{MinP}[t, l] == 0$, i.e., the minimum power of all units is zero (which means, each unit is a turbine). If the first *dimension* has size 1 then the entry $\text{MinP}[0, l]$ is assumed to contain the minimum power of arc l for all time instants. Otherwise, $\text{MinPower}[i, l]$ is the fixed value of $\text{MinP}[t, l]$ for arc l and all time t in the interval $[\text{ChangeIntervals}[i - 1], \text{ChangeIntervals}[i]]$, with the assumption that $\text{ChangeIntervals}[-1] = 0$. If `NumberIntervals` ≤ 1 or `NumberIntervals` $\geq \text{TimeHorizon}$, then the mapping clearly does not require `NumberIntervals`, which in fact is not loaded.
- The *variable* `MaxPower`, of type `netCDF::NcDouble` and indexed over both *dimensions* `NumberIntervals` and `NumberArcs`. The first *dimension* may have either size 1 or size `NumberIntervals`, whereas the second one always has size `NumberArcs` (if it is provided at all). This is meant to represent the matrix $\text{MaxP}[t, l]$ which, for each time instant t at each arc l contains the maximum power value of the unit; it must be that $\text{MinP}[t, l] < \text{MaxP}[t, l]$ for each time instant t and each arc l . This variable is optional; if it is not provided then it is assumed that $\text{MaxP}[t, l] == 0$, i.e., the maximum power of all units is zero (which means, each unit is a turbine). If the first *dimension* has size 1 then the entry $\text{MaxP}[0, l]$ is assumed to contain the maximum power of arc l for all time instants. Otherwise, $\text{MaxPower}[i, l]$ is the fixed value of $\text{MaxP}[t, l]$ for arc l and all time t in the interval $[\text{ChangeIntervals}[i - 1], \text{ChangeIntervals}[i]]$, with the assumption that $\text{ChangeIntervals}[-1] = 0$. If `NumberIntervals` ≤ 1 or `NumberIntervals` $\geq \text{TimeHorizon}$, then the mapping clearly does not require `NumberIntervals`, which in fact is not loaded.
- The *variable* `DeltaRampUp`, of type `netCDF::NcDouble` and indexed over both *dimensions* `NumberIntervals` and `NumberArcs`. The first *dimension* may have either size 1 or size `NumberIntervals`, whereas the second one always has size `NumberArcs` (if it is provided at all). This is meant to represent the matrix $\text{DP}[t, l]$ which contains the maximum possible increase of the flow rate at time instant t for arc l . This variable is optional; if it is not provided then it is assumed that $\text{DP}[t, l] == \text{MaxP}[t, l] - \text{MinP}[t, l]$, i.e., all units can ramp up by an arbitrary amount, i.e., there are no ramp-up constraints. If the first *dimension* has size 1 then the entry $\text{DP}[0, l]$ is assumed to contain the maximum possible increase of the flow rate of arc l for all time instants. Otherwise, $\text{DeltaRampUp}[i, l]$ is the fixed value of $\text{DP}[t, l]$ for arc l and all time t in the interval $[\text{ChangeIntervals}[i - 1], \text{ChangeIntervals}[i]]$, with the assumption that $\text{ChangeIntervals}[-1] = 0$. If `NumberIntervals` ≤ 1 or `NumberIntervals` $\geq \text{TimeHorizon}$, then the mapping clearly does not require `NumberIntervals`, which in fact is not loaded.
- The *variable* `DeltaRampDown`, of type `netCDF::NcDouble` and indexed over both *dimensions* `NumberIntervals` and `NumberArcs`. The first *dimension* may have either size 1 or size `NumberIntervals`, whereas the second one always has size `NumberArcs` (if it is provided at all). This is meant to represent the matrix $\text{DM}[t, l]$ which contains the maximum possible decrease of the flow rate at each time instant t of each arc l . This variable is optional; if it is not provided then it is assumed that $\text{DM}[t, l] == \text{MaxP}[t, l] - \text{MinP}[t, l]$, i.e., the unit can ramp down by an arbitrary amount, i.e., there are no ramp-down constraints. If first *dimension* has size 1 then the entry $\text{DM}[0, l]$ is assumed to contain the maximum possible decrease of the flow rate of arc l for all time instants. Otherwise, $\text{DeltaRampDown}[i, l]$ is the fixed value of $\text{DM}[t, l]$ for arc l and all time t in the interval $[\text{ChangeIntervals}[i - 1], \text{ChangeIntervals}[i]]$, with the assumption that $\text{ChangeIntervals}[-1] = 0$. If `NumberIntervals` ≤ 1 or `NumberIntervals` $\geq \text{TimeHorizon}$, then the mapping clearly does not require `NumberIntervals`, which in fact is not loaded.
- The *variable* `PrimaryRho`, of type `netCDF::NcDouble` and indexed both over the *dimensions* `NumberIntervals` and `NumberArcs`. The first *dimension* may have either size 1 or size `NumberIntervals`, whereas the second one always has size `NumberArcs` (if it is provided at all). This is meant to represent the matrix $\text{PR}[$

$t, l]$ which, for each time instant t and arc l , contains the maximum possible fraction of active power that can be used as primary reserve. This variable is optional, when it's not present then $PR[t, l] == 0$ for all t and l , i.e., the unit is not capable of producing any primary reserve. Note that only turbines can produce primary reserve, i.e., $PR[t, l] > 0 \implies \text{MaxP}[t, l] > 0$. If the first *dimension* has size 1 then the entry $PR[0, l]$ is assumed to contain the maximum possible fraction of active power that can be used as primary reserve by arc l for all time instants. Otherwise, $\text{PrimaryRho}[i, l]$ is the fixed value of $PR[t, l]$ for arc l and all t in the interval $[\text{ChangeIntervals}[i - 1], \text{ChangeIntervals}[i]]$, with the assumption that $\text{ChangeIntervals}[-1] = 0$. If $\text{NumberIntervals} \leq 1$ or $\text{NumberIntervals} \geq \text{TimeHorizon}$, then the mapping clearly does not require NumberIntervals , which in fact is not loaded.

- The *variable* `SecondaryRho`, of type `netCDF::NcDouble` and indexed both over the *dimensions* `NumberIntervals` and `NumberArcs`. The first *dimension* may have either size 1 or size `NumberIntervals`, whereas the second one always has size `NumberArcs` (if it is provided at all). This is meant to represent the matrix $SR[t, l]$ which, for each time instant t and arc l , contains the maximum possible fraction of active power that can be used as secondary reserve. This variable is optional, when it's not present then $SR[t, l] == 0$ for all t and l , i.e., the unit is not capable of producing any secondary reserve. Note that only turbines can produce secondary reserve, i.e., $SR[t, l] > 0 \implies \text{MaxP}[t, l] > 0$. If the first *dimension* has size 1 then the entry $SR[0, l]$ is assumed to contain the maximum possible fraction of active power that can be used as secondary reserve by arc l for all time instants. Otherwise, $\text{SecondaryRho}[i, l]$ is the fixed value of $SR[t, l]$ for arc l and all t in the interval $[\text{ChangeIntervals}[i - 1], \text{ChangeIntervals}[i]]$, with the assumption that $\text{ChangeIntervals}[-1] = 0$. If $\text{NumberIntervals} \leq 1$ or $\text{NumberIntervals} \geq \text{TimeHorizon}$, then the mapping clearly does not require NumberIntervals , which in fact is not loaded.
- The *variable* `NumberPieces`, indexed over the *dimension* `NumberArcs`. `NumberPieces[l]` tells how many pieces the concave flow-to-active-power function has for unit (arc) l . Note that pumps must necessarily have exactly one piece. The sum over all i of `NumberPieces[i]` is the total number of pieces (say, `TotalNumberPieces`). Clearly, $\text{TotalNumberPieces} \geq \text{NumberArcs}$; if the flow-to-active-power function for all turbines only have one piece (those of pumps necessarily are so), then $\text{TotalNumberPieces} == \text{NumberArcs}$ and there is no need to define this variable. If, instead, if it is defined, then should always be such that $\text{TotalNumberPieces} \geq \text{NumberArcs}$.
- The *variable* `LinearTerm`, of type `netCDF::NcDouble` and indexed over the set $\{0, \dots, \text{TotalNumberPieces} - 1\}$ (see `NumberPieces`). `LinearTerm[h]` gives the linear term a_h of the linear function $a_h f + b_h$ that defines the concave flow-to-active-power function for some unit; the total function is $F2AP(f) = \min\{a_h f + b_h, h \in H\}$ for some finite set H that depends on the individual unit. It is then necessary to be able to assign a unique index $h = 0, 1, \dots, \text{TotalNumberPieces} - 1$ to each pair (unit, linear function $a_h f + b_h$). When $\text{TotalNumberPieces} == \text{NumberArcs}$, the index is the same as $i = 0, 1, \dots, \text{NumberArcs} - 1$ (there is a one-to-one correspondence between each (unit) arc and each piece). When, instead, $\text{TotalNumberPieces} > \text{NumberArcs}$, a mapping must be defined. The mapping is the obvious one: each index of $i = 0, 1, \dots, \text{NumberArcs} - 1$, corresponds with a unit (arc), and the linear functions for each unit (arc) also have some natural ordering. Thus, in general the mapping is:

- piece 0 = first piece of unit (arc) 0
- piece 1 = second piece of unit (arc) 0
- ...
- piece `NumberPieces[0] - 1` = last piece of unit (arc) 0
- piece `NumberPieces[0]` = first piece of unit (arc) 1
- piece `NumberPieces[0] + 1` = second piece of unit (arc) 1
- ...

which of course boils down to $h = i$ when each arc has exactly one piece.

- The *variable* `ConstantTerm`, of type `netCDF::NcDouble` and indexed over the set $\{0, \dots, \text{TotalNumberPieces} - 1\}$. `ConstantTerm[h]` gives the constant term b_h of the linear function $a_h f + b_h$ that defines the concave flow-to-active-power function for some unit; see the comments to `LinearTerm` for details.

- The *variable* `InertiaPower`, of type `netCDF::NcDouble` and indexed both over the *dimensions* `NumberIntervals` and `NumberArcs`. The first *dimension* may have either size 1 or size `NumberIntervals` whereas the second one always has size `NumberArcs` (if it is provided at all). This is meant to represent the matrix $IP[t, l]$ which, for each time instant t and arc l , contains the contribution that the unit can give to the inertia constraint which depends on the active power that it is currently generating (basically, the constant to be multiplied to the active power variable) at time t for arc l . The variable is optional; if it is not defined, $IP[t, l] == 0$ for each time instants t and arc l . If the first *dimension* has size 1 then the entry $IP[0, l]$ is assumed to contain the the inertia power value for arc l and all time instants t . Otherwise, $InertiaPower[i, l]$ is the fixed value of $IP[t, l]$ for all t in the interval $[ChangeIntervals[i - 1], ChangeIntervals[i]]$, with the assumption that $ChangeIntervals[-1] = 0$ and all l . If `NumberIntervals` ≤ 1 or `NumberIntervals` $\geq TimeHorizon$ then the mapping clearly does not require `ChangeIntervals`, which in fact is not loaded.
- The *variable* `InitialFlowRate`, of type `netCDF::NcDouble` and indexed over the *dimension* `NumberArcs`. Each entry $InFR[i]$ indicates the amount of the flow that was going along arc i at time instant -1 . This is necessary to compute ramp-up and ramp-down limits (cf. `DeltaRampUp` and `DeltaRampDown`), and therefore it is useless if there are no ramp constraints on **any** unit (arc), in which case it is not loaded.
- The *variable* `InitialVolumetric`, of type `netCDF::NcDouble` and indexed over the *dimension* `NumberReservoirs`. Each entry $InV[r]$ indicates the volumes of water in reservoir r at time instant -1 .
- The negative or positive scalar *variable* `UphillFlow`, of type `netCDF::NcInt64` and indexed over the *dimension* `NumberArcs`. Each entry $UpF[l]$ indicates the uphill flow delay for each unit (arc) l ; the nontrivial concept is detailed below. This variable is optional, if it is not provided it is taken to be $UpF[l] == 0$.
- The positive scalar *variable* `DownhillFlow`, of type `netCDF::NcUInt64` and indexed over the *dimension* `NumberArcs`. Each entry $DnF[l]$ indicates the downhill flow delay for each arc (unit) l . This variable is optional, if it is not provided it is taken to be $DnF[l] == 0$.

The last two quantities require some comment. Let us assume that we have any arc l , with $(StartArc[l] = n, EndArc[l] = n')$, which corresponds to (say) a turbine. This means that a positive flow along l implies that water is being taken away from n and delivered to n' , passing through the turbine to produce flow, as graphically depicted below:

$$\backslash n / \Longrightarrow [TURBINE] \Longrightarrow \backslash n' /$$

$$UpF[l] \qquad \qquad \qquad DnF[l]$$

The issue here is that the turbine can be geographically far enough from both n and n' that the water can take a long time (one or more time instant, especially if these are short such as 15 or 5 minutes) to reach the turbine from n and n' from the turbine.

A particular note of caution has to be mentioned regarding the fact that $UpF[l]$ can be **negative**: this means that the water is used in the turbine **before** it goes out of reservoir n . This is counter-intuitive, but can be explained by the fact that the pipe between n and the turbine can be full, and therefore works as a mini reservoir in itself. When the turbine is started, a bubble (depression) is created uphill the turbine; this bubbles up the $n \Rightarrow TURBINE$ pipe until it reaches n , and it is only at that point that the water in n starts flowing away. Hence, there is a negative temporal delay between the water starting flowing in the turbine and it starting flowing away from n . Note that if the $n \Rightarrow TURBINE$ pipe is rather empty, the delay is positive in that one have to start sending the water, which may take some time before filling the pipe and therefore starting the turbine. Of course these are all somewhat crude approximations of the true physical behavior, but they are accurate enough for this setting. Yet, the case $UpF[l] < 0$ cannot be disregarded.

4.5.3 BatteryUnitBlock

The `BatteryUnitBlock` class derives from `UnitBlock` and implements a reasonably standard Battery storage, E-mobility, Centralized demand response, Distributed load management, Distributed storage, and Power to gas units in a single class at Unit commitment problem. This can be the case of actual physical batteries, either large (battery storage) or small (e-mobility, distributed storage), of methods that use some intermediate energy vector with limited local storage/production (power-to-gas units), as well as of logical mechanisms that allow to temporally shift production/consumption in a limited way, thereby acting like an energy storage (centralized demand response, distributed load management). `BatteryUnitBlock` provides a quite general concept of battery that covers different units which mostly fit the same mathematical equations pattern. For instance, a `BatteryUnitBlock` may or may not have a fixed demand (e-mobility has, other units have not) and it may or may not provide primary and secondary reserve (battery storage may do, but other units don't). Battery storage provide an additional flexibility to the system by shifting a surplus of electric energy (e.g. due to high renewable feeding) to times with high demand or lower renewable generation. The distributed battery storage can be aggregated in the energy cells or directly placed in a single node of the network. We will therefore not stress this dependency in the subsequent equations. We emphasize that potential contribution of batteries to inertia is still a subject of active research and should be considered as optional. Besides, since the transport sector is moving towards electrification, electric mobility will have a rising impact on the electricity system. First, electricity demand is growing due to a higher amount of electric vehicles that need to be charged. On the other hand, vehicles are used only a small amount of time while being charged over a much longer timespan (e.g. at night). This allows to shift the charging process in time and provide this flexibility to the overall energy system by means of an additional generator (vehicle-to-grid) or an additional load (power-to-vehicle). Two main differences between battery storages unit and other existing units in this class are:

- battery storages unit can do primary and secondary reserve, while other units cannot;
- some of the units may have a fixed demand that battery storages unit has not.

Moreover, as the considered storage cycle is small w.r.t. the EUC time horizon, distributed storage is not considered as seasonal storage. Hence, the associated mathematical description follows the same equations as the one provided for battery storages unit. The specificity of distributed storage only relies on the fact that it is connected to a distribution grid node.

- The *variable* `MinStorage`, of type `netCDF::NcDouble` and either of size 1 or indexed over the *dimension* `NumberIntervals`. This is meant to represent the vector `MinS[ t ]` that, for each time instant t , contains the minimum storage level of the unit for the corresponding time step. If `MinStorage` has length 1 then `MinS[ t ]` contains the same value for all t . Otherwise, `MinStorage[ i ]` is the fixed value of `MinS[ t ]` for all t in the interval `[ ChangeIntervals[ i - 1 ] , ChangeIntervals[ i ] ]`, with the assumption that `ChangeIntervals[ - 1 ] = 0`. Note that it must always be $0 \leq \text{MinS}[t] < \text{MaxS}[t]$ for all t . If `NumberIntervals` ≤ 1 or `NumberIntervals` $\geq \text{TimeHorizon}$, then the mapping clearly does not require `ChangeIntervals`, which in fact is not loaded.
- The *variable* `MaxStorage`, of type `netCDF::NcDouble` and either of size 1 or indexed over the *dimension* `NumberIntervals`. This is meant to represent the vector `MaxS[ t ]` that, for each time instant t , contains the maximum storage level of the unit for the corresponding time step. If `MaxStorage` has length 1 then `MaxS[ t ]` contains the same value for all t . Otherwise, `MaxStorage[ i ]` is the fixed value of `MaxS[ t ]` for all t in the interval `[ ChangeIntervals[ i - 1 ] , ChangeIntervals[ i ] ]`, with the assumption that `ChangeIntervals[ - 1 ] = 0`. Note that it must always be $0 \leq \text{MinS}[t] < \text{MaxS}[t]$ for all t . If `NumberIntervals` ≤ 1 or `NumberIntervals` $\geq \text{TimeHorizon}$, then the mapping clearly does not require `ChangeIntervals`, which in fact is not loaded.
- The *variable* `MinPower`, of type `netCDF::NcDouble` and either of size 1 or indexed over the *dimension* `NumberIntervals`. This is meant to represent the vector `MinP[ t ]` that, for each time instant t , contains the minimum active power output value of the unit for the corresponding time step. If `MinPower` has length 1 then `MinP[ t ]` contains the same value for all t . Otherwise, `MinPower[ i ]` is the fixed value of `MinP[ t ]` for all t in the interval `[ ChangeIntervals[ i - 1 ] , ChangeIntervals[ i ] ]`, with the assumption that `ChangeIntervals[ - 1 ] = 0`. Note that it must be $\text{MinP}[t] \leq 0$ for all t . If `NumberIntervals` ≤ 1 or `NumberIntervals` $\geq \text{TimeHorizon}$, then the mapping clearly does not require `ChangeIntervals`, which in fact is not loaded.

- The *variable* `MaxPower`, of type `netCDF::NcDouble` and either of size 1 or indexed over the *dimension* `NumberIntervals`. This is meant to represent the vector `MaxP[ t ]` that, for each time instant t , contains the maximum active power output value of the unit for the corresponding time step. If `MaxPower` has length 1 then `MaxP[ t ]` contains the same value for all t . Otherwise, `MaxPower[ i ]` is the fixed value of `MaxP[ t ]` for all t in the interval `[ ChangeIntervals[ i - 1 ] , ChangeIntervals[ i ] ]`, with the assumption that `ChangeIntervals[ - 1 ] = 0`. Note that it must be `MinP[ t ] <= 0` for all t . If `NumberIntervals <= 1` or `NumberIntervals >= TimeHorizon`, then the mapping clearly does not require `ChangeIntervals`, which in fact is not loaded.
- The scalar *variable* `InitialPower`, of type `netCDF::NcDouble` and not indexed over any *dimension*. This variable indicates the amount of the power that the unit was producing at time instant -1 , i.e., before the start of the time horizon; this is necessary to compute the ramp-up and ramp-down constraints. This variable is optional; if `DeltaRampUp` and `DeltaRampDown` are not present, `InitialPower` should not be read, since there are no ramping constraints. If `DeltaRampUp` and `DeltaRampDown` are present but `InitialPower` is not provided, its value is taken to be 0.
- The *variable* `MaxPrimaryPower`, of type `netCDF::NcDouble` and either of size 1 or indexed over the *dimension* `NumberIntervals`. This is meant to represent the vector `MaxPP[ t ]` that, for each time instant t , contains the maximum active power that can be used as primary reserve of the unit for the corresponding time step. If `MaxPrimaryPower` has length 1 then `MaxPP[ t ]` contains the same value for all t . Otherwise, `MaxPrimaryPower[ i ]` is the fixed value of `MaxPP[ t ]` for all t in the interval `[ ChangeIntervals[ i - 1 ] , ChangeIntervals[ i ] ]`, with the assumption that `ChangeIntervals[ - 1 ] = 0`. This variable is optional, if is not provided then `MaxPP[ t ] == 0` for all t . If `NumberIntervals <= 1` or `NumberIntervals >= TimeHorizon`, then the mapping clearly does not require `ChangeIntervals`, which in fact is not loaded.
- The *variable* `MaxSecondaryPower`, of type `netCDF::NcDouble` and either of size 1 or indexed over the *dimension* `NumberIntervals`. This is meant to represent the vector `MaxSP[ t ]` that, for each time instant t , contains the maximum active power that can be used as secondary reserve of the unit for the corresponding time step. If `MaxSecondaryPower` has length 1 then `MaxSP[ t ]` contains the same value for all t . Otherwise, `MaxSecondaryPower[ i ]` is the fixed value of `MaxSP[ t ]` for all t in the interval `[ ChangeIntervals[ i - 1 ] , ChangeIntervals[ i ] ]`, with the assumption that `ChangeIntervals[ - 1 ] = 0`. This variable is optional, if is not provided then `MaxSP[ t ] == 0` for all t . Note that `MaxPP[ t ] == 0` implies `MaxSP[ t ] == 0` (that is, if `MaxPrimaryPower` is not defined then neither should `MaxSecondaryPower`). If `NumberIntervals <= 1` or `NumberIntervals >= TimeHorizon`, then the mapping clearly does not require `ChangeIntervals`, which in fact is not loaded.
- The *variable* `DeltaRampUp`, of type `netCDF::NcDouble` and either of size 1 or indexed over the *dimension* `NumberIntervals`. This is meant to represent the vector `DP[ t ]` that, for each time instant t , contains the ramp-up value of the unit for the corresponding time step, i.e., the maximum possible increase of active power production w.r.t. the power that had been produced in time instant $t - 1$, if any. If `DeltaRampUp` has length 1 then `DP[ t ]` contains the same value for all t . Otherwise, `DeltaRampUp[ i ]` is the fixed value of `DP[ t ]` for all t in the interval `[ ChangeIntervals[ i - 1 ] , ChangeIntervals[ i ] ]`, with the assumption that `ChangeIntervals[ - 1 ] = 0`. This variable is optional; if it is not provided then it is assumed that `DP[ t ] == MaxP[ t ]`, i.e., the unit can ramp up by an arbitrary amount, i.e., there are no ramp-up constraints. If `NumberIntervals <= 1` or `NumberIntervals >= TimeHorizon`, then the mapping clearly does not require `ChangeIntervals`, which in fact is not loaded.
- The *variable* `DeltaRampDown`, of type `netCDF::NcDouble` and either of size 1 or indexed over the *dimension* `NumberIntervals`. This is meant to represent the vector `DM[ t ]` that, for each time instant t , contains the ramp-down value of the unit for the corresponding time step, i.e., the maximum possible decrease of active power production w.r.t. the power that had been produced in time instant $t - 1$, if any. If `DeltaRampDown` has length 1 then `DM[ t ]` contains the same value for all t . Otherwise, `DeltaRampDown[ i ]` is the fixed value of `DM[ t ]` for all t in the interval `[ ChangeIntervals[ i - 1 ] , ChangeIntervals[ i ] ]`, with the assumption that `ChangeIntervals[ - 1 ] = 0`. This variable is optional; if it is not provided then it is assumed that `DM[ t ] == MaxP[ t ]`, i.e., the unit can ramp down an arbitrary amount, i.e., there are no ramp-down constraints. If `NumberIntervals <= 1` or `NumberIntervals >= TimeHorizon`, then the mapping clearly does not require `ChangeIntervals`, which in fact is not loaded.

- The *variable* `StoringBatteryRho`, of type `netCDF::NcDouble` and to be either of size 1 or indexed over the *dimension* `NumberIntervals`. This is meant to represent the vector `SBR[ t ]` that, for each time instant t , contains the inefficiency of storing energy of the unit for the corresponding time step. This variable is optional; if it is not provided then it is assumed that `SBR[ t ] == 1` for all t , i.e., no (significant) energy is spent just for storing it in the battery (this simplifies the model somewhat, see below). If `StoringBatteryRho` has length 1 then `SBR[ t ]` contains the same value for all t . Otherwise, `StoringBatteryRho[ i ]` is the fixed value of `SBR[ t ]` for all t in the interval `[ ChangeIntervals[ i - 1 ] , ChangeIntervals[ i ] ]` with the assumption that `ChangeIntervals[ - 1 ] = 0`. Note that it must be always such that $1 \leq \text{SBR}[t]$ for all t (as `SBR[ t ]` is the amount of energy that has to be used to store 1 unit of energy in the battery). If `NumberIntervals <= 1` or `NumberIntervals >= TimeHorizon`, then the mapping clearly does not require `ChangeIntervals`, which in fact is not loaded.
- The *variable* `ExtractingBatterRho`, of type `netCDF::NcDouble` and to be either of size 1 or indexed over the *dimension* `NumberIntervals`. This is meant to represent the vector `EBR[ t ]` that, for each time instant t , contains the inefficiency of extracting energy of the unit for the corresponding time step. This variable is optional; if it is not provided, then it is assumed that `EBR[ t ] == 1` for all t , i.e., no (significant) energy is spent just for extracting it from the battery (this simplifies the model somewhat, see below). If `ExtractingBatterRho` has length 1 then `EBR[ t ]` contains the same value for all t . Otherwise, `ExtractingBatterRho[ i ]` is the fixed value of `EBR[ t ]` for all t in the interval `[ ChangeIntervals[ i - 1 ] , ChangeIntervals[ i ] ]` with the assumption that `ChangeIntervals[ - 1 ] = 0`. Note that it must be always such that $\text{EBR}[t] \leq 1$ [$\leq \text{SBR}[t]$] for all t (as `EBR[ t ]` is the amount of energy that is obtained when 1 unit of energy is removed from the battery). If `NumberIntervals <= 1` or `NumberIntervals >= TimeHorizon`, then the mapping clearly does not require `ChangeIntervals`, which in fact is not loaded.

Note: the special case in which `EBR[ t ] == SBR[ t ] == 1` for all t , i.e., no energy is spent for storing it in / retrieving it from the battery, leads to significantly simpler mathematical models. In particular, one single variable can be used to represent both storing and retrieving, rather than requiring two separate ones (unless primary and secondary reserve are allowed and/or the cost is defined, since this also requires using two), and the binary variables need not to be defined. For details, see the comments to `generate_abstract_variable()` and `generate_abstract_constraints()`.

- The scalar *variable* `InitialStorage`, of type `netCDF::NcDouble` and not indexed over any *dimension*. This variable indicates the amount of the storage level that the unit was producing at time instant -1 , i.e., before the start of the time horizon; this is necessary to compute the storage level connection with intake and outtake constraints.
- The *variable* `Cost`, of type `netCDF::NcDouble` and either of size 1 or indexed over the *dimension* `NumberIntervals`. This is meant to represent the vector `C[ t ]` that, for each time instant t , contains the monetary cost of storing one unit or energy into, or extracting it from, the battery (the cost is the same in both cases) at the corresponding time step. This variable is optional; if it is not provided then it's taken to be zero. If `Cost` has length 1 then `C[ t ]` contains the same value for all t . Otherwise, `Cost[ i ]` is the fixed value of `C[ t ]` for all t in the interval `[ ChangeIntervals[ i - 1 ] , ChangeIntervals[ i ] ]`, with the assumption that `ChangeIntervals[ - 1 ] = 0`. If `NumberIntervals <= 1` or `NumberIntervals >= TimeHorizon`, then the mapping clearly does not require `ChangeIntervals`, which in fact is not loaded.
- The *variable* `Demand`, of type `netCDF::NcDouble` and indexed over the *dimension* `TimeHorizon`: the entry `Demand[ t ]` is assumed to contain the amount of energy that must be discharged from the battery and sent away for some other purpose (say, driving your e-car) at time t . This variable is optional; if it isn't defined, then `Demand[ t ] == 0`. Otherwise the `Demand[ t ]` contains the demand value for each time instant t .

4.5.4 IntermittentUnitBlock

The `IntermittentUnitBlock` class derives from `UnitBlock` and implements a Intermittent Generation for a unit representing generation (be it centralized or distributed) by intermittent (= unreliable) sources in the unit commitment problem, such as wind farms, solar parks and run-of-the-river hydroelectricity. Each unit is supposed to be connected to a specific node of the clustered network (which means that the distributed case refers to distributed in a small region, where of course small depends on the granularity of the network description). The model relies mainly on historical data of local generation of wind and solar at each node of the grid; these data are used to develop normalized generation profiles associated with wind and solar generators. Intermittent generators are supposed to be able to contribute to primary and secondary reserves. Contribution to the system inertia concerns more specifically run of river generators. The potential contribution of solar or wind generation to inertia is still the subject of active research. Reserve requirements are specified in order to be symmetrically available to increase or decrease power injected into the grid. Besides the mandatory type attribute of any `:Block`, the group must contain all the data required by the base `UnitBlock`, as described in the comments to `UnitBlock::deserialize( netCDF::NcGroup )`. In particular, we refer to that description for the crucial *dimensions* `TimeHorizon`, `NumberIntervals` and `ChangeIntervals`. The `netCDF::NcGroup` must then also contain:

- The *variable* `MinPower`, of type `netCDF::NcDouble` and either of size 1 or indexed over the *dimension* `NumberIntervals`. This is meant to represent the vector $\text{MinP}[t]$ that, for each time instant t , contains the minimum potential production value of the unit for the corresponding time step. If `MinPower` has length 1 then $\text{MinP}[t]$ contains the same value for all t . Otherwise, $\text{MinPower}[i]$ is the fixed value of $\text{MinP}[t]$ for all t in the interval $[\text{ChangeIntervals}[i-1], \text{ChangeIntervals}[i]]$, with the assumption that $\text{ChangeIntervals}[-1] = 0$. If `NumberIntervals` ≤ 1 or `NumberIntervals` $\geq \text{TimeHorizon}$, then the mapping clearly does not require `ChangeIntervals`, which in fact is not loaded. Note that it must be $\text{MinP}[t] \geq 0$ for all t .
- The *variable* `MaxPower`, of type `netCDF::NcDouble` and either of size 1 or indexed over the *dimension* `NumberIntervals`. This is meant to represent the vector $\text{MaxP}[t]$ that, for each time instant t , contains the maximum potential production value of the unit for the corresponding time step. If `MaxPower` has length 1 then $\text{MaxP}[t]$ contains the same value for all t . Otherwise, $\text{MaxPower}[i]$ is the fixed value of $\text{MaxP}[t]$ for all t in the interval $[\text{ChangeIntervals}[i-1], \text{ChangeIntervals}[i]]$, with the assumption that $\text{ChangeIntervals}[-1] = 0$. If `NumberIntervals` ≤ 1 or `NumberIntervals` $\geq \text{TimeHorizon}$, then the mapping clearly does not require `ChangeIntervals`, which in fact is not loaded. Note that it must be $\text{MaxP}[t] \geq \text{MinP}[t] [\geq 0]$ for all t . Yet, $\text{MaxP}[t] == \text{MinP}[t]$ is possible: it means that (at time instant t) the unit cannot be curtailed and cannot provide any reserve.
- The *variable* `InertiaPower`, of type `netCDF::NcDouble` and either indexed over the *dimension* `NumberIntervals` or has size 1. This is meant to represent the vector $\text{IP}[t]$ which, for each time instant t , contains the contribution that the unit can give to the inertia constraint which depends on the active power that it is currently generating (basically, the constant to be multiplied to the active power variable) at time t for this unit. The variable is optional; if it is not defined, $\text{IP}[t] == 0$ for each time instants t . If it has size 1 then the entry $\text{IP}[0]$ is assumed to contain the the inertia power value for this unit and all time instants t . Otherwise, $\text{InertiaPower}[i]$ is the fixed value of $\text{IP}[t]$ for all t in the interval $[\text{ChangeIntervals}[i-1], \text{ChangeIntervals}[i]]$, with the assumption that $\text{ChangeIntervals}[-1] = 0$. If `NumberIntervals` ≤ 1 or `NumberIntervals` $\geq \text{TimeHorizon}$ then the mapping clearly does not require `ChangeIntervals`, which in fact is not loaded.
- The scalar *variable* `Gamma`, of type `netCDF::NcDouble` and not indexed over any *dimension*. This *variable* is used to take into account an uncertainty on the maximal potential production. Note that it must be $0 \leq \text{Gamma} \leq 1$; when `Gamma` $== 0$, the unit does not provide any reserve.
- The scalar *variable* `Kappa`, of type `netCDF::NcDouble` and not indexed over any *dimension*. This *variable* is used to multiply to the minimum and maximum power at each time instant t . This is optional, if it is not provided it is taken to be `Kappa` $== 1$.

4.5.5 SlackUnitBlock

The `SlackUnitBlock` class derives from `UnitBlock` and implements implements a slack unit; a (typically, fictitious) unit capable of producing (typically, a large amount of) active power and/or primary/secondary reserve and/or inertia at any time period completely independently from each other and from all other time periods, albeit at a (typically, huge) cost. Such a unit is typically added to a Unit Commitment problem to ensure that it has a (fictitious) feasible solution, which may help solution methods. At the very least such a modified UC would produce a “least unfeasible” solution which can be used to identify the parts of the system that lack capacity/resources. Besides the mandatory `type` attribute of any `:Block`, the group must contain all the data required by the base `UnitBlock`, as described in the comments to `UnitBlock::deserialize( netCDF::NcGroup )`. In particular, we refer to that description for the crucial *dimensions* `TimeHorizon`, `NumberIntervals` and `ChangeIntervals`. The `netCDF::NcGroup` must then also contain:

- The *variable* `MaxPower`, of type `netCDF::NcDouble` and either of size 1 or indexed over the *dimension* `NumberIntervals`. This is meant to represent the vector `MxP[ t ]` that, for each time instant t , contains the maximum active power output value of the unit for the corresponding time step. If `MaxPower` has length 1 then `MxP[ t ]` contains the same value for all t . Otherwise, `MaxPower[ i ]` is the fixed value of `MxP[ t ]` for all t in the interval `[ ChangeIntervals[ i - 1 ] , ChangeIntervals[ i ] ]`, with the assumption that `ChangeIntervals[ - 1 ] = 0`. This *variable* is optional, if is not provided then `MxP[ t ] == 0` for all t . Note that it must be `MxP[ t ] >= 0` for all t . If `NumberIntervals <= 1` or `NumberIntervals >= TimeHorizon` then the mapping clearly does not require `ChangeIntervals`, which in fact is not loaded.
- The *variable* `MaxPrimaryPower`, of type `netCDF::NcDouble` and either of size 1 or indexed over the *dimension* `NumberIntervals`. This is meant to represent the vector `MaxPP[ t ]` that, for each time instant t , contains the maximum amount of primary reserve that the unit can produce in the corresponding time step. If `MaxPrimaryPower` has length 1 then `MaxPP[ t ]` contains the same value for all t . Otherwise, `MaxPrimaryPower[ i ]` is the fixed value of `MaxPP[ t ]` for all t in the interval `[ ChangeIntervals[ i - 1 ] , ChangeIntervals[ i ] ]`, with the assumption that `ChangeIntervals[ - 1 ] = 0`. This *variable* is optional, if is not provided then `MaxPP[ t ] == 0` for all t . If `NumberIntervals <= 1` or `NumberIntervals >= TimeHorizon` then the mapping clearly does not require `ChangeIntervals`, which in fact is not loaded.
- The *variable* `MaxSecondaryPower`, of type `netCDF::NcDouble` and either of size 1 or indexed over the *dimension* `NumberIntervals`. This is meant to represent the vector `MaxSP[ t ]` that, for each time instant t , contains the maximum amount of secondary reserve that the unit can produce in the corresponding time step. If `MaxSecondaryPower` has length 1 then `MaxSP[ t ]` contains the same value for all t . Otherwise, `MaxSecondaryPower[ i ]` is the fixed value of `MaxSP[ t ]` for all t in the interval `[ ChangeIntervals[ i - 1 ] , ChangeIntervals[ i ] ]`, with the assumption that `ChangeIntervals[ - 1 ] = 0`. This *variable* is optional, if is not provided then `MaxSP[ t ] == 0` for all t . If `NumberIntervals <= 1` or `NumberIntervals >= TimeHorizon` then the mapping clearly does not require `ChangeIntervals`, which in fact is not loaded.
- The *variable* `MaxInertia`, of type `netCDF::NcDouble` and either indexed over the *dimension* `NumberIntervals` or has size 1. This is meant to represent the vector `MaxI[ t ]` which, for each time instant t , contains the maximum “amount of inertia” (contribution that the `SlackUnit` can give to the inertia constraint) at time t . This *variable* is optional; if it is not defined, `MaxI[ t ] == 0` for all time instants. If it has size 1, then `MaxI[ t ] == MaxInertia[ 0 ]` for all t , regardless to what `NumberIntervals` says. Otherwise, `MaxInertia[ i ]` is the fixed value of `MaxI[ t ]` for all t in the interval `[ ChangeIntervals[ i - 1 ] , ChangeIntervals[ i ] ]`, with the assumption that `ChangeIntervals[ - 1 ] = 0`. If `NumberIntervals <= 1` or `NumberIntervals >= TimeHorizon` then the mapping clearly does not require `ChangeIntervals`, which in fact is not loaded.
- The *variable* `ActivePowerCost`, of type `netCDF::NcDouble` and either indexed over the *dimension* `NumberIntervals` or has size 1. This is meant to represent the vector `APC[ t ]` that, for each time instant t , contains the cost of producing one unit of active power at the corresponding time step. This *variable* is optional, if it is not provided then it’s taken to be zero(although this is a very strange setting, as it would typically imply that all the demand, or at least as much as possible of it, is satisfied by the fictitious `SlackUnit` rather than from “real” ones). If `ActivePowerCost` has length 1 then `APC[ t ]` contains the same value for t . Otherwise,

$\text{ActivePowerCost}[i]$ is the fixed value of $\text{APC}[t]$ for all t in the interval $[\text{ChangeIntervals}[i-1], \text{ChangeIntervals}[i]]$, with the assumption that $\text{ChangeIntervals}[-1] = 0$. If $\text{NumberIntervals} \leq 1$ or $\text{NumberIntervals} \geq \text{TimeHorizon}$ then the mapping clearly does not require ChangeIntervals , which in fact is not loaded.

- The *variable* PrimaryCost , of type `netCDF::NcDouble` and either indexed over the *dimension* NumberIntervals or has size 1. This is meant to represent the vector $\text{PC}[t]$ that, for each time instant t , contains the cost of producing one unit of primary reserve at the corresponding time step. This *variable* is optional; if it is not provided then it's taken to be zero (but this is a very strange setting, cf. the discussion in ActivePowerCost). If PrimaryCost has length 1 then $\text{PC}[t]$ contains the same value for t . Otherwise, $\text{PrimaryCost}[i]$ is the fixed value of $\text{PC}[t]$ for all t in the interval $[\text{ChangeIntervals}[i-1], \text{ChangeIntervals}[i]]$, with the assumption that $\text{ChangeIntervals}[-1] = 0$. If $\text{NumberIntervals} \leq 1$ or $\text{NumberIntervals} \geq \text{TimeHorizon}$ then the mapping clearly does not require ChangeIntervals , which in fact is not loaded.
- The *variable* SecondaryCost , of type `netCDF::NcDouble` and either indexed over the *dimension* NumberIntervals or has size 1. This is meant to represent the vector $\text{SC}[t]$ that, for each time instant t , contains the cost of producing one unit of secondary reserve at the corresponding time step. This *variable* is optional; if it is not provided then it's taken to be zero (but this is a very strange setting, cf. the discussion in ActivePowerCost). If SecondaryCost has length 1 then $\text{SC}[t]$ contains the same value for t . Otherwise, $\text{SecondaryCost}[i]$ is the fixed value of $\text{SC}[t]$ for all t in the interval $[\text{ChangeIntervals}[i-1], \text{ChangeIntervals}[i]]$, with the assumption that $\text{ChangeIntervals}[-1] = 0$. If $\text{NumberIntervals} \leq 1$ or $\text{NumberIntervals} \geq \text{TimeHorizon}$ then the mapping clearly does not require ChangeIntervals , which in fact is not loaded.
- The *variable* InertiaCost , of type `netCDF::NcDouble` and either indexed over the *dimension* NumberIntervals or has size 1. This is meant to represent the vector $\text{IC}[t]$ that, for each time instant t , contains the cost of producing “one unit” of inertia. Since the inertia-producing variable is $u[t]$, which is in the interval $[0, 1]$, the cost of $u[t]$ is $\text{MaxI}[t] * \text{IC}[t]$; in other words, $u[t]$ represents the fraction of the maximum possible amount of inertia ($\text{MaxI}[t]$) that can be produced at time step t . This *variable* is optional; if it is not provided then it's taken to be zero (but this is a very strange setting, cf. the discussion in ActivePowerCost). If InertiaCost has length 1 then $\text{IC}[t]$ contains the same value for t . Otherwise, $\text{InertiaCost}[i]$ is the fixed value of $\text{IC}[t]$ for all t in the interval $[\text{ChangeIntervals}[i-1], \text{ChangeIntervals}[i]]$, with the assumption that $\text{ChangeIntervals}[-1] = 0$. If $\text{NumberIntervals} \leq 1$ or $\text{NumberIntervals} \geq \text{TimeHorizon}$ then the mapping clearly does not require ChangeIntervals , which in fact is not loaded.

4.6 HydroSystemUnitBlock

The class `HydroSystemUnitBlock`, which derives from the `Block`, in order to define a base class for any possible hydro unit plus the linking `PolyhedralFunctionBlock` to describe the future value of water function in a `UCBlock`. Besides the mandatory type attribute of any `:Block`, the group should contain the following:

- The *dimension* `NumberHydroUnits` containing the number of hydro units (`HydroUnitBlock`) in the problem.
- The *groups* `HydroUnitBlock_0`, `HydroUnitBlock_1`, ..., `HydroUnitBlock_(n-1)`, with $n == \text{NumberHydroUnits}$, containing each one `HydroUnitBlock`.
- The *group* `PolyhedralFunctionBlock` which contains a `PolyhedralFunctionBlock`, whose `PolyhedralFunction` represents the future value of the water (a.k.a. Bellman values) left at the end of the time horizon in all the reservoirs of all the `HydroUnitBlock` of the `HydroSystemUnitBlock`.

The future value of water function is represented by the single `PolyhedralFunction` which lives inside the `PolyhedralFunctionBlock`. The vector of active variable of `PolyhedralFunction` is therefore in a one-to-one correspondence with the set of `ColVariable` in the `HydroUnitBlock` that represent the amount of water left in each reservoir at the end of the time horizon. Thus, it is necessary to specify the order of the active `ColVariable` of the `PolyhedralFunction`. Let us denote by $X[0]$, $X[1]$, ..., $X[R-1]$ the vector of active `ColVariable` (i.e., $X[i]$ is the one returned by `get_active_var(i)` and $R = \text{get_num_active_var}()$). Since each `HydroUnitBlock` can have more than one reservoir (cf. `HydroUnitBlock::get_number_reservoirs()`), R is just the total number of reservoir, which is computed by just calling `get_number_reservoirs()` on each of the `HydroUnitBlock` and summing all the results. Clearly, $R \geq \text{NumberHydroUnits}$. Some of the `HydroUnitBlock` may have just one reservoir; if this happens for all the hydro unit blocks (but this is not likely), then $R == \text{NumberHydroUnits}$. In this case the mapping is obvious: $X[i]$ is the `ColVariable` that represent the amount of water left in the only reservoir of `HydroUnitBlock_i` at the end of the time horizon. When, instead, $R > \text{NumberHydroUnits}$, a mapping must be defined. The mapping is the obvious one: `HydroUnitBlock` have an ordering $n = 0, 1, \dots, \text{NumberHydroUnits} - 1$ (cf. the *groups* `HydroUnitBlock_0`, `HydroUnitBlock_1`, ... above), and the reservoirs into each `HydroUnitBlock` also have a natural ordering. Thus, in general the mapping is:

- $X[0] = \text{ColVariable}$ representing the amount of water left in the first reservoir of `HydroUnitBlock_0` at the end of the time horizon
- $X[1] = \text{ColVariable}$ representing the amount of water left in the second reservoir of `HydroUnitBlock_0` at the end of the time horizon
- ...
- $X[k] = \text{ColVariable}$ representing the amount of water left in the k -th reservoir of `HydroUnitBlock_0` at the end of the time horizon, with $k = \text{HydroUnitBlock_0} \rightarrow \text{get_number_reservoirs}()$
- $X[k+1] = \text{ColVariable}$ representing the amount of water left in the first reservoir of `HydroUnitBlock_1` at the end of the time horizon
- $X[k+2] = \text{ColVariable}$ representing the amount of water left in the second reservoir of `HydroUnitBlock_1` at the end of the time horizon
- ...

This must be the format of the data (linear inequalities) that define the `PolyhedralFunction`: the i -th entry of each vector is related to the future value of the water stored in the reservoir identified by the above mapping. See `PolyhedralFunction::deserialize()` for details about how the data must be stored in the `PolyhedralFunctionBlock` *group*.

4.7 PolyhedralFunctionBlock

The class `PolyhedralFunctionBlock` derives from `AbstractBlock` to define the following concept: a `Block` whose sole distinguishing feature is a `PolyhedralFunction` as objective, but which can otherwise contain any kind of “abstract” `Variable` and `Constraint` (provided these are handled by the base `AbstractBlock` class). The rationale for the `PolyhedralFunctionBlock` class is that a `PolyhedralFunction` is a perfectly fine object in itself, but one that several solvers cannot easily deal with in its “natural” form. However, a `PolyhedralFunction` can also be represented in a very natural way by means of one extra continuous `ColVariable` plus a finite set of (dynamic) `FRowConstraint` with `LinearFunction` inside (a.k.a., linear constraints). Thus, the main feature that `PolyhedralFunctionBlock` implements is the ability to “present” itself (construct an “abstract representation”) as either having a `FRealObjective` with a `PolyhedralFunction` inside, or having a set of (continuous) `ColVariable` and (linear) `RowConstraint`. We call the first the “natural representation” of a `PolyhedralFunctionBlock`, and the latter its “linearized representation”. Note that most individual changes in the `PolyhedralFunction` result in many changes to the “linearized representation”, that are properly bunched into appropriate `GroupModification`. Conversely, many individual changes to the “linearized representation” cannot (or would be too complex to) be implemented in the `PolyhedralFunction`, because each one of them individually would lead it to end in a partly inconsistent state, and only a co-ordinated set of them (say, properly bunched into an appropriate `GroupModification`) would work. Hence, a number of changes to the “linearized representation” are not allowed; see the comments to the protected method `guts_of_addModification_LR()` for details. Other than that, `PolyhedralFunctionBlock` entirely relies on the machinery provided by `AbstractBlock` to handle all the rest of the `Block`, and therefore is subject to the limitations of that class regarding what kind of `Constraint`, `Variable` and `Objective` are supported. Besides the mandatory type attribute of any `:Block`, the group should contain the following:

- all the data necessary to describe a `PolyhedralFunction`; see §4.7.1 for details;
- any other data necessary to represent the “arbitrary” part of the `AbstractBlock`; see `AbstractBlock::deserialize()` for details.

4.7.1 PolyhedralFunction

The `PolyhedralFunction` class derives from `C05Function`, and defines a simple implementation of a convex (or concave) `Function` defined by the maximum (or minimum) of a “small” number of explicitly provided affine forms. In other words, if the `PolyhedralFunction` depends on a set of n `ColVariable`, its input data is a $m \times n$ matrix A and a $m \times 1$ vector b (with m given and “small”). Serialize a `PolyhedralFunction` into a `netCDF::NcGroup`, with the following format:

- The *dimension* `PolyFunction_NumVar` containing the number of columns of the A matrix, i.e., the number of active variables.
- The *dimension* `PolyFunction_NumRow` containing the number of rows of the A matrix. The dimension is optional, if it is not provided than 0 (no rows) is assumed.
- The *variable* `PolyFunction_A`, of type `netCDF::NcDouble` and indexed over both the *dimensions* `NumRow` and `NumVar` (in this order); it contains the (row-major) representation of the matrix A . The variable is only optional if `NumRow == 0`.
- The *variable* `PolyFunction_b`, of type `netCDF::NcDouble` and indexed over the *dimension* `NumRow`, which contains the vector b . The variable is only optional if `NumRow == 0`.
- The *dimension* `PolyFunction_sign` (actually a `bool`), which contains the “verse” of the `PolyhedralFunction`, i.e., true for a convex max-function and false for a concave min-function (encoded in the obvious way, i.e., zero for false, nonzero for true). The *variable* is optional, if it is not provided true is assumed.
- The scalar *variable* `PolyFunction_lb`, of type `netCDF::NcDouble` and not indexed over any *dimension*, which contains the global lower (if `PolyFunction_sign == true`, upper otherwise) bound on the value of the function over all the space. The *variable* is optional; if it is not provided it means that no finite lower (upper) bound exists, i.e., the lower (upper) bound is `-inf (+inf)`.

4.8 NetworkBlock

The class `NetworkBlock`, which derives from the `Block`, defines the basic interface for the constraints/optimization problems which describe the behaviour of the transmission network in a specific time instant in the Unit Commitment (UC) problem, as represented in `UCBlock`. The base class handles only basic information: it allows to read/set the topology (and capacity/susceptances) of the network, and the active power demand at the different nodes in the given time instant. This information is actually bunched together in a small “passive” `NetworkData` object (no methods, just a data repository) that can be either de-serialized or passed ready-made (typically, by the `UCBlock`). Details of the kind of network that is implemented (“bus”, DC equations, AC equations, OPF, ...) are entirely demanded to derived objects. The interface between a `NetworkBlock` and the rest of the UC is just the vector of node injection variables, which will have to satisfy the technical constraints of the transmission network. Besides the mandatory type attribute of any `:Block`, the group should contain the following:

- Optionally, the *dimensions* and variables necessary to a `NetworkData` object, that describe the transmission network; see `NetworkData` in §4.8.1 for details. All that is optional, because the `NetworkData` object can alternatively be passed to the `NetworkBlock` via a call to `set_NetworkData()`. Note that if `set_NetworkData()` is called, but the representation of a `NetworkData` object is found in the `NcGroup`, then the `NetworkData` passed by `set_NetworkData()` is ignored, and a new `NetworkData` object is read from the `NcGroup` and used instead.
- The `ActiveDemand`, of type `netCDF::NcDouble`, and of size “number of nodes”. If the `NetworkData` object description is present in the `NcGroup` this is the *dimension* `NumberNodes`, but the `NetworkData` object is optional and it may not be there. Thus if `NumberNodes` is not there and `ActiveDemand` is, then the `NetworkData` object must have been passed by `set_NetworkData()`, and the number of nodes can be read via `NetworkData::get_number_nodes()`. However, `ActiveDemand` itself is optional. If it is not found in the `NcGroup`, then it **must** be passed (either before or after the call to `deserialize()`) by calling `set_ActiveDemand()`. Since both groups of data are optional, the `NcGroup` can actually be empty which implies that all the data will be (or have been) passed by the in-memory interface. In this case, it would clearly be preferable to **entirely avoid the `NcGroup` to be there**, and in fact `UCBlock` has provisions for the `NcGroup` describing the `NetworkBlock` to be optional (see `UCBlock` in §4.4).

4.8.1 NetworkData

The `NetworkData` class is a nested sub-class which only serves to have a quick way to load all the basic data (topology and electrical characteristics) that describe the transmission network. The rationale is that while often the network does not change during the (short) time horizon of UC, it makes sense to allow for this to happen. This means that individual `NetworkBlock` objects may in principle have different `NetworkData`, but most often they can share the same. By bunching all the information together we make it easy for this sharing to happen. Deserialize a `NetworkData` out of a `netCDF::NcGroup`, which should contain the following:

- the *dimension* `NumberNodes` containing the number of nodes in the problem; this *dimension* is optional, if it is not provided then it is taken to be `== 1`; If `NumberNodes == 1` (equivalently, it is not provided), the network is a “bus” formed of only one node, and therefore all the subsequent information need not to be present since it is not loaded. If `NumberNodes > 1`, then all the subsequent information is mandatory;
- the *dimension* `NumberLines` containing the number of lines in the transmission network;
- the *variable* `StartLine`, of type `int` and indexed over the *dimension* `NumberLines`; the i -th entry of the variable is the starting point of the line (a number in $0, \dots, \text{NumberLines} - 1$). Note that lines are not oriented, but the flow of energy is; that is, a positive flow along line i means that energy is being taken away from `StartLine[ i ]` and delivered to `EndLine[ i ]` (see next), a negative flow means vice-versa. Note that lines names here go from `0` to `NLines.getSize() - 1`;
- the *variable* `EndLine`, of type `int` and indexed over the *dimension* `NumberLines`; the i -th entry of the variable is the ending point of the line (a number in $0, \dots, \text{NumberLines} - 1$. Note that lines are not oriented, but see above). `StartLine[ i ] == EndLine[ i ]` (a self-loop) is not allowed, but multiple lines between the same pair of nodes are. Note that lines names here go from `0` to `NLines.getSize() - 1`;
- the *variable* `MinPowerFlow`, of type `netCDF::NcDouble` and indexed over the *dimension* `NumberLines`; the i -th entry of the variable is assumed to contain the minimum power flow on line i (note that this is typically a negative number as lines are bi-directional, see above);
- the *variable* `MaxPowerFlow`, of type `netCDF::NcDouble` and indexed over the *dimension* `NumberLines`; the i -th entry of the variable is assumed to contain the maximum power flow at line i (a non-negative number);
- the *variable* `Susceptance`, of type `netCDF::NcDouble` and indexed over the *dimension* `NumberLines`; the i -th entry of this variable is assumed to contain the susceptance of line i . Note that this is strictly a positive value.

4.8.2 BusNetworkBlock

The `BusNetworkBlock` class, which derives from `NetworkBlock` [see `NetworkBlock.h`] implements the `Block` concept [see `Block.h`] in order to define a bus transmission network in the Unit Commitment problem for a given instant in the time horizon. There is actually precious little that this class has to do that is not done already by the base `NetworkBlock` class.

4.8.3 DCNetworkBlock

The `DCNetworkBlock` class derives from `NetworkBlock`, and defines the standard linear constraints corresponding to the “DC model” of the transmission network in the Unit Commitment problem.

4.9 HeatBlock

The `HeatBlock` class implements the `Block` concept [see `Block.h`] for a set of “heat units” as required in the `plan4res` project. The constraints regarding heat management in the `plan4res` project can be described separately for each `HeatBlock`. In the parlance of `plan4res`, a `HeatBlock` is the intersection of an `Energy Cell` and a `heat-ID`: a set of heat-producing units that are geographically close enough to exchange heat of the same type, together with possibly a (single) storage for this type of heat, in order to satisfy a given heat demand. `HeatBlocks` are only connected to each other because some heat-producing units are also electricity-producing ones, where heat is a by-product or a co-product of electricity. However, this linking “happens at the level of the main UC model”, and therefore it is not required for the description of the internal constraints of a `HeatBlock`. All this on a given time horizon, as in the UC problem. The operations of the heat generating unit are described on a discrete time horizon where there may exist some heat-producing units, possibly of a single heat storage, and of the demand that has to be satisfied. Besides the mandatory type attribute of any `:Block`, the group should contain all the following:

- the *dimension* `NumberHeatUnits` containing the number heat-producing units. This is not optional, and has to be ≥ 1 ;
- the *dimension* `TimeHorizon` containing the time horizon. The *dimension* is optional because the same information may be passed via the method `set_time_horizon()`, or directly retrieved from the father if it is a `UCBlock`; see the comments to `set_time_horizon()` for details;
- the *dimension* `NumberIntervals` that is provided to allow that all time-dependent data in the `HeatBlock` can only change at a subset of the time instants of the time interval, being therefore piecewise-constant (possibly, constant). `NumberIntervals` should therefore be $\leq \text{TimeHorizon}$, with three distinct cases:
 - `NumberIntervals` ≤ 1 , which is taken to mean `NumberIntervals` $= 1$; this is what is assumed if the *dimension*, that is optional, is not there. This means that the value of each relevant data in the `UCBlock` (see e.g. `CostHeatUnit`, `MinHeatProduction` and `MaxHeatProduction` below) is the same for each time instant $0, \dots, \text{TimeHorizon} - 1$ in the time horizon. In this case, the variable `ChangeIntervals` (see below) is ignored;
 - $1 < \text{NumberIntervals} < \text{TimeHorizon}$, which means that in some time instants, **but not all of them**, the values of some of the relevant data are changing; the intervals are then described in variable `NumberIntervals`;
 - `NumberIntervals` $= \text{TimeHorizon}$, which means that values of the relevant data changes at every time interval (in principle; of course there is nothing preventing the same value to be repeated in the `netCDF` input). Also in this case the variable `NumberIntervals` is ignored, since it is useless.

Note that this (together with `NumberIntervals`, if defined) obviously sets the maximum frequency at which data can change; if some data changes less frequently (say, it is constant), then the same value will have to be repeated. Individual data can also have specific provisions for the case where the data is all equal despite `NumberIntervals` saying differently.

- the *variable* `ChangeIntervals`, of type `integer` and indexed over the *dimension* `NumberIntervals`. The time horizon is subdivided into `NumberIntervals = k` of the form $[0, i_1], [i_1+1, i_2], \dots, [i_{k-1}+1, \text{TimeHorizon}-1]$; `ChangeIntervals` then has to contain $[i_1, i_2, \dots, i_{k-1}]$. Note that, therefore, `ChangeIntervals` has one significant value less than `NumberIntervals`, which means that `ChangeIntervals[ NumberIntervals - 1 ]` is ignored. Anyway, the whole variable is ignored if either `NumberIntervals` ≤ 1 (such as if it is not defined), or `NumberIntervals` $\geq \text{TimeHorizon}$;
- the *variable* `TotalHeatDemand`, of type `netCDF::NcDouble` and indexed over the *dimension* `TimeHorizon`: entry `TotalHeatDemand[ t ]` is assumed to contain the total heat demand of this heat block to be satisfied for the corresponding time instant t . Note that this variable does **not** use the `NumberIntervals` / `ChangeIntervals` system, as demand is usually changing from one time instant to the next;
- the *variable* `CostHeatUnit`, of type `netCDF::NcDouble` and indexed over two *dimensions*. The first *dimension* can have size 1 or `NumberIntervals`. The second *dimension* has size `NumberHeatUnits`. This is meant to represent the matrix `CHU[ t , i ]` which is assumed to contain the unitary cost of heat production of heat-producing unit i for time instant t . If the first *dimension* has size 1, then the cost is the same for all time

instants (for the same unit). Otherwise, $\text{CostHeatUnit}[h, i]$ is the fixed value of $\text{CHU}[t, i]$ for all t in the interval $[\text{ChangeIntervals}[h - 1], \text{ChangeIntervals}[h]]$, with the assumption that $\text{ChangeIntervals}[-1] = 0$;

- the *variable* `MinHeatProduction`, of type `netCDF::NcDouble` and indexed over two *dimensions*. The first *dimension* can have size 1 or `NumberIntervals`. The second *dimension* has size `NumberHeatUnits`. This is meant to represent the matrix $\text{MinHP}[t, i]$ which is assumed to contain the minimum heat production of heat-producing unit i for time instant t . The variable is optional; if it is not provided at all, it is intended $\text{MinHP}[t, i] == 0$ for all i and t . If the first *dimension* has size 1, then the minimum heat production is the same for all time instants (for the same unit). Otherwise, $\text{MinHeatProduction}[h, i]$ is the fixed value of $\text{MinHP}[t, i]$ for all t in the interval $[\text{ChangeIntervals}[h - 1], \text{ChangeIntervals}[h]]$, with the assumption that $\text{ChangeIntervals}[-1] = 0$;
- the *variable* `MaxHeatProduction`, of type `netCDF::NcDouble` and indexed over two *dimensions*. The first *dimension* can have size 1 or `NumberIntervals`. The second *dimension* has size `NumberHeatUnits`. This is meant to represent the matrix $\text{MaxHP}[t, i]$ which is assumed to contain the maximum heat production of heat-producing unit i for time instant t . It is assumed $\text{MaxHP}[t, i] \geq \text{MinHP}[t, i] \geq 0$ for all i and t , with strict inequality holding for at least some t for each unit i (otherwise the production of unit i is fixed and there is nothing to decide). The variable is optional; if it is not provided at all, it is intended $\text{MaxHP}[t, i] == 0$ for all i and t . If the first *dimension* has size 1, then the maximum heat production is the same for all time instants (for the same unit). Otherwise, $\text{MaxHeatProduction}[h, i]$ is the fixed value of $\text{MaxHP}[t, i]$ for all t in the interval $[\text{ChangeIntervals}[h - 1], \text{ChangeIntervals}[h]]$, with the assumption that $\text{ChangeIntervals}[-1] = 0$;
- the *variable* `MinHeatStorage`, of type `netCDF::NcDouble` and either indexed over the *dimension* `NumberIntervals` or has size 1. This is meant to represent the vector $\text{MinHS}[t]$ which, for each time instant t , contains the minimum heat storage of this `HeatBlock`. The variable is optional, if it is not provided at all it is intended that $\text{MinHS}[t] == 0$ for all t . If the variable has size 1, then the minimum heat storage is the same for all time instants. Otherwise, $\text{MinHeatStorage}[h]$ is the fixed value of $\text{MinHS}[t]$ for all t in the interval $[\text{ChangeIntervals}[h - 1], \text{ChangeIntervals}[h]]$, with the assumption that $\text{ChangeIntervals}[-1] = 0$;
- the *variable* `MaxHeatStorage`, of type `netCDF::NcDouble` and either indexed over the *dimension* `NumberIntervals` or has size 1. This is meant to represent the vector $\text{MaxHS}[t]$ which, for each time instant t , contains the maximum heat storage of this `HeatBlock`. The variable is optional, if it is not provided at all it is intended that $\text{MaxHS}[t] == 0$ for all t and since it's assumed that $\text{MaxHS}[t] \geq \text{MinHS}[t] \geq 0$ for all t , this means that there is no heat storage in this `HeatBlock`. If the variable has size 1, then the maximum heat storage is the same for all time instants. Otherwise, $\text{MaxHeatStorage}[h]$ is the fixed value of $\text{MaxHS}[t]$ for all t in the interval $[\text{ChangeIntervals}[h - 1], \text{ChangeIntervals}[h]]$, with the assumption that $\text{ChangeIntervals}[-1] = 0$;
- the scalar *variable* `InitialHeatAvailable`, of type `netCDF::NcDouble` and not indexed over any *dimension*, which indicates the the amount of heat in the storage at the beginning of the first time instant in this `HeatBlock`. It is assumed that $\text{MaxHS}[0] \geq \text{InitialHeatAvailable} \geq \text{MinHS}[0]$. This variable is optional, if it is not provided it is taken to be $\text{InitialHeatAvailable} == \text{MinHS}[0]$. If there is no heat storage (say, `MaxHeatStorage` is not defined) then this variable is not read, because it is not used;
- the scalar *variable* `StoringHeatRho`, of type `netCDF::NcDouble` and not indexed over any *dimension*, which indicates the inefficiency of storing heat in the heat storage (if any) in this `HeatBlock`. This variable is optional and it must always be $\text{StoringHeatRho} \leq 1$, if it is not provided it is taken to be $\text{StoringHeatRho} == 1$. If there is no heat storage (say, `MaxHeatStorage` is not defined) then this variable is not read, because it is not used;
- the scalar *variable* `ExtractingHeatRho`, of type `netCDF::NcDouble` and not indexed over any *dimension*, and which indicates the inefficiency of extracting heat from the heat storage (if any) in this `HeatBlock`. This variable is optional and it must always be $\text{ExtractingHeatRho} \geq 1$, if it is not provided it is taken to be $\text{ExtractingHeatRho} == 1$. If there is no heat storage (say, `MaxHeatStorage` is not defined) then this variable is not read, because it is not used;

- the scalar *variable* `KeepingHeatRho`, of type `netCDF::NcDouble` and not indexed over any *dimension*, which indicates the double of keeping heat in the heat storage (if any) in this `HeatBlock`. This variable is optional and it must always be `KeepingHeatRho <= 1`, if it is not provided it is taken to be `KeepingHeatRho == 1`. If there is no heat storage (say, `MaxHeatStorage` is not defined) then this variable is not read, because it is not used;

4.10 SDDPBlock

The SDDPBlock is a class that derives from Block and represents a multistage stochastic programming problem of the form

$$\min_{x_0 \in \mathcal{X}^{n_0}} f_0(x_0) + \mathbb{E} \left[\min_{x_1 \in \mathcal{X}^{n_1}} f_1(x_1) + \mathbb{E} \left[\cdots + \mathbb{E} \left[\min_{x_{T-1} \in \mathcal{X}^{n_{T-1}}} f_{T-1}(x_{T-1}) \right] \right] \right],$$

where T is called the time horizon, $\mathcal{X}^{n_t} \equiv \mathcal{X}^{n_t}(x_{t-1}, \xi_t) \subseteq \mathbb{R}^{n_t}$ for each $t \in \{0, \dots, T-1\}$, and $\xi = \{\xi_t\}_{t \in \{1, \dots, T-1\}}$ is a stochastic process. Notice that x_{-1} and ξ_0 are deterministic. For each $t \in \{0, \dots, T-1\}$, we call

$$\min_{x_t \in \mathcal{X}^{n_t}} f_t(x_t) + \mathcal{V}_{t+1}(x_t, \xi_t)$$

the problem associated with stage t , where

$$\mathcal{V}_{t+1}(x_t, \xi_t) = \mathbb{E} [V_{t+1}(x_t, \xi_{t+1}) \mid \xi_t]$$

is the (expected value) cost-to-go function (also called value function, future value function, future cost function), with $\mathcal{V}_T \equiv 0$ and

$$V_t(x_{t-1}, \xi_t) = \min_{x_t \in \mathcal{X}^{n_t}} f_t(x_t) + \mathcal{V}_{t+1}(x_t, \xi_t)$$

with given x_{-1} and (deterministic) ξ_0 . We consider an approximation to the problem associated with stage $t \in \{0, \dots, T-1\}$ as the problem

$$\min_{x_t \in \mathcal{X}^{n_t}} f_t(x_t) + \mathcal{P}_{t+1}(x_t) \quad (2)$$

where $\mathcal{P}_{t+1}(x_t)$ is a polyhedral function, i.e., it is a function of the form

$$\mathcal{P}_{t+1}(x_t) = \max_{i \in \{1, \dots, k_t\}} \{d_{t,i}^\top x_t + e_{t,i}\}$$

with $d_{t,i} \in \mathbb{R}^{n_t}$ and $e_{t,i} \in \mathbb{R}$ for each $i \in \{1, \dots, k_t\}$.

An SDDPBlock is then characterized by the following:

1. It has a time horizon T .
2. It has T sub-Blocks, each one being a StochasticBlock. The t -th sub-Block represents an approximation to the problem associated with stage t as defined in (2).
3. It has pointers to $T-1$ PolyhedralFunction. The t -th PolyhedralFunction represents the function $\mathcal{P}_{t+1}$ in (2) and, therefore, must be defined in the t -th sub-Block of this SDDPBlock or in any of the sub-Blocks of that sub-Block, recursively.
4. It has a set of scenarios $\mathcal{S}$. Each scenario in $\mathcal{S}$ is represented by a vector of double and spans all the time horizon T . Each vector is divided into T parts, each one being associated with a stage of the multistage problem. Let S denote a vector representing a scenario in $\mathcal{S}$. Then, S is defined as

$$S = (S_0, \dots, S_{T-1})$$

where S_t is a sub-vector of S with size s_t , for each $t \in \{0, \dots, T-1\}$, and is associated with the sub-problem at stage t , i.e., it provides data for the t -th sub-Block of this SDDPBlock. We say that S_t represents the t -th sub-scenario of the scenario represented by S .

We assume that the sub-scenarios are organized in such a way that related random data appear in contiguous areas of the sub-scenario. For instance, suppose that the random data is associated with demand, inflow, and wind power. In this case, the data related to demand should be a contiguous sub-vector D_t of the sub-scenario associated with stage t , as well as that related to inflow (F_t) and wind power (W_t). In this example, the sub-scenario S_t could be organized as

$$S_t = (D_t, F_t, W_t).$$

We say that this sub-scenario has three groups of related random data. The order in which the groups of related random data appear in S_t is not relevant. We could have, for instance,

$$S_t = (W_t, D_t, F_t).$$

But the sub-scenario associated with stage t must respect the same order for each scenario in $\mathcal{S}$.

Besides the mandatory type attribute of any Block, an SMS++ NETCDF *group* for an SDDPBlock contains the following:

- The TimeHorizon *dimension*, containing the time horizon.
- The description of the sub-Blocks of the SDDPBlock. This is given by the sub-groups StochasticBlock and StochasticBlock.t, for each $t \in \{0, \dots, \text{TimeHorizon} - 1\}$. These sub-groups are optional, but they cannot be all absent. If StochasticBlock.t is not provided for some $t \in \{0, \dots, \text{TimeHorizon} - 1\}$, then the StochasticBlock *group* must be provided and contain a complete description of the t -th sub-Block of this SDDPBlock. If the StochasticBlock *group* is not provided, then StochasticBlock.t must be provided for each $t \in \{0, \dots, \text{TimeHorizon} - 1\}$ and contain a complete description of the t -th sub-Block of this SDDPBlock.

If StochasticBlock.t is provided but the description of its inner Block is not provided, then the StochasticBlock *group* must be provided and contain the description of an inner Block of a StochasticBlock. In this case, the description of the inner Block provided in the StochasticBlock *group* will be used to construct the inner Block of the StochasticBlock described by the StochasticBlock.t *group*.

If StochasticBlock.t is provided but the description of its vector of DataMapping is not provided, then if the StochasticBlock *group* is provided and contains a description of a vector of DataMapping, then it is used to construct the vector of DataMapping of the StochasticBlock described by the StochasticBlock.t *group*.

- The AbstractPath *group* containing the description of a vector of AbstractPath as described in §3.3.1. The number of AbstractPath must be equal to either 1 or TimeHorizon-1. If the number of AbstractPath is TimeHorizon-1 then the i -th AbstractPath in this vector must be the path to the PolyhedralFunction associated with the i -th sub-Block of this SDDPBlock. The i -th AbstractPath is taken with respect to the inner Block of the i -th sub-Block of this SDDPBlock. If the number of AbstractPath in this vector is 1, then all paths are assumed to be equal: for each $i \in \{0, \dots, \text{TimeHorizon} - 1\}$, the provided AbstractPath will be the path to the PolyhedralFunction associated with the i -th sub-Block of this SDDPBlock (taken with respect to the inner Block of this i -th sub-Block).
- The NumberScenarios *dimension* specifying the number of scenarios.
- The ScenarioSize *dimension* containing the size of a single scenario, which spans all stages.
- The SubScenarioSize *variable*, of type netCDF::NcUint64 and indexed over *dimension* TimeHorizon. This dimension is optional. If it is not provided, then all sub-scenarios are assumed to have the same size, i.e., $s_t = (\text{ScenarioSize} / \text{TimeHorizon})$ for all $t \in \{0, \dots, \text{TimeHorizon} - 1\}$, and ScenarioSize is a multiple of TimeHorizon. If this *dimension* is provided, then SubScenarioSize[t] is the size of the sub-scenario associated with stage t , i.e., $s_t = \text{SubScenarioSize}[t]$, for each $t \in \{0, \dots, \text{TimeHorizon} - 1\}$. In the latter case, the following must hold:

$$\text{ScenarioSize} = \sum_{t=0}^{\text{TimeHorizon}-1} \text{SubScenarioSize}[t].$$

- The two-dimensional *variable* Scenarios of type netCDF::NcDouble and indexed over the dimensions NumberScenarios and ScenarioSize, containing the scenarios. The i -th row of Scenarios contains the i -th scenario, so that Scenarios[i][j] is the j -th component of the i -th scenario.
- The NumberRandomDataGroups *dimension* containing the number of groups of related random data within each sub-scenario. This *dimension* is optional. If it is not provided, then we assume that there is a single group of related random data. Also, this *dimension* is meaningful only if all sub-scenarios have the same size.
- The SizeRandomDataGroups *variable*, of type netCDF::Uint64 and indexed over the NumberRandomDataGroups dimension, containing the size of each group of related random data in each sub-scenario. For each $i \in \{0, \dots, \text{NumberRandomDataGroups} - 1\}$, NumberRandomDataGroups[i] is the size of the i -th group of a sub-scenario. This *variable* is optional. It is required only if NumberRandomDataGroups is provided and NumberRandomDataGroups > 1.

- The *StateSize* *variable*, of type `netCDF::UInt64` and being either a scalar or a one-dimensional array indexed over *TimeHorizon* dimension, specifying the sizes of the states at each stage. If this *variable* is a scalar, then all states are assumed to have the size given by *StateSize*. If it is an array then, for each $t \in \{0, \dots, \text{TimeHorizon} - 1\}$, *StateSize*[*t*] contains the size of the state at stage *t*. The state being a vector, its size is the dimension of the space in which it lies.
- The *AdmissibleState* *variable*, of type `netCDF::NcDouble`, containing an admissible state for each stage. An admissible state for a stage is an state which makes the problem at that stage feasible. If *StateSize* is scalar and *AdmissibleState* has *dimension* *StateSize*, then all stages are assumed to have the same admissible state given by *AdmissibleState*. Otherwise, *AdmissibleState* contains the concatenation of the states for all stages as follows.

- If *StateSize* is scalar then, for each $t \in \{0, \dots, \text{TimeHorizon} - 1\}$, an admissible state for stage *t* is given by

$$(\text{AdmissibleState}[s_t], \dots, \text{AdmissibleState}[s_t + \text{StateSize} - 1]),$$

where $s_t = t \times \text{StateSize}$. In this case, *AdmissibleState* must have size $\text{TimeHorizon} \times \text{StateSize}$.

- If *StateSize* is a one-dimensional array then, for each $t \in \{0, \dots, \text{TimeHorizon} - 1\}$, an admissible state for stage *t* is given by

$$(\text{AdmissibleState}[s_t], \dots, \text{AdmissibleState}[s_t + \text{StateSize}[t] - 1]),$$

where $s_t = \sum_{i=0}^{t-1} \text{StateSize}[i]$. In this case, *AdmissibleState* must have size equal to

$$\sum_{i=0}^{\text{TimeHorizon}-1} \text{StateSize}[i].$$

4.11 StochasticBlock

Besides the mandatory `type` attribute of any `Block`, an SMS++ NETCDF *group* for a `StochasticBlock` contains the following:

- The *group* `Block`, containing the description of the inner `Block`. This *group* is optional. If it is not provided, then the inner `Block` must be provided by other means.
- The description of a vector of `SimpleDataMapping`, containing the `DataMappings` associated with this `StochasticBlock`. This is optional. For the time being, each provided `DataMapping` is expected to be a `SimpleDataMapping`. Moreover, the inner `Block` of this `StochasticBlock` will serve as the reference `Block` for both the serialization and deserialization of each `SimpleDataMapping`.

In order to better understand what the different elements, a comment about `StochasticBlock` is in order. The `StochasticBlock` class is meant to represent a `Block` whose data may be stochastic. The idea is that any `Block` could have its stochastic version without any changes in its implementation. The `StochasticBlock` comes to facilitate this process. A `StochasticBlock` is characterized by the following:

1. It has exactly one sub-`Block` (called here as the inner `Block`), which is the `Block` that is becoming stochastic. This can be any `:Block`.
2. It is aware that some data of the inner `Block` is stochastic (and may be subject to changes) and that the value for this data is represented by a vector of `double`. An instance of this vector is called a scenario for the stochastic data.
3. It has a set of `DataMapping`. This is used to both identify the stochastic data in the inner `Block` and modify the values of this data. The inner `Block` may have different types of stochastic data, located in different sub-`Blocks`. A `DataMapping` is meant to represent one piece of the stochastic data.

To understand how it could be used, we present the following simple example. Consider a `Block` B which would be the inner `Block` of a `StochasticBlock`. This `Block` B may have all sorts of data. We could have a stochastic version of B by selecting some of its data to become uncertainties. Let us suppose that B has a vector that represents some demand over time (from time 0 to $T - 1$). One could establish that the demand at times 0, 3, and 8 is stochastic. Suppose also that B has a sub-`Block` B_1 that has a vector representing costs to produce certain goods. One could also establish that the cost to produce good number 6 is stochastic. In this example, the stochastic data is formed by the demand of `Block` B at times 0, 3, and 8, and the cost to produce number 6 of `Block` B_1 . `DataMapping` can then be used to identify each of those uncertainties.

We could say that a vector of length 4 is a scenario for this stochastic data such that the first three elements of this vector are values for the demands and the fourth element is a value for the cost of producing good number 6. We could use a `DataMapping` to identify that demand as stochastic data. To this end, we need two sets. One set identifies what part of the scenario (indices of that vector) is associated with the demand and the second set identifies what part of the demand of B is stochastic. In this example, the first set would be $S_1 = \{0, 1, 2\}$, which states that the values for the demand are present at positions 0, 1, and 2 of the scenario vector, and the second set would be $S_2 = \{0, 3, 8\}$, which states that the demands that are stochastic are those at positions 0, 3, and 8. We also need a way of modifying this data in B . For this, it suffices a method in B that can be used to update its demand (see `SimpleDataMapping` in §3.3.2), let us say one called `set_demand(S2, vector)`.

Another `DataMapping` could be used to identify the cost of producing good number 6 as stochastic. This `DataMapping` would have $S_1 = \{4\}$ as the first set (stating that the value for this data is at position 4 of the scenario vector) and $S_2 = \{5\}$ as the second one (indicating which good cost is stochastic). Again, we need a way to update this cost in B_1 , which could be done, for instance, if B_1 provides a method like `set_cost(S2, vector)`.

4. It provides a method called `set_data()` which has a scenario (a vector) as parameter and that sets the data of the inner `Block` according to the set of `DataMapping` present in the `StochasticBlock`. In the example above, when the `set_data()` method of `StochasticBlock` is called, the values for the demands and the good cost are updated with the given values in the scenario.

In fact, the data of the inner `Block` does not need to be stochastic in any sense. What this class provides is a means to set the value of the data of its inner `Block`.

A `StochasticBlock` should have a probability distribution (or some kind of partial stochastic process) that describes the uncertainty in it. However, for the moment, it is not supported by this class and this feature will be implemented later on. Typically, an object of this class would be used in conjunction with a scenario generator and the `set_data()` method of this object would be called to consider a particular scenario.

4.12 BendersBlock

A BendersBlock is a Block whose Objective is an FRealObjective whose Function is a BendersBFunction. Moreover, it has a vector of ColVariable which are the active Variable of that BendersBFunction.

Besides the mandatory type attribute of any :Block, an SMS++ NETCDF *group* for a BendersBlock contains the following:

- The *dimension* NumVar containing the number of ColVariable of the BendersBlock.
- The *group* BendersBFunction containing the description of the BendersBFunction that is the Function of the FRealObjective of the BendersBlock.

4.13 BendersBFunction

The BendersBFunction is a convenience class implementing the “abstract” concept of a Benders function of “any” Block. BendersBFunction derives from **both** C05Function and Block.

The main ingredients of a BendersBFunction are the following:

1. A “base” Block B , representing an optimization problem of the form

$$\min\{c(y) : w \leq E(y) \leq z, y \in Y\}, \quad (\text{B})$$

where c , w , and z are vectors of appropriate sizes, E is a function of y , and Y is a convex set. Let E_i denote the i -th component of E . If E_i is nonlinear, then it must be either convex (in which case we must have $w = -\infty$) or concave (in which case we must have $z = +\infty$). This will be the one, and only, sub-Block of BendersBFunction (when “seen” as a Block).

2. A matrix A with m rows and n columns, a vector b with m rows, and a vector of pairs $[(C_i, S_i)]_{i \in I}$, each pair being formed by a pointer to a RowConstraint of Block B and a ConstraintSide, where $I = \{1, \dots, m\}$.

Problem (B) would typically be associated with an original problem

$$\min\{d(x) + c(y) : g \leq Fx + E(y) \leq h, x \in X, y \in Y\} \quad (\text{O})$$

defined in terms of variables x and y . By reformulating problem (O) as

$$\min\{d(x) + \phi(x) : x \in X\}, \quad (\text{O}')$$

where

$$\phi(x) = \min\{c(y) : (g - Fx) \leq E(y) \leq (h - Fx), y \in Y\}, \quad (\text{P})$$

we see that (P) assumes the form of (B) with $w = g - Fx$ and $z = h - Fx$. The BendersBFunction represents the function ϕ whose underlying optimization problem is given by (B). The variables x are the active Variables of this BendersBFunction. As can be seen in (P), the left- and right-hand sides of (some or all) the constraints may depend on x . The Block B , however, does not depend on x . In order to have the left- and right-hand sides of the constraints in (B) dependent on x , we consider the mappings $x \rightarrow g - Fx$ and $x \rightarrow h - Fx$, which are used to update the left- and right-hand sides w and z of the constraints in (B) according to the values of the variables x .

These mappings are bunched together into a single mapping M defined by the matrix A and the vector b such that

$$M(x) = Ax + b. \quad (\text{3})$$

The i -th component of this mapping, $M_i(x) = [Ax + b]_i$, is associated with the left- or right-hand side (or both) of the RowConstraint of Block B pointed by C_i . The ConstraintSide S_i indicates which sides of the RowConstraint (pointed by) C_i are affected, as follows:

- If $S_i = \text{'L'}$, then $M_i(x)$ gives the value of the left-hand side of the RowConstraint (pointed by) C_i .
- If $S_i = \text{'R'}$, then $M_i(x)$ gives the value of the right-hand side of the RowConstraint (pointed by) C_i .
- If $S_i = \text{'B'}$, then $M_i(x)$ gives the value of both the left- and right-hand side of the RowConstraint (pointed by) C_i . This is the case, for example, of an equality constraint, in which the left- and right-hand sides are equal.

Notice that the affected constraints must necessarily be RowConstraint. Moreover, if a RowConstraint in Block B has finite bounds that can be different from each other and are affected by the values of x , then it would be listed twice (once for each of its bounds). That is, there would be indices $i, j \in I$ such that $i \neq j$ and $C_i = C_j$, $S_i = \text{'L'}$, and $S_j = \text{'R'}$.

Important: The vector of pairs $[(C_i, S_i)]_{i \in I}$ must not contain duplicate entries (i.e., there must not exist indices $i, j \in I$ such that $i \neq j$, $C_i = C_j$ and $S_i = S_j$). Moreover, if there is $i \in I$ such that $S_i = \text{'B'}$, there must not exist $j \in I$ such that $C_i = C_j$.

We stress that the i -th active Variable of this BendersBFunction is associated with the i -th column of the mapping matrix A . In other words, the i -th active Variable serves as the coefficient of the i -th column of the A matrix in the mapping M .

Note that the `BendersBFunction` is not supposed to have any `Variable` or `Constraint` (besides those defined in its only inner `Block B`).

An SMS++ `NETCDF group` for a `BendersBFunction` contains the following:

- The *dimension* `NumVar` containing the number of columns of the A matrix, i.e., the number of active variables.
- The *dimension* `NumRow` containing the number of rows of the A matrix. This dimension is optional; if it is not provided then 0 (no rows) is assumed.
- The *dimension* `NumNonzero`, of type `netCDF::NcUInt64`, containing the number of nonzero entries in the A matrix. This dimension is optional and determines in which format the matrix A is given. If this dimension is present, then `NumRow` must also be. If `NumNonzero` is not present, then the A matrix is given as a dense matrix. If it is present, then the A matrix is given in a sparse format as defined by the variables `Row`, `Column`, and `A`. During serialization, the following criterion is used to decide the format in which the A matrix is stored. If at most 25% of its elements are nonzero, it is stored in sparse format; otherwise, it is stored in dense format.
- The *variable* `NumNonzeroAtRow`, of type `netCDF::NcUInt64` and indexed over the dimension `NumRow`, containing the number of nonzero elements of the A matrix in each row. If the dimension `NumNonzero` is not present, then this variable is ignored. If `NumNonzero` is present, then `NumNonzeroAtRow` is optional only if `NumVar`, `NumRow`, and `NumNonzero` are equal. In this case, if `NumNonzeroAtRow` is not provided, then the A matrix is assumed to be the identity matrix. If `NumVar`, `NumRow`, and `NumNonzero` are not all equal, then `NumNonzeroAtRow` must be provided. For each $i \in \{0, \dots, \text{NumRow} - 1\}$, `NumNonzeroAtRow[i]` is the number of nonzero elements in the i -th row of the A matrix.
- The *variable* `Column`, of type `netCDF::NcUInt64` and indexed over the dimension `NumNonzero`, containing the column indices of the entries of the matrix. If `NumNonzero` or `NumNonzeroAtRow` is not present, then this variable is ignored. For each $k \in \{0, \dots, \text{NumNonzero} - 1\}$, `Column[k]` is the column index of the k -th nonzero entry of the A matrix, whose value is given by `A[k]`. `Column` stores the column indices in row-major order, that is, if (i, j) and (p, q) are the entries of the k -th and l -th nonzero elements of A , respectively, with $k < l$, then $i \leq p$.
- The *variable* `A`, of type `netCDF::NcDouble`. This variable stores the values of the elements of the A matrix of the mapping. If the dimension `NumNonzero` is provided but `NumNonzeroAtRow` is not, then the A matrix is assumed to be the identity matrix and this *variable* is ignored. If both the dimensions `NumNonzero` and `NumNonzeroAtRow` are provided, then this variable is indexed over this dimension and contains the values of the (potentially) nonzero entries of the matrix. In this case, `A[k]` is the value of the k -th nonzero entry, whose column index is given by `Column[k]`. The nonzero elements of the matrix are given in left-to-right top-to-bottom (“row-major”) order. If the dimension `NumNonzero` is not present, then `A` is indexed over both the `NumRow` and `NumVar` dimensions (in this order); it contains the (row-major) representation of the matrix A . This variable is optional only if `NumRow == 0`.
- The *variable* `b`, of type `netCDF::NcDouble` and indexed over the dimension `NumRow`, which contains the vector b . This variable is optional. If it is not provided, then we assume `b[i] = 0` for each $i \in \{0, \dots, \text{NumRow} - 1\}$.
- The *variable* `ConstraintSide`, of type `netCDF::NcChar` and indexed over the dimension `NumRow`, indicating, at position i , which side of the i -th `Constraint` is affected. The possible values are ‘L’ for the left-hand (or lower bound) side, ‘R’ for the right-hand (or upper bound) side, and ‘B’ for both sides. This variable is optional. If it is not provided, then all entries of this array are assumed to be ‘B’.
- The *group* `AbstractPath` containing the description of a vector of `AbstractPath` to the affected `RowConstraints`. The number of `AbstractPath` in that vector must be equal to the number of rows of the A matrix and, therefore, the dimension associated with the number of `AbstractPath` is `NumRow`. The i -th `AbstractPath` in this vector must be the path to the i -th affected `RowConstraint` (which is associated with the i -th row of the A matrix). This *group* is optional only if `NumRow == 0`.
- The *group* `Block`, containing the description of the inner `Block`.
- The *group* `BlockConfig`, containing the `BlockConfig` of the inner `Block`. This *group* is optional.
- The *group* `BlockSolver`, containing the `BlockSolverConfig` of the inner `Block`.

5 BlockConfig for existing :Block

As described in §3.2.1, each Block can have several Configuration parameters, which can be dealt with by a BlockConfig object. It makes therefore sense to collect here all possible Configuration parameters for existing :Block, since these may end up in a SMS++ NETCDF file.

5.1 Parameters of MCFBlock

The MCFBlock class defines the following Configuration parameters:

- `f_static_constraints.Configuration`. The static constraint of a MCF are the flow conservation equations and the bound constraints. The latter are typically implemented via a `std::vector<LB0Constraint>` with exactly m entries. However, If *all* the upper bound are $+\infty$, it is possible to skip the `std::vector<LB0Constraint>` entirely and just use the fact that the ColVariable can be defined to be non-negative. If `f_static_constraints.Configuration` is set, it is a `SimpleConfiguration<int>` whose `f_value` is not 0, the bound constraints are not implemented if it is possible to do so; otherwise they are always implemented. Note that this makes it impossible to change any upper bound.
- `f_is_feasible.Configuration`. If `f_is_feasible.Configuration` is a `SimpleConfiguration<double>`, its `f_value` is taken as the parameter for deciding what “approximately feasible” exactly means while checking satisfaction of both flow conservation constraint and flow upper/lower bounds; the value is taken as the *relative* tolerance for those checks.
- `f_is_optimal.Configuration`. Checking optimality of a primal-dual solution of the MCF problems means checking that both are approximately feasible, and that they approximately satisfy the Complementary Slackness Conditions. This requires two parameters for deciding what “approximately feasible” means, one for the primal (ϵ_f) and one for the dual (ϵ_c). If `f_is_optimal.Configuration` is a `SimpleConfiguration<std::pair<double, double>>`, then $\epsilon_c = \text{f_value.first}$ and $\epsilon_f = \text{f_value.second}$.
- `f_solution.Configuration`. For all the methods dealing with solutions, i.e., `get_Solution()` and `map_[back/forward]_solution()`, if `f_solution.Configuration` is a `SimpleConfiguration<int>`, then its `f_value` decides what is mapped, with the following encoding: 1 means “only map the primal solution”, 2 means “only map the dual solution”, everything else (e.g., 0) means “map everything”.

6 ComputeConfig for existing :Solver

As described in §3.2.3, each Block can have several ComputeConfig parameters, which can be dealt with by a BlockSolverConfig object. It makes therefore sense to collect here all possible ComputeConfig parameters for the existing objects that define actual values for their algorithmic parameters, since these may end up in a SMS++ NETCDF file.

6.1 Standard parameters in Solver

The base Solver class defines a set of “general” algorithmic parameters, described in the following table.

name	type	description
intMaxIter	int	The algorithmic parameter for setting the maximum number of iterations that the next call to solve() is allowed to execute for trying to solve the Block. The concept of “what exactly an iteration is” is clearly Solver-dependent, and the user of the Solver need supposedly be aware of which concrete :Solver it is actually using to be able to sensibly set this parameter; however, because most Solver will actually be iterative processes, it makes sense to offer support for this notion in the base class.
intMaxSol	int	The algorithmic parameter for setting the maximum number of different solutions to the Block that the Solver should attempt to obtain and store. Mathematical models can have (very) many solutions: an objective function precisely helps in selecting among them, but even that may not be enough to narrow the choice down to a single solution (multiple optima may exist, or quasi-optimal solutions may also be sought for for any number of reasons). On the other hand, a solution can be a “big” object, and storing it may be costly. It may be therefore helpful for a Solver to know beforehand how many different solutions the user would like to get. Among the reasonable values for this parameter, 1 says “I don’t care of multiple solutions, give me only the best one”, and 0 says “I don’t care of solutions at all, just tell me if there is any, and what its value is”. The default is 1.
intLogVerb	int	An integer parameter dictating how “verbose” the log of the Solver has to be. The specific meaning of each value is Solver-dependent, but it is intended that 0 means “no log at all”, and increasing values correspond to increasing verbosity. The default value is 0 (no log).
dblMaxTime	dbl	The algorithmic parameter for setting the maximum time limit that the next call to solve() can expend in trying to solve the Block. The value is assumed to be in seconds, and it’s a double (so both very quick and very slow solvers are supported). The base Solver class does not explicitly distinguish between “wall-clock time” and “CPU time”, which may be rather different especially in a parallel environment, but this concept can be easily added by derived classes.
dblRelAcc	dbl	The algorithmic parameter for setting the <i>relative</i> accuracy required to the solution of the Block. This is geared towards single-objective optimization problems, and it is defined as follows: a solution for a <i>minimization</i> problem has a relative accuracy of ε if a feasible (to within the required tolerance for each Block) solution has been obtained, an upper bound ub on its objective function value has been found, a lower bound lb on the optimal value of the problem has been found, and $ub - lb \leq \varepsilon \max(lb , 1)$. The roles of ub and lb are suitably reversed for a maximization problem. The default is $1e-6$.
dblAbsAcc	dbl	The algorithmic parameter for setting the <i>absolute</i> accuracy required to the solution of the model. This is geared towards single-objective optimization problems, and it is defined as follows: a solution for a <i>minimization</i> problem has a absolute accuracy of ε if a feasible (to within the required tolerance for each Block) solution has been obtained, an upper bound ub on its objective function value has been found, a lower bound lb on the optimal value of the problem has been found, and $ub - lb \leq \varepsilon$. The roles of ub and lb are suitably reversed for a maximization problem. The default is $+\infty$, which is intended to mean that the only working accuracy is the relative one.

dblUpCutOff	dbl	The algorithmic parameter for setting the <i>upper cut off</i> of the solution process. This is a value basically telling the Solver “when enough is enough”. In particular, if the Solver obtains a certified lower bound lb on the optimal value such that $lb \geq \varepsilon$ with the provided parameter ε , then it can stop. This is a certificate that the optimal value is at least ε . For a maximization problem the condition says: I actually needed a solution with objective function value at least as good as (i.e., larger than) ε , now that I have found one the problem is as good as solved to me. Hence this parameter is analogous to the maximum absolute accuracy dblAbsAcc, but “weaker” in that it does not require any upper bound to work. For a minimization problem, instead, the condition says: I actually needed a solution with objective function value at least as good as (i.e., smaller than) ε , now that I have know for sure that this is never going to happen the problem is as good as unfeasible to me. The default is $+\infty$.
dblLwCutOff	dbl	The algorithmic parameter for setting the <i>lower cut off</i> of the solution process. This is a value basically telling the Solver “when enough is enough”. In particular, if the Solver obtains a certified upper bound ub on the optimal value such that $ub \leq \varepsilon$ with the provided parameter ε , then it can stop. This is a certificate that the optimal value is at most ε . For a minimization problem the condition says: I actually needed a solution with objective function value at least as good as (i.e., smaller than) ε , now that I have found one the problem is as good as solved to me. Hence this parameter is analogous to the maximum absolute accuracy dblAbsAcc, but “weaker” in that it does not require any lower bound to work. For a maximization problem, instead, the condition says: I actually needed a solution with objective function value at least as good as (i.e., larger than) ε , now that I have know for sure that this is never going to happen the problem is as good as unfeasible to me. The default is $-\infty$.
dblRAccSol	dbl	The algorithmic parameter for setting the relative accuracy of the accepted solutions. It instructs the Solver not to even consider a solution among the ones to be reported intMaxSol if its objective function value is “too” bad. For a minimization problem, the objective function value of a feasible solution provides an upper bound ub on the optimal value. Assuming a lower bound lb on the optimal value of the problem has been found, a solution is deemed acceptable with the provided parameter ε if $ub - lb \leq \varepsilon \max(ub , lb , 1)$. Note that if no lb is available, the above formula can be replaced with $ub - f_{best} \leq \varepsilon \max(ub , f_{best} , 1)$, where f_{best} is the value of the best (with smallest objective value) solution found so far. The roles of ub and lb are suitably reversed for a maximization problem. The default is $+\infty$.
dblAAccSol	dbl	Similar to dblRAccSol but for an <i>absolute</i> accuracy; that is, a solution is deemed acceptable with the provided parameter ε if $ub - lb \leq \varepsilon$ or $ub - f_{best} \leq \varepsilon$ with the same notation as in dblRAccSol and the same provisions about the case of a maximization problem. The default is $+\infty$.
dblFAccSol	dbl	The algorithmic parameter for setting the maximum relative allowed violation of constraints. Whenever the Solver is incapable of finding feasible solutions (maybe because there is none), it may still be useful that it returns the “least unfeasible” ones. This parameter instructs the Solver not to even consider a solution among the ones to be reported (see intMaxSol) if its violation is “too” bad. The actual meaning of this parameter may be Solver-dependent, but it can be thought to work as the “relative constraint violation”. A setting of 0 may be taken as a way to tell the Solver not to bother to produce unfeasible solutions at all, which is why this is the default value of the parameter.

6.2 Standard parameters in CDASolver

The CDASolver class extends the parameters of Solver with the ones described in the following table.

name	type	description
------	------	-------------

intMaxDSol	int	The algorithmic parameter for setting the maximum number of different <i>dual</i> solutions that the Solver should attempt to obtain and store. Since dual solutions can be a “big” objects (in some cases, much bigger than primal ones), storing them may be costly. It may be therefore helpful for a CDASolver to know beforehand how many different dual solutions the user would like to get. Among the reasonable values for this parameter, 1 says “I don’t care of multiple dual solutions, give me only the best one”, and 0 says “I don’t care of dual solutions at all, just tell me if there is any, and what its value is”. The default is 1.
dblRAccDSol	dbl	The algorithmic parameter for setting the relative accuracy of the accepted <i>dual</i> solutions. It instructs the CDASolver not to even consider a solution among the ones to be reported (see intMaxDSol) if its objective function value is “too” bad. If the original problem is a minimization one, its dual is a maximization one. Hence, the objective function value of a dual solution provides a lower bound lb on the optimal dual value (which may, or may not, be equal to that of the primal problem). Assuming an upper bound ub on the optimal value of the <i>dual</i> problem has been found (note that this can be obtained by means of the objective value of a feasible <i>primal</i> solution), a solution is deemed acceptable with the provided parameter ε if $ub - lb \leq \varepsilon \max(ub , lb , 1)$. Note that if no ub is available, the above formula can be replaced with $f_{best} - lb \leq \varepsilon \max(f_{best} , lb , 1)$ where f_{best} is the value of the best (with largest objective value) solution found so far. The roles of ub and lb are suitably reversed if the primal is a maximization problem, so that the dual is a minimization one. The default is $+\infty$.
dblAAccDSol	dbl	Similar to dblRAccDSol but for an <i>absolute</i> accuracy; that is, a dual solution is deemed acceptable with the provided parameter ε if $ub - lb \leq \varepsilon$ or $f_{best} - lb \leq \varepsilon$ with the same notation as in dblRAccDSol and the same provisions about the case of a maximization problem. The default is $+\infty$.
dblFAccDSol	dbl	The algorithmic parameter for setting the maximum relative allowed violation of <i>dual</i> constraints, assuming of course something like that exists in the specific dual that is being dealt with. Whenever the CDASolver is incapable of finding feasible dual solutions (maybe because there is none), it may still be useful that it returns the “least unfeasible” ones. This parameter instructs the CDASolver not to even consider a solution among the ones to be reported (see intMaxDSol) if its violation is “too” bad. The actual meaning of this parameter is necessarily be CDASolver-dependent; intuitively, it may be thought to work as the “relative constraint violation” <code>feas_epsilon</code> of Block if the concept of “dual constraint” is applicable. A setting of 0 may be taken as a way to tell the CDASolver not to bother to produce unfeasible solutions at all, which is why this is the default value of the parameter.

6.3 Standard parameters in Function

The base Function class defines a set of “general” algorithmic parameters, described in the following table.

name	type	description
intMaxIter	int	The algorithmic parameter for setting the maximum number of iterations in the next call to <code>compute()</code> . The concept of “what exactly an iteration is” is clearly dependent on the Function, and it may not even make sense for all Function. However, some Function will actually be iterative processes, and therefore it makes sense to offer support for this notion in the base class. The default is $+\infty = \text{no limits}$.
dblMaxTime	dbl	The parameter for setting the maximum time limit for the next call to <code>compute()</code> . The value is assumed to be in seconds, and it’s a double (so both very fast and very slow computations are supported). The Function class (so far) does not explicitly distinguish between “wall-clock time” and “CPU time”, which may be rather different especially in a parallel environment. The default is $+\infty$.
dblRelAcc	dbl	The parameter for setting the <i>relative</i> accuracy required to the function value. That is, if both an upper bound ub and a lower bound lb on the value have been found, then <code>compute()</code> can stop as soon as $ub - lb \leq \text{dblRelAcc} \max(lb , 1)$. The default is $1e-6$.

dblAbsAcc	dbl	The parameter for setting the <i>absolute</i> accuracy required to the function value. That is, if both an upper bound ub and a lower bound lb on the value have been found, then <code>compute()</code> can stop as soon as $ub - lb \leq \text{dblRelAcc}$. The default is $+\infty$, which is intended to mean that the only working accuracy is the relative one.
dblUpCutOff	dbl	The parameter for setting the “upper cut off” of the computation; that is, if a lower bound lb on the value has been found, then <code>compute()</code> can stop as soon as $lb \geq \text{dblUpCutOff}$. This is a certificate that the value is at least <code>dblUpCutOff</code> . The default is $+\infty$, i.e., no upper cut off.
dblLwCutOff	dbl	The parameter for setting the “lower cut off” of the computation; that is, if an upper bound ub on the value has been found, then <code>compute()</code> can stop as soon as $ub \leq \text{dblLwCutOff}$. This is a certificate that the value is at most <code>dblLwCutOff</code> . The default is $-\infty$, i.e., no lower cut off.

6.4 Standard parameters in C05Function

The C05Function class extends the parameters of Function with the ones described in the following table.

name	type	description
intLPMaxSz	int	The algorithmic parameter for setting the size of the “local pool”, that is, the maximum number of linearizations that should be stored in the local pool. The default is 1, which corresponds to the fact that the Function can only produce a single linearization at a time (for it is, say, smooth).
intGPMaxSz	int	The algorithmic parameter for setting the size of the “global pool”, that is, the maximum number of linearizations that should be stored in the local pool. The default is 0, which corresponds to the fact that the Function cannot store any linearization (for it is, say, smooth and therefore there is no need to).
dblRAccLin	dbl	A linearization (g, α) computed at the point x is “accurate” if the value of the linearization coincides with the value of the function at x , i.e., $\alpha = f(x)$. In general linearizations that are not “completely accurate” can still be useful: for instance, in the Lagrangian case an ε -optimal solution to the Lagrangian problem gives rise to a valid linearization (g, α) with $\varepsilon \geq f(x) - \alpha$. This can be deemed interesting if ε is “small”, but not if ε is “large”. This parameter instructs the C05Function not to bother reporting (and therefore storing in the “local pool”) any linearization having a relative error with $f(x)$ larger than <code>dblRAccLin</code> . This would generally mean $ f(x) - \alpha \leq \text{dblRAccLin} \max(f(x) , \alpha , 1)$, except that the value $f(x)$ may not be known exactly, with only lower and/or upper bounds on it available. The actual formula therefore depends on what information is actually available: for instance, in the Lagrangian case one knows that $f(x) \geq \alpha$, and therefore typically an upper estimate $ub \geq f(x)$ is used in the formula instead of $f(x)$. The default is 0, i.e., “only perfect linearizations are allowed”.
dblAAccLin	dbl	Similar to <code>dblRAccLin</code> but for an <i>absolute</i> accuracy; that is, a linearization is deemed acceptable if $ f(x) - \alpha \leq \text{dblAAccLin}$, except that the value $f(x)$ may not be known exactly, with only lower and/or upper bounds on it available. The actual formula therefore depends on what information is actually available: for instance, in the Lagrangian case one knows that $f(x) \geq \alpha$, and therefore typically an upper estimate $ub \geq f(x)$ is used in the formula instead of $f(x)$. The default is 0, i.e., “only perfect linearizations are allowed”.

6.5 Parameters of MCFSolver<MCFC>

The MCFSolver<> class derives from CDASolver, and therefore it extends the parameters of with the ones described in the following table.

name	type	description
kReopt	int	Whether the MCF algorithm (whatever it is) should re-optimize after changes in the data, as opposed to restarting from scratch.

However, `MCFsolver<>` is a template class, whose template parameter actually is a concrete MCF solver object derived from `MCFClass`. Each of these has its own algorithmic parameters, which are described in the next subsections.

6.5.1 Parameters of `MCFsolver<MCFsimplex>`

name	type	description
<code>kAlgPrimal</code>	<code>int</code>	If > 0 it instructs the <code>MCFsimplex</code> solver to use the primal (network) simplex method, otherwise the dual one.
<code>kAlgPricing</code>	<code>int</code>	Selects which pricing rule is used within the algorithm. Possible values are <code>kDantzig = 0</code> (Dantzig's rule, most violated constraint), <code>kFirstEligibleArc = 1</code> (first eligible arc in round-robin), and <code>kCandidateListPivot = 2</code> (Candidate List Pivot Rule).
<code>kNumCandList</code>	<code>int</code>	Parameter to set the number of candidate list for the Candidate List Pivot method.
<code>kHotListSize</code>	<code>int</code>	Parameter to set the size of Hot List for the Candidate List Pivot method.

7 ComputeConfig for existing :Function

7.1 Parameters of BendersBFunction

The BendersBFunction class extends the parameters of C05Function with the ones described in the following table. Setting any of these parameters causes the corresponding parameter of the CDASolver of the inner Block of the BendersBFunction to be overwritten (if the BendersBFunction has an inner Block and this Block has a CDASolver attached to it).

name	type	description
intMaxIter	int	This parameter is associated with the Solver::intMaxIter.
intLPMaxSz	int	This parameter is associated with the CDASolver::intMaxDSol parameter.
intGPMaxSz	int	The algorithmic parameter for setting the size of the “global pool”, that is, the maximum number of linearizations that should be stored in the global pool.
dblMaxTime	double	This parameter is associated with the Solver::dblMaxTime.
dblRelAcc	double	This parameter is associated with the Solver::dblRelAcc.
dblAbsAcc	double	This parameter is associated with the Solver::dblAbsAcc.
dblUpCutOff	double	This parameter is associated with the Solver::dblUpCutOff.
dblLwCutOff	double	This parameter is associated with the Solver::dblLwCutOff.
dblRAccLin	double	This parameter is associated with the CDASolver::dblRAccDSol.
dblRAccLin	double	This parameter is associated with the CDASolver::dblAAccDSol.

8 Examples

8.1 Seasonal Storage Valuation

The multistage stochastic problem in the Seasonal Storage Valuation (SSV) is represented by an `SDDPBlock` (see §4.10). The `SDDPBlock` has T nested `Blocks`, where T is the time horizon and, therefore, we have `TimeHorizon` = T in the specification of `SDDPBlock`. For each $t \in \{0, \dots, T-1\}$, the t -th nested `Block` of the `SDDPBlock` is a `StochasticBlock` (see §4.11) which is associated with stage t . The `TimeHorizon` of an `SDDPBlock` could be, for instance, 52, the number of weeks of a year. Then, each sub-`Block` of this `SDDPBlock` could be associated with a particular week of the year.

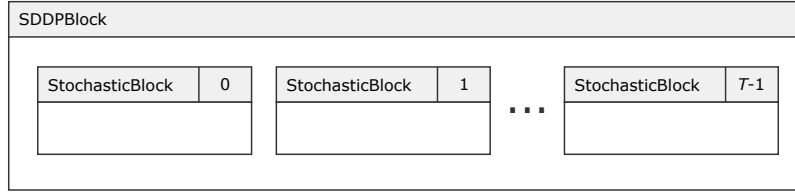


Figure 1 An `SDDPBlock` and its nested `Blocks`.

The set of scenarios in `SDDPBlock` is given by a matrix (see the *variable* `Scenarios` in §4.10), in which each row represents a scenario that spans the time horizon. If there are m scenarios (see *dimension* `NumberScenarios` in §4.10) and each scenario is a (row) vector in $\mathbb{R}^n$ (in which case n would be the size of a scenario; see *dimension* `ScenarioSize` in §4.10), a matrix $\mathcal{S} \in \mathbb{R}^{m \times n}$ with scenarios would have the form

$$\mathcal{S} = \begin{bmatrix} s_{0,0} & s_{0,1} & \dots & s_{0,n-1} \\ s_{1,0} & s_{1,1} & \dots & s_{1,n-1} \\ \vdots & \vdots & \vdots & \vdots \\ s_{m-1,0} & s_{m-1,1} & \dots & s_{m-1,n-1} \end{bmatrix}.$$

For each $i \in \{0, \dots, m-1\}$, the i -th scenario, represented by the vector

$$S_i = (s_{i,0}, s_{i,1}, \dots, s_{i,n-1}),$$

is assumed to be organized in such a way that

$$S_i = (S_{i,0}, S_{i,1}, \dots, S_{i,T-1}),$$

where $S_{i,t}$, the t -th sub-scenario of S_i , contains the data for the sub-problem at stage t (which is represented by the t -th `StochasticBlock` of the `SDDPBlock`). The sub-scenarios for different stages may have different sizes. For each $i \in \{0, \dots, m-1\}$ and $t \in \{0, \dots, T-1\}$, $S_{i,t} \in \mathbb{R}^{n_t}$, so that

$$n = \sum_{t=0}^{T-1} n_t.$$

In the specification of `SDDPBlock`, the vector $(n_0, \dots, n_{T-1})$ containing the sizes of the sub-scenarios is given by the *variable* `SubScenarioSize`. This *variable* is optional and, in the case it is not provided, all sub-scenarios are assumed to have the same size, i.e., $n_t = n/T$ for each $t \in \{0, \dots, T-1\}$, and n is a multiple of T .

If all sub-scenarios have the same size, their data must be organized so that related random data are in contiguous parts of the vector representing a sub-scenario. For instance, suppose that each sub-scenario contains data related to demand, inflow, and wind power. In this example, we would say that there are three groups of data (the *dimension* `NumberRandomDataGroups` in `SDDPBlock` would then be 3) and, for each $i \in \{0, \dots, m-1\}$ and $t \in \{0, \dots, T-1\}$, the sub-scenario $S_{i,t}$ would be written as

$$S_{i,t} = (D_{i,t}, F_{i,t}, W_{i,t}).$$

The *variable* `SizeRandomDataGroups` of `SDDPBlock` would then be an array containing the sizes of the sub-vectors $D_{i,t}$, $F_{i,t}$, and $W_{i,t}$. For instance, if $D_{i,t}$ is a vector with size 7, $F_{i,t}$ is one with size 4, and $W_{i,t}$ is one with size 11,

the *variable* SizeRandomDataGroups would contain the array (7, 4, 11). Hence, the matrix $\mathcal{S}$ would then have the following form:

$$\begin{array}{c}
 \begin{array}{ccccccc}
 \text{Stage 0} & & \dots & & \text{Stage } t & & \dots & & \text{Stage } T-1 \\
 \hline
 D_{0,0} & F_{0,0} & W_{0,0} & \cdots & D_{0,t} & F_{0,t} & W_{0,t} & \cdots & D_{0,T-1} & F_{0,T-1} & W_{0,T-1} \\
 \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \vdots \\
 D_{i,0} & F_{i,0} & W_{i,0} & \cdots & D_{i,t} & F_{i,t} & W_{i,t} & \cdots & D_{i,T-1} & F_{i,T-1} & W_{i,T-1} \\
 \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \vdots \\
 D_{m-1,0} & F_{m-1,0} & W_{m-1,0} & \cdots & D_{m-1,t} & F_{m-1,t} & W_{m-1,t} & \cdots & D_{m-1,T-1} & F_{m-1,T-1} & W_{m-1,T-1}
 \end{array}
 \end{array}
 \begin{array}{l}
 \text{Scenario 0} \\
 \vdots \\
 \text{Scenario } i \\
 \vdots \\
 \text{Scenario } m-1
 \end{array}$$

In this example, if $T = 52$, the *dimension* ScenarioSize would be $52(7 + 4 + 11)$. The light gray box indicate a scenario while a dark gray box indicates a sub-scenario.

Each sub-Block of the SDDPBlock represents a short-term problem that is associated with a UCBlock as follows. For each $t \in \{0, \dots, T-1\}$, the inner Block of the t -th StochasticBlock of the SDDPBlock is a BendersBlock. The BendersBlock has a set of Variables and an Objective. Its Objective is an FRealObjective whose Function is a BendersBFunction. The Variables of the BendersBlock are the ones that are active in the BendersBFunction. The inner Block of the BendersBFunction is a UCBlock (or a Lagrangian relaxation of a UCBlock) which depends on the values of the Variables that are active in the BendersBFunction. In the SSV, the Variables defined in the BendersBlock (which are active in the BendersBFunction) represent the initial volumes of the reservoirs. These affect the values of the left- or right-hand side of some Constraints of the UCBlock. The affected Constraints and how they are affected are determined by the BendersBFunction. In the BendersBFunction, the affected Constraints are indicated by a vector of AbstractPath (each path being relative to the BendersBlock) and the affected sides of the Constraints are indicated by the one-dimensional *variable* ConstraintSide. The way the Constraints are affected by those Variables is defined by the linear mapping given by the NETCDF *variables* A and b (together with their auxiliary *variables* and *dimensions*). Figure 2 summarizes the structure of an StochasticBlock in which x_t represent the Variables of the BendersBlock.

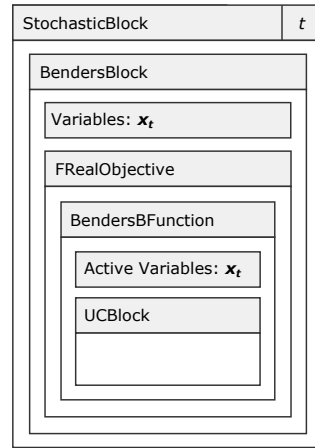


Figure 2 Structure of an StochasticBlock in the SSV.

8.1.1 BendersBlock

Before giving an example on how to construct the BendersBlock, we shall look into some details of the UCBlock. Following the example above, in which the SDDPBlock represents a multistage problem over 52 weeks, there would be one UCBlock associated with each of those 52 weeks. Each UCBlock could be, for instance, a deterministic multistage problem with 168 stages (one stage for each hour of the week). In this case, the dimension TimeHorizon in the specification of the UCBlock would be 168. As you can see, there are two independent time horizons: one at the level

of the SDDPBlock (the mid-term problem) and one at the level of the UCBlock (the short-term problem). To avoid confusion, we will refer to the time horizon and stages of the SDDPBlock as the outer time horizon and outer stages, respectively, and the time horizon and stages of the UCBlock as the inner time horizon and inner stages, respectively. We will also use the notations T^{out} and T^{in} to refer to the outer and inner time horizons when necessary.

A UCBlock has two sets of sub-Blocks. The first one is a set of UnitBlocks whose size is given by the `NumberUnits` *dimension*. The second one is a set of NetworkBlocks whose size is (inner) `TimeHorizon`. Figure 3 illustrates the UCBlock and its nested Blocks. In this figure, N is the number of UnitBlocks (*dimension* `NumberUnits`) and T^{in} is the inner time horizon (*dimension* `TimeHorizon` in the specification of UCBlock).

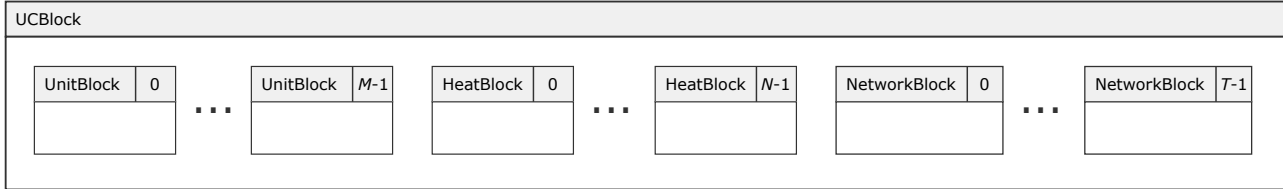


Figure 3 UCBlock and its nested Blocks.

There can be many types of UnitBlocks in the UCBlock (e.g., `ThermalUnitBlock`, `BatteryUnitBlock`, ...). In the SSV, exactly one of those UnitBlocks is a `HydroSystemUnitBlock`. The UnitBlocks of the UCBlock can be given in any order the user may wish. But to simplify the example, we assume that the `HydroSystemUnitBlock` is the first of the UnitBlocks (in Figure 3, it would be the UnitBlock with index 0). Hence, the description of the `HydroSystemUnitBlock` should be given in the *group* `UnitBlock_0` in the specification of the UCBlock.

The `HydroSystemUnitBlock` (see §4.6) is a UnitBlock that has a list of `HydroUnitBlocks` (see §4.5.2) and one `PolyhedralFunctionBlock` as nested Blocks. The `PolyhedralFunctionBlock` is the last among the nested Blocks of the `HydroSystemUnitBlock`. Figure 4 depicts the `HydroSystemUnitBlock` with its nested Blocks in which H denotes the number of `HydroUnitBlocks` (*dimension* `NumberHydroUnits` in the specification of the `HydroSystemUnitBlock`).

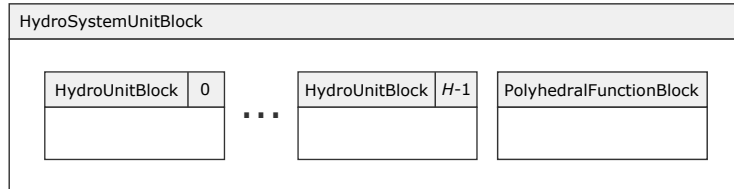


Figure 4 `HydroSystemUnitBlock` and its nested Blocks.

The Variables of the `BendersBlock` represent the dependence between consecutive UnitBlocks. In the SSV, this dependence is related to the volumes of the reservoirs: the initial volume of a reservoir of a `HydroUnitBlock` at the outer stage $t \in \{1, \dots, T^{\text{out}} - 1\}$ must be equal to the volume of that reservoir at the last inner stage of the `HydroUnitBlock` at the outer stage $t - 1$. The Variables of the `BendersBlock` at the outer stage $t \in \{0, \dots, T^{\text{out}} - 1\}$ thus represent the initial volumes of the reservoirs of the `HydroUnitBlocks` at that stage.

Let r_i denote the number of reservoirs (*dimension* `NumberReservoirs` in the specification of `HydroUnitBlock`) of the i -th `HydroUnitBlock`. For each $i \in \{0, \dots, H - 1\}$ and $j \in \mathcal{N}^i = \{0, \dots, r_i - 1\}$, let $\tilde{v}_{i,j}$ denote the initial volume of the j -th reservoir of the i -th `HydroUnitBlock`. These initial volumes appear in the “final volumes constraints” of the `HydroUnitBlock`. For the inner stage $t = 0$ and the i -th `HydroUnitBlock`, these constraints are

$$v_{j,t}^i = \tilde{v}_{i,j} + F_{j,t}^i + \left(\sum_{l=(d,j) \in \mathcal{L}^i} f_{t-\tau_l^{i,\text{dn}},l}^i - \sum_{l=(j,d) \in \mathcal{L}^i} f_{t-\tau_l^{i,\text{up}},l}^i \right), \quad \forall j \in \mathcal{N}^i \text{ and } t = 0. \quad (4)$$

In these constraints, $v_{j,t}^i$ denotes the volume of the j -th reservoir at the (inner) stage t , $F_{j,t}^i$ denotes the inflow at the j -th reservoir at the (inner) stage t , $f_{k,l}^i$ is the flow rate at time k for arc l , and $\tau_l^{i,\text{dn}}$ and $\tau_l^{i,\text{up}}$ denote the uphill and downhill flow rates at arc l , respectively. (See the documentation of the `HydroUnitBlock` class for details.)

Therefore, we could define the Variables x of the BendersBlock as that representing the following vector:

$$x = \left[\underbrace{\tilde{v}_{0,0} \quad \dots \quad \tilde{v}_{0,r_0-1}}_{\text{HydroUnitBlock } 0} \quad \dots \quad \underbrace{\tilde{v}_{k,0} \quad \dots \quad \tilde{v}_{k,r_k-1}}_{\text{HydroUnitBlock } k} \quad \dots \quad \underbrace{\tilde{v}_{H-1,0} \quad \dots \quad \tilde{v}_{H-1,r_{H-1}-1}}_{\text{HydroUnitBlock } H-1} \right]^\top. \quad (5)$$

As aforementioned, these are the active Variables of the BendersBFunction. Based on the values of these Variables, the BendersBFunction is responsible for updating the Constraints of the HydroUnitBlocks. By isolating the independent terms in constraints (4), we can rewrite them as

$$v_{j,t}^i - \left(\sum_{l=(d,j) \in \mathcal{L}^i} f_{t-\tau_l^i, \text{dn}, l}^i - \sum_{l=(j,d) \in \mathcal{L}^i} f_{t-\tau_l^i, \text{up}, l}^i \right) = \tilde{v}_{i,j} + F_{j,t}^i, \quad \forall j \in \mathcal{N}^i \text{ and } t = 0. \quad (6)$$

Notice that the independent terms appear in the right-hand side of this constraint. Since it is an equality constraint, the left- and right-hand sides of the RowConstraint representing this constraint must be equal to $\tilde{v}_{i,j} + F_{j,t}^i$ and, therefore, are identically affected by the initial volumes of the reservoir. This means that we could define the mapping M in the BendersBFunction (see §4.13) as

$$M(x) = x + F,$$

where

$$F = \left[\underbrace{F_{0,0}^0 \quad \dots \quad F_{r_0-1,0}^0}_{\text{HydroUnitBlock } 0} \quad \dots \quad \underbrace{F_{0,0}^k \quad \dots \quad F_{r_k-1,0}^k}_{\text{HydroUnitBlock } k} \quad \dots \quad \underbrace{F_{0,0}^{H-1} \quad \dots \quad F_{r_{H-1}-1,0}^{H-1}}_{\text{HydroUnitBlock } H-1} \right]^\top. \quad (7)$$

Since the mapping matrix A is very sparse (it is the identity matrix), it would be convenient to use the sparse representation of the mapping. The number of rows and columns of the mapping matrix are equal to the total number of reservoirs. Also the number of nonzero entries in the mapping matrix is equal to that number. Therefore, in the specification of the BendersBFunction, we would have

$$\text{NumVar} = R, \quad \text{NumRow} = R, \quad \text{and} \quad \text{NumNonzero} = R,$$

where R is the total number of reservoirs, i.e., $R = \sum_{i=0}^{H-1} r_i$. Since the A matrix is the identity matrix, the *variables* NumNonzeroAtRow, Column, and A do not need to be provided. But just as an exercise, we explain how the A matrix could be provided in terms of the *variables* NumNonzeroAtRow, Column, and A. Since there is a single nonzero entry at each row, the *variable* NumNonzeroAtRow would be a vector of size NumRow (i.e., R) with all entries equal to 1:

$$\text{NumNonzeroAtRow} = (1, \dots, 1).$$

The *variable* Column would be the vector

$$\text{Column} = (0, 1, \dots, R-1)$$

and the *variable* A, containing the nonzero entries of the mapping matrix, would be the vector of size NumNonzero (i.e., R) given by

$$A = (1, \dots, 1).$$

Finally, the *variable* b, containing the constant term of the vector, would be the vector of size NumRow (i.e., R) given by

$$b = F,$$

with F given by (7).

Now, to each row of the mapping, we must associate the corresponding RowConstraint. In a HydroUnitBlock, constraints (6) belong to a two-dimensional group of RowConstraints which is indexed over the dimensions (inner) TimeHorizon and NumberReservoirs. Thus, in that group of RowConstraints, the j -th constraint in (6) is represented by the RowConstraint whose (two-dimensional) index is $(0, j)$. In the specification of the BendersBFunction, each of those RowConstraints would be indicated by an AbstractPath starting from the BendersBlock, i.e., having the BendersBlock as the reference Block.

Considering the structure of the BendersBlock illustrated by Figure 2 and the structure of the UCBlock depicted in Figure 3, assuming the HydroSystemUnitBlock is the first UnitBlock in the UCBlock, an AbstractPath from the BendersBlock to the “final volume” Constraint with (two-dimensional) index $(0, j)$ of the i -th HydroUnitBlock would be given by

$$O \rightarrow B(0) \rightarrow B(i) \rightarrow C(11, j), \quad (8)$$

where $B(k)$ indicated the k -th nested Block of a Block and $C(k_1, k_2)$ indicates the Constraint whose element index is k_2 within the group of Constraints whose index is k_1 .

To understand this path, we can start from the RowConstraint of interest and navigate up to the reference Block. Since the two-dimensional index of the target RowConstraint is $(0, j)$, its element index is j according to the definition of “element index” of a Constraint in an AbstractPath (see (1) for details). Moreover, that RowConstraint belongs to the group of Constraints whose index is 11 (the “final volume” constraints are the twelfth group of Constraints in a HydroUnitBlock; see the comments of the HydroUnitBlock class for details). The target RowConstraint belongs to the i -th HydroUnitBlock, i.e., the i -th nested Block of the HydroSystemUnitBlock. Hence, we have the node $B(i)$. The HydroSystemUnitBlock is the first nested Block of the UCBlock. Hence, we have the node $B(0)$. Finally, the UCBlock is the inner Block of the BendersBFunction, which is the Function of the FRealObjective of the BendersBlock. Hence, we have the node O .

The AbstractPaths of those RowConstraints must be given as a vector of AbstractPath. From the general path (8) to a RowConstraint, it is straightforward to construct a vector of AbstractPaths to those RowConstraints. By concatenating the sequence of nodes of all AbstractPaths, we would have the following vectors of size $4R$:

```
PathNodeTypes = ('O', 'B', 'B', 'C', ..., 'O', 'B', 'B', 'C')
PathGroupIndices = (inf, 0, 0, 11, ..., inf, 0, i, 11, ..., inf, 0, R - 1, 11)
PathElementIndices = (inf, inf, inf, 0, ..., inf, inf, inf, r0 - 1,
                      inf, inf, inf, 0, ..., inf, inf, inf, r1 - 1,
                      ...,
                      inf, inf, inf, 0, ..., inf, inf, inf, rH-1 - 1).
```

Since each AbstractPath has four nodes and there are a total of R AbstractPaths, we would have

$$\text{PathDim} = R, \quad \text{TotalLength} = 4R, \quad \text{and} \quad \text{PathStart} = (0, 4, 8, \dots, 4(R - 1)).$$

Lastly, since all those RowConstraints represent equality constraints, the ConstraintSide *variable* would be a vector of size NumRow (i.e., R) with all elements equal to ‘B’, meaning that both sides of the RowConstraint are equally affected:

$$\text{ConstraintSide} = ('B', \dots, 'B').$$

But since all elements are equal to ‘B’, the ConstraintSide *variable* does not need to be provided.

8.1.2 StochasticBlock

The *group* Block of the StochasticBlock must contain the description of the associated BendersBlock. Moreover, the StochasticBlock also requires a vector of SimpleDataMapping which, in the case of the SSV, defines how the data of the UCBlock is updated considering a given sub-scenario. A sub-scenario is a vector of double containing a realization of the stochastic data of the UCBlock. Each SimpleDataMapping (see §3.3.2) would be associated with a particular part of the sub-scenario and a particular stochastic data of some :Block. A SimpleDataMapping requires five elements:

- A :Block whose data is affected.
- The name of the function that will be responsible for updating the data of that :Block.
- An ordered multiset of indices that specifies which elements of the sub-scenario should be considered. This multiset must therefore contain elements between 0 and $n - 1$, where n is the size of the sub-scenario. We call this multiset the SetFrom multiset.
- An ordered multiset of indices that specifies which elements of the :Block will be updated. We call this multiset the SetTo multiset.

- The type of the data that will be updated.

(A multiset is a set that can contain multiple instances of the same element. For instance, $\{2, 2, 4, 4, 7\}$ is a multiset with cardinality 5. An ordered multiset is a multiset in which the order of the elements matter. For instance, $\{2, 2, 4, 4, 7\}$ and $\{2, 4, 4, 2, 7\}$ are different multisets. In most cases, if not all, the `SetFrom` and `SetTo` multisets will be simple sets.) The `SetFrom` multiset defines a mapping $\mathcal{M} : \mathbb{R}^n \rightarrow \mathbb{R}^m$, where n is the size of the sub-scenario and m is the cardinality of the `SetFrom` multiset. Let s_i denote the i -th element of the `SetFrom` multiset. For a given sub-scenario ξ , $\mathcal{M}(\xi)$ is the vector whose i -th element is ξ_{s_i} . The `SetTo` multiset will be used to associate each element of $\mathcal{M}(\xi)$ to a particular element of in the data structure of the `:Block`.

Consider, for example, a `UCBlock` that has N `UnitBlocks`, one of them being a `HydroSystemUnitBlock` and two of them being `ThermalUnitBlocks`. Suppose that the maximum power of the first `ThermalUnitBlock` is stochastic for the inner stages 2, 3, and 5 and that the maximum power of the second `ThermalUnitBlock` is stochastic for the inner stages 0, 1, 2, and 3. Suppose also that the `NetworkBlock` associated with the inner stage 1 has 10 nodes, indexed from 0 to 9, and that the active demand at the node with index 4 is stochastic. Finally, suppose that the fourth `HydroUnitBlock` of the `HydroSystemUnitBlock` has some stochastic inflows. The inflow occurs at each reservoir of a `HydroUnitBlock` at each inner stage. To simplify, let us suppose that the inflows at the second reservoir at the inner stage 0 and the inflows at the fifth reservoir at the inner stage 15 are stochastic.

We could define a sub-scenario ξ as a vector in $\mathbb{R}^{10}$ such that

- (ξ_0, ξ_1, ξ_2) represent the maximum power of the first `ThermalUnitBlock` for the inner stages 2, 3, and 5.
- $(\xi_3, \xi_4, \xi_5, \xi_6)$ represent the maximum power of the second `ThermalUnitBlock` for the inner stages 0, 1, 2, and 3.
- (ξ_7) represent the active demand of the `NetworkBlock` associated with the inner stage 1 at node with index 4.
- (ξ_8, ξ_9) represent the inflows at the second and fifth reservoirs of the fourth `HydroUnitBlock` at the inner stages 0 and 15, respectively.

In this case, we can construct four `SimpleDataMappings` to update the stochastic data of these `Blocks` and one `SimpleDataMapping` to update the `BendersBFunction` (which depends on the inflow at the first inner stage). First of all, here is a list of functions that are registered in the methods factory. In the context of the SSV, they can be used to define the stochastic data (through the `SimpleDataMapping`) as we will see in the sequel.

- `NetworkBlock::set_active_demand`
This function updates the active demand of a `NetworkBlock`. The active demand is represented by a one-dimensional array whose i -th element is the active demand at the i -th node.
- `ThermalUnitBlock::set_maximum_power`
This function updates the maximum active power output of a `ThermalUnitBlock`. The maximum active power output is represented by a one-dimensional array whose t -th element is the maximum active power output at the inner stage t .
- `HydroUnitBlock::set_inflow`
This function updates the inflow of a `HydroUnitBlock`. The inflow is represented by a two-dimensional array whose element at position (i, t) is the inflow at node i at the inner stage t .
- `IntermittentUnitBlock::set_maximum_power`
This function updates the maximum potential power production of a `IntermittentUnitBlock`. The maximum potential power production is represented by a one-dimensional array whose t -th element is the maximum potential power production at the inner stage t .
- `BendersBFunction::modify_constants`
This function updates the constant b of the mapping of the `BendersBFunction` (see (3) in §4.13).

Let us begin with the first `ThermalUnitBlock`. Its data is given in the first two entries of the sub-scenario and is associated with the maximum power at the inner stages 2, 3, and 5. Therefore, we have $\{0, 1, 2\}$ as the `SetFrom` set and $\{2, 3, 5\}$ as the `SetTo` set. Since the maximum power is of type `double`, its data type is 'D'. The name of the function responsible for updating the maximum power is `ThermalUnitBlock::set_maximum_power`. Finally, we need to specify that this data is associated with our first `ThermalUnitBlock`. This is done by providing an

AbstractPath to that Block. Suppose that this ThermalUnitBlock is the second sub-Block of the UCBLOCK so that its group index is 1 (this ThermalUnitBlock would be the UnitBlock with index 1 in Figure 3 and associated with *group* UnitBlock_0 in the specification of the UCBLOCK; see §4.4). The path from the BendersBlock to this ThermalUnitBlock would then be given by

$$O \rightarrow B(0) \rightarrow B(1).$$

Analogously, for the second ThermalUnitBlock, we would have $\{3, 4, 5, 6\}$ as the SetFrom set, $\{0, 1, 2, 3\}$ as the SetTo set and, assuming it is the fourth sub-Block of the UCBLOCK (and thus its group index is 5), the path to this Block would be

$$O \rightarrow B(0) \rightarrow B(5).$$

For the NetworkBlock, we would have $\{7\}$ as the SetFrom set and $\{4\}$ (the index of the node) as the SetTo set. The name of the function that updates this data is `NetworkBlock::set_active_demand` and the data is also of type double (indicated by ‘D’). Finally, we need to specify the path from the BendersBlock to this NetworkBlock. As we see in Figure 3, the NetworkBlocks come after the UnitBlocks. Since in our example the UCBLOCK has N UnitBlocks and the NetworkBlock of interest is associated with stage 1 (the stages being indexed from 0 to T^{in}), the group index of this NetworkBlock would be $N + 1$. Therefore, the path from the BendersBlock to this NetworkBlock would be given by

$$O \rightarrow B(0) \rightarrow B(N + 1).$$

For the HydroUnitBlock, we would have $\{8, 9\}$ as the SetFrom. Since the inflow of a HydroUnitBlock is associated with a two-dimensional data structure, the specification of the SetTo is a little bit more complicated (but not too much). This data structure, let us call it *inflow*, is such that `inflow[i][t]` gives the inflow at reservoir i at the inner stage t . As you may notice, a single inflow is identified by two indices (the index of the reservoir and the inner stage) while SetTo is a set of simple indices. So, we must convert a pair of indices into a single index. The conversion is made considering that a multi-dimensional structure is vectorized by making its last dimension varying faster. This means that for a two-dimensional matrix A with m rows $a_0, \dots, a_{m-1}$ and n columns, its vectorized form is given by arranging its rows in sequence: $(a_0, a_1, \dots, a_{m-1})$. Then, the element at position (i, j) in the matrix A is the element at position $i \times n + j$ in its vector representation. See (1) for the index of an element in an arbitrarily multi-dimensional array. The pairs of indices that we need to convert are $(1, 0)$ (the second reservoir, whose index is 1, and the inner stage 0) and $(4, 15)$ (the fifth reservoir, whose index is 4, and the inner stage 15). Since the fourth HydroUnitBlock has r_3 reservoirs and the inner time horizon is T^{in} , the data structure representing the inflows is a $r_3 \times T^{\text{in}}$ two-dimensional array. Thus, the single indices associated with $(1, 0)$ and $(4, 15)$ are T^{in} and $4T^{\text{in}} + 15$, respectively. Thus, we would have $\{T^{\text{in}}, 4T^{\text{in}} + 15\}$ as the SetTo set. The name of the function that updates the inflow is `HydroUnitBlock::set_inflow` and the data is of type double (indicated by ‘D’). Finally, we need to provide the path from the BendersBlock to this HydroUnitBlock. We are assuming that the HydroSystemUnitBlock is the first Block in the UCBLOCK. Therefore, since our HydroUnitBlock of interest has index 3 (it is the fourth HydroUnitBlock of the HydroSystemUnitBlock), the path would be given by

$$O \rightarrow B(0) \rightarrow B(0) \rightarrow B(3).$$

As we saw in §8.1.1, the mapping of the BendersBFunction depends on the inflows at the first inner stage. In our example, the inflow at the second reservoir of the fourth HydroUnitBlock at the inner stage 0 is stochastic. Therefore, we also need a SimpleDataMapping to update the mapping of the BendersBFunction. In that mapping, the inflows appear in the vector F defined in (7). In F , the initial inflow at the second reservoir of the fourth HydroUnitBlock is linked with index $r_0 + r_1 + r_2 + 1$. Thus, in the SimpleDataMapping used to update the BendersBFunction we would have $\{8\}$ as the SetFrom set (since the initial inflow at the second reservoir of the fourth HydroUnitBlock is given by ξ_8) and $\{r_0 + r_1 + r_2 + 1\}$ as the SetTo set. The name of the function that updates the constant part of the mapping of the BendersBFunction is `BendersBFunction::modify_constants` and the data is of type double (indicated by ‘D’). The path from the BendersBlock to the BendersBFunction is given by

$$O.$$

In the specification of the StochasticBlock, these SimpleDataMappings must be provided as a vector of SimpleDataMapping (see §3.3.2.2). Since there are five SimpleDataMappings, we have `NumberDataMappings = 5`. All stochastic data have a type double, so `DataType` does not need to be provided. Anyway, it could be specified as

$$\text{DataType} = (\text{'D'}, \text{'D'}, \text{'D'}, \text{'D'}, \text{'D'}).$$

SetFrom	SetTo
$\{0, 1, 2\}$	$\{2, 3, 5\}$
$\{3, 4, 5, 6\}$	$\{0, 1, 2, 3\}$
$\{7\}$	$\{4\}$
$\{8, 9\}$	$\{T^{\text{in}}, 4T^{\text{in}} + 15\}$
$\{8\}$	$\{r_0 + r_1 + r_2 + 1\}$

Table 7 The **SetFrom** and **SetTo** sets.

All multisets can be provided as Subsets. However, since the representation of a Range is more compact for sets with at least three elements, we will use it whenever is convenient. Table 7 shows the sets that we need to provide. The first two SetFrom sets and the second SetTo set have at least three elements, so we will use a Range to represent each of them. We begin with the SetSize *variable*. This one-dimensional *variable* indicates the size or the type of each set. For a Subset, we must include its size in SetSize. For a Range, we must include the number 0 in SetSize. Thus, in our example, this *variable* would be given as

$$\text{SetSize} = (0, 3, \quad 0, 0, \quad 1, 1, \quad 2, 2, \quad 1, 1).$$

Recall that the elements at positions $2i$ and $2i + 1$ correspond to the sizes (or types) of the i -th SetFrom and SetTo sets. Now, we must specify the elements that belong to each set. For a Range representing the set $[a, b)$, we must provide a and b . For a Subset, we must provide its elements. Thus, the SetElements *variable* would be given as

$$\text{SetElements} = (0, 3, 2, 3, 5, \quad 3, 7, 0, 4, \quad 7, 4, \quad 8, 9, T^{\text{in}}, 4T^{\text{in}} + 15, \quad 8, r_0 + r_1 + r_2 + 1).$$

(The first SimpleDataMapping has $\{0, 1, 2\}$ as its SetFrom set and $\{2, 3, 5\}$ as its SetTo set. In this case, the SetFrom can be represented as the Range $[0, 3)$. The SetTo cannot be represented as a Range (it is not an interval of integers), so we need to explicitly list all its elements. Thus, the first elements of

SetSize are 0 and 3 (indicating that the first SetFrom set is a Range and the first SetTo set is a Subset with three elements). Moreover, the first elements of SetElements are 0 and 3 (describing the first SetFrom) and 2, 3, and 5 (describing the first SetTo). The next elements of SetSize and SetElements are those describing the SetFrom and SetTo sets of the second SimpleDataMapping. In the second SimpleDataMapping, we have $\{3, 4, 5, 6\}$ as the SetFrom set and $\{0, 1, 2, 3\}$ as the SetTo set. These sets can be represented as the Ranges $[3, 7)$ and $[0, 4)$, respectively. So, the next elements of SetSize are 0, 0 (indicating that both sets are Ranges) and the next elements of SetElements are 3, 7, 0, 4. And so on.)

The FunctionName *variable* must contain the names of the functions (registered in the “methods factory”) responsible for updating the data of the Blocks. In our example, we would have

```
FunctionName = ("ThermalUnitBlock::set_maximum_power",
               "ThermalUnitBlock::set_maximum_power",
               "NetworkBlock::set_active_demand",
               "HydroUnitBlock::set_inflow",
               "BendersBFunction::modify_constants").
```

We must also provide a vector of AbstractPath to the affected Blocks and the BendersBFunction. Since we have five SimpleDataMappings, we also have five paths. So, we have PathDim = 5. In our example, the first three paths have size 3, the fourth path has size 4, and the last path has size 1. So, we have

$$\text{TotalLength} = 14 \quad \text{and} \quad \text{PathStart} = (0, 3, 6, 9, 13).$$

We also have

```
PathNodeTypes = ('O', 'B', 'B', 'O', 'B', 'B', 'O', 'B', 'B', 'O', 'B', 'B', 'B', 'O')
PathGroupIndices = (inf, 0, 1, inf, 0, 5, inf, 0, N + 1, inf, 0, 1, 3, inf)
PathElementIndices = (inf, inf, inf, inf, inf, inf, inf, inf, inf, inf, inf, inf, inf, inf).
```

Since all elements of PathElementIndices are equal to inf, this *variable* does not need to be provided. Finally, we need to inform the type of each affected object. The first four objects are Blocks while the last one is a Function. Thus, we have

$$\text{Caller} = ('B', 'B', 'B', 'B', 'F').$$

8.1.3 SDDPBlock

We now describe the remaining part of the SDDPBlock. In the beginning of §8.1, we have seen how the scenarios are specified in the SDDPBlock. According to our notation, the *dimension* TimeHorizon of SDDPBlock would be equal to T^{out} . Following the example of §8.1.2, each sub-scenario (a part of the scenario that is associated with a particular outer stage) has size 10. Assuming all sub-scenarios have the same size, there would be no need to provide the *dimension* SubScenarioSize and the *dimension* ScenarioSize would be equal to $10T^{\text{out}}$ (for each stage $t \in \{0, \dots, T^{\text{out}} - 1\}$, the size of a sub-scenario associated with stage t would be 10). Also, assuming there is a total of m scenarios, the *dimension* NumberScenarios would be equal to m . In our example, we can identify three groups of related random data: maximum power of a ThermalUnitBlock, active demand of a NetworkBlock, and inflows of a HydroUnitBlock. Thus, the *dimension* NumberRandomDataGroups would be equal to 3. The first group would then have size 6 (the maximum power of both affected ThermalUnitBlocks) and the second and third groups would have each one size 2. Therefore, the SizeRandomDataGroups *variable* would be given by

$$\text{SizeRandomDataGroups} = (6, 2, 2).$$

For instance, if we had $T^{\text{out}} = 3$ and $m = 5$, the Scenarios *variable* could be the following matrix:

	Stage 0	Stage 1	Stage 2	
	1 2 5 7 6 8 7 2 4 7	3 2 7 6 7 2 8 2 3 2	0 2 7 4 4 4 6 3 7 0	Scenario 0
	7 6 6 2 3 0 0 1 1 4	7 7 7 5 3 8 2 0 4 6	2 5 6 1 7 1 3 7 0 7	Scenario 1
	1 5 8 7 0 7 5 6 7 5	3 0 2 3 1 5 7 6 3 2	6 1 3 1 2 7 2 1 4 7	Scenario 2
	8 0 4 0 1 0 0 7 1 5	6 1 5 5 0 3 1 1 0 7	5 0 5 8 0 8 8 3 4 5	Scenario 3
	5 3 0 2 3 8 8 3 5 1	8 3 6 1 7 5 6 0 0 2	3 2 1 4 1 0 4 4 1 6	Scenario 4

The green, red, and blue boxes indicate the maximum power of the ThermalUnitBlocks, the active demand of the NetworkBlock, and the inflows of the HydroUnitBlock, respectively. The gray box indicates a sub-scenario.

The last sub-Block of a HydroSystemUnitBlock is a PolyhedralFunctionBlock (see §4.7) which contains a PolyhedralFunction (see §4.7.1). For each $t \in \{0, \dots, T^{\text{out}} - 2\}$, the PolyhedralFunction associated with the outer stage t represents an approximation to the (unknown) expected future cost function. This is the function that is updated at each step of the SDDP method. This function depends on the volumes of all reservoirs at the last inner stage ($T^{\text{in}} - 1$). Following the same notation used to describe constraints (6), this function depends on the variables $v_{j, T^{\text{in}} - 1}^i$ for each HydroUnitBlock $i \in \{0, \dots, H - 1\}$ and for each reservoir $j \in \mathcal{N}^i$ of that HydroUnitBlock. The only thing that must be provided in the specification of the PolyhedralFunction is the *dimension* PolyFunction.NumVar. This *dimension* must be equal to the total number of reservoirs (which is R in our example).

If one wishes to provide an initial approximation of the future cost function, then other *variables* and *dimensions* must also be provided. The PolyhedralFunction represents a function f of the form

$$f(x) = \max_{k \in \{0, \dots, p-1\}} a_k^\top x + b_k \quad (9)$$

or

$$f(x) = \min_{k \in \{0, \dots, p-1\}} a_k^\top x + b_k. \quad (10)$$

The *dimension* PolyFunction.sign indicates whether the PolyhedralFunction has the form (9), and therefore is a convex, or (10), and thus is concave. In the SSV, the future cost function is convex and has the form (9), so that PolyFunction.sign should be true. The *dimension* PolyFunction.NumRow indicates the number of affine functions that define the polyhedral function. In (9) and (10), this number is p . The vectors a_k and the constants b_k are provided in the PolyFunction.A and PolyFunction.b *variables*, respectively. PolyFunction.A must contain a matrix $A \in \mathbb{R}^{p \times R}$ such that, for each $k \in \{0, \dots, p - 1\}$, a_k is its k -th row, and PolyFunction.b is a one-dimensional array whose k -th element is b_k . Finally, it is also possible to indicate a global lower (or upper) bound

on the value of the function by setting the `PolyFunction_lb` *variable*. In the SSV, a valid global lower bound on the value of the future cost function is zero.

In the specification of the `SDDPBlock`, it is necessary to specify a path to each of these `PolyhedralFunctions`. For each outer stage $t \in \{0, \dots, T^{\text{out}} - 2\}$, the required path is an `AbstractPath` from the `BendersBlock` associated with stage t to the `PolyhedralFunction`. Recall that we assumed that the `HydroSystemUnitBlock` is the first sub-Block of the `UCBlock`, so that its group index is 0. Thus, the path from the `BendersBlock` to the `PolyhedralFunction` would be the following:

$$O \rightarrow B(0) \rightarrow B(0) \rightarrow B(H). \quad (11)$$

We have seen in §8.1.2 an example on how to specify a vector of `AbstractPaths`. Here we assume that all paths are equal (that is, we assume that the `HydroSystemUnitBlock` is the first sub-Block of `UCBlock` and that the `PolyhedralFunctionBlock` is the last sub-Block of the `HydroSystemUnitBlock`). In this case, a vector with a single `AbstractPath` can be provided. Thus, the `PathDim` *dimension* would be 1 and, according (11), we would have `TotalLength` = 4, `PathStart` = (0) and

$$\begin{aligned} \text{PathNodeTypes} &= ('O', 'B', 'B', 'B') \\ \text{PathGroupIndices} &= (\text{inf}, 0, 0, H) \\ \text{PathElementIndices} &= (\text{inf}, \text{inf}, \text{inf}, \text{inf}). \end{aligned}$$

Again, since all elements of `PathElementIndices` are equal to `inf`, this *variable* does not need to be provided.

The `SDDPBlock` also requires an admissible state for each outer stage. In the SSV, a state for a given stage is formed by the initial volumes of all the reservoirs at that stage. Thus, the `StateSize` would be equal to R , the total number of reservoirs. An admissible state for stage $t \in \{0, \dots, T^{\text{out}} - 1\}$ is a state for which the sub-problem at stage t is feasible. Trivial examples of non-admissible states include those in which the volumes of the reservoirs are too low (so that the demand may not be satisfied) or too high (so that the maximum volume limit of the reservoir may be exceeded if there is enough inflow). The state is a vector that must follow the same convention as that chosen for the vector x defined in (5). So, a state s^t for the outer stage $t \in \{0, \dots, T^{\text{out}} - 1\}$ must be a vector in $\mathbb{R}^R$ with the form

$$s^t = \left[\underbrace{s_{0,0}^t \dots s_{0,r_0}^t}_{\text{HydroUnitBlock } 0} \dots \underbrace{s_{i,0}^t \dots s_{i,r_i}^t}_{\text{HydroUnitBlock } i} \dots \underbrace{s_{H-1,0}^t \dots s_{H-1,r_{H-1}}^t}_{\text{HydroUnitBlock } H-1} \right] \quad (12)$$

where $s_{i,j}^t$ is the initial volume for the j -th reservoir of the i -th `HydroUnitBlock`. The `AdmissibleState` *variable* would then be a one-dimensional array of size RT^{out} containing the states for all outer stages arranged as follows:

$$\text{AdmissibleState} = (s^0, \dots, s^{T^{\text{out}}-1}),$$

where $s^t \in \mathbb{R}^R$ is given by (12) for each $t \in \{0, \dots, T^{\text{out}} - 1\}$.

9 Conclusion

This document is intended as a quick reference manual for anybody who is required to construct SMS++ NETCDF input files. It should be remarked that basically all the `:Block` and `:Configuration` objects are supposed to have an in-memory interface to initialize the data. Thus, a workable way to produce NETCDF input files for those is to use the in-memory interface, and then the `serialize()` method to save it into a file. Yet, NETCDF has interfaces for all primary programming languages, hence files with the required format can be written independently by any SMS++ object.

References

- [1] Unidata “The NETCDF software”, <http://doi.org/10.5065/D6H70CW6>, 2019.